PRESERVING NETWORK TRAFFIC CLASSIFICATION USING MACHINE LEARNING TECHNIQUES

By

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Declaration / Statement of Originality

I, Toluwalase Emmanuel Oludipe, declare that the work presented in this thesis, titled "PRESERVING PRIVACY NETWORK TRAFFIC CLASSIFICATION USING MACHINE LEARNING TECHNIQUES", is entirely my own and has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution.

All sources of information, data, figures, and ideas from other authors have been properly cited. Where this research involved collaboration, the exact nature and extent of my contribution is clearly indicated.

Ethics and Compliance: Any dataset used was processed in accordance with

University policies and applicable regulations. Necessary approvals (if any) were obtained and

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Abstract

This dissertation explores if the encrypted traffic in network can be properly classified by supervised machine-learning models trained purely based on the metadata without exposing the user's privacy. This paper follows a privacy-by-design quantitative research approach and compares three classifiers, Logistic Regression, Random Forest, and XGBoost across two benchmark data sets, namely CICIDS2017 and CSE-CIC-IDS2018. In order to compare the results, the models were trained and validated using identical preprocessing and evaluation conditions with only flow-level metadata (packet sizes, durations, and timings).

Results show that ensemble and boosting algorithms attain state-of-the-art results with an accuracy over 99% with good cross-dataset generalisation. A noise injection Gaussian experiment was performed to quantify the trade-off of privacy-utility and found that light anonymisation ($\sigma \leq 0.05$) maintained predictive ability within 0.5% of baseline performance. Interpretability was explored using SHAP analysis which highlighted key metadata descriptors affecting classification and reducing the opacity of the complex models.

The research provides a single assessment framework of combined performance optimisation, explainable AI and regulatory compliance. It empirically demonstrates that privacy preserving intrusion detection is operational and scalable under the guidance and requirements of GDPR and ISO 27001. By showing the potential of improving the use of metadata-only models to provide precise, understandable and compliant threat detection, the research contributes to the practice of responsible Artificial Intelligence and to the broader efforts for a safe and ethical functioning digital infrastructure, as outlined in UN Sustainable Development Goals 9 and 16 respectively.

Chapter 1

Introduction

1.1 Background and Context

Encrypted internet traffic has increased exponentially, which essentially has revolutionized network security. Encryption protocols like HTTPS and TLS are increasingly being utilized globally to protect web traffic since they are now used by more than 95 percent of web traffic as compared to less than 50 percent a decade ago (Hancock, 2023; Google, 2024). The requirement to ensure the security of online payment, protect personal information and meet privacy legislations causes this spurt. Although a guarantee of confidentiality, encryption poses a serious problem to network defenders.

Deep Packet Inspection (DPI) is a form of traditional traffic analysis based on the analysis of the packet payload to detect signatures or anomalies. Nevertheless, encrypted protocols, such as TLS 1.3, and QUIC, make packet content invisible, and in many cases, DPI is ineffective (Chowdhury et al., 2024). This transition has posed a cybersecurity paradox where, on the one hand, encryption can shield users against surveillance, and blinds security tools that may fail to detect malware, phishing, and data exfiltration buried inside encrypted streams on the other hand (Shen et al., 2023).

The legal requirements restrict this disparity of visibility even more. Laws like the General Data Protection Regulation (GDPR) have imposed strict limitations on the accessibility and the procedure of personal data without authorization. Invasive surveillance that includes inspection of the user content can go against the supposed principles of data protection, that is, privacy by design and data minimisation (European Data Protection Board, 2024). The security teams are thus compelled to come up with mechanisms of detecting the threats without decryption and scrutinising of sensitive information.

This situation creates an immediate requirement to create new privacy-security balancing strategies. An emerging area is analysing the flow level, based on the metadata information including packet sizes, timing, and duration that can be still observed even in encryption. Such metadata may permit the detection of threats when used in conjunction with machine learning without violating privacy conventions.

1.2 Research Significance

The fact that there is a high level of encryption on one hand and a rapid growth of cyber threats on the other hand makes the subject of privacy-preserving encrypted traffic classification not only important but also urgent. With most of internet traffic being encrypted, security teams have a crucial visibility gap: when they are unable to differentiate between malicious traffic and legitimate ones, threats can pass through. This is no longer hypothetical, as more than 85 percent of the attacks on cyberspace today take place in encrypted channels (Desai et al., 2022). Deployment of systems that identify the threats without decryption of the traffic is hence the highest priority in modern cybersecurity plans.

This issue is more than technical. With enterprises, telecom providers and cloud platforms switching to zero-trust architectures and experiencing increasing compliance pressure, they have to secure their networks and demonstrating user privacy is intact. Although this space is driven by solutions with immediate real world use cases, specifically in next generation intrusion detection systems, secure network analytics and governance, which can promote security and compliance with regulations.

In contrast to signature-based solutions, ML models can detect unperceivable patterns in flow-level metadata (packet size, packet timing, and direction of sequence without accessing packet payloads) (Alwhbi, Zou and Alharbi, 2024). More recent work demonstrates that we are now able to infer the type of encrypted traffic as well as the presence of malware based entirely on side-channel features on both neural nets (deep learning) and classical models (Akbari et al., 2022). Nominal classification can be done without content inspection, which goes in line with privacy by design.

This research is expected to contribute to studying privacy-wise ML-based classifier development and testing that will bring a scalable, flexible model of current network defence. It is of interest to academic and industry stakeholders, and it provides a potential plan towards the possibility of effective cybersecurity in encrypted environments that do not interfere with user privacy and compliance with any regulations.

1.3 Problem Statement and Research Gap

Even though research is currently being done regarding the encrypted traffic analysis, there is still a considerable gap in the practical experience and knowledge. The available solutions lack in either of the two ways, i.e., they are either privacy-invasive or otherwise, not practically tested in real-life scenario. On the one hand, other network security strategies attempt to evade brute-force encryption rearrangement e.g. tapping traffic through company proxies or requiring clients to load custom certificates- TLS break-and-inspect. These methods undermine the privacy and trust of the users by basically engaging in a covert form of decryption, which completely conflicts with the privacy-preserving principle and in most cases is not possible in unrestricted settings (Minarik, 2019). Otherwise, lab experiments or simulations rarely demonstrate scalable deployability of proposed academic ideas that do not include payload inspection. Most of the literature reports excellent accuracy rates when the same task (classification of encrypted traffic) is performed via machine learning, however, dig deeper and some key limitations are shared among them. As an example, they often conduct an assessment on pre-selected data or with the simplification of the scenario, which does not display the uniqueness of real network traffic (Akbari et al., 2022). It is easy to hear that classes in such experiments are defined to make the task easier (e.g. by combining extremely different protocols with easily distinguishable information, such as protocol metadata; e.g. TLS handshake metadata containing the server name indication) or that models are based on aspects that may not generalize to the next generation of encrypted protocols (e.g. protocol metadata, new privacy features will conceal this information). This study therefore fills a dual gap: the lack of experimentally validated machine learning models for encrypted traffic classification that are both practically deployable and compliant with privacy regulations. Rather than inspecting payloads, it proposes a scalable, metadata-driven classification framework that aligns with privacy by design.

Research Question: This research paper seeks to design and test a privacy-preserving framework of network traffic classification that in a bid to provide cybersecurity will rely on machine learning to detect and classify network encrypted flows by identifying and classifying them without having to inspect packet payloads. Put simply, the project aims at illustrating a method capable of identifying malicious/ target traffic patterns by analysing encrypted streams of data in a way that

preserves the privacy of a user and conforms to the data protection principles applicable in the context. This goal will be realised by a set of objectives as outlined below.

1.4 Research Aim and Objectives

Research Aim: This dissertation aims to design and evaluate a privacy-preserving framework for encrypted network traffic classification using machine learning techniques that rely solely on metadata features, without payload inspection. The goal is to detect and classify malicious traffic in a way that supports cybersecurity goals while aligning with legal and ethical privacy standards.

Objectives: To fulfil the research aim, the dissertation is guided by the following SMART objectives (Specific, Measurable, Achievable, Relevant, and Time-bound):

- **Objective 1: Literature Review and Analysis** – To make a review of the existing network traffic classification methodologies under encryption (both academic and industry method), their approaches, their accuracy, and the privacy concerns. The measure of this goal can be in creating a summarised analysis of no less than 30 recent sources, which underscore gaps and unresolved dilemmas that give the research focus.
- **Objective 2: Design of Privacy-Preserving Classification Method** – To design a novel approach to classification of encrypted traffic based on supervised machine learning. This involves specifying the flow-level metadata characteristics that will be employed (e.g. the distribution of packet sizes, timing, protocol information) and the impact of privacy requirements on this set of features. The design will not explicitly examine its payload, and a documented method/architecture fit to be implemented will be a measure of success
- **Objective 3: Dataset Collection and Prototype Implementation** – To obtain or create an appropriate dataset of network traffic flows (including both benign and malicious, or multi-application) traffic and train a prototype ML classification model. This will include processing raw traffic to isolate the selected metadata characteristics, labelling flows (e.g. by application or attack type), and training the ML model. This must be implemented within project timeframe and will be declared successful once a working classifier is acquired to evaluate it.

- **Objective 4: Experimental Evaluation of Performance** – To measure the accuracy of the classification and performance of the proposed ML model using encrypted traffic data. These involve evaluating such metrics as precision, recall, F1-score, and the overall accuracy of traffic classification (i.e. differentiating between malware and normal traffic or detecting the type of application) with only encrypted-flow features. The objective will be met by conducting offline experiments (e.g. cross-validation on the dataset) and comparing results against at least one baseline (such as a simpler classifier or known approach) to demonstrate improvement or competitiveness.
- **Objective 5: Privacy Impact Assessment and Simulation** – To assess the privacy-preserving nature of the solution by analysing what sensitive information (if any) is inferred or exposed through the classification process. This may involve a qualitative discussion of GDPR compliance and a simple simulation or case study to ensure no personal data need be decrypted for the model to work. For example, verifying that the model's features do not include any content or personal identifiers and may simulate an attacker attempt to reverse-engineer user information from the available metadata. Success for this objective is achieved if the method can be demonstrated to align with privacy requirements (i.e. it uses only anonymized or non-identifying data) while still functioning effectively.

The combination of these objectives encompasses the entire potential of the research including its foundation and implementation. Every aim can be related to the particular work of the research: objective 1 is directed to knowing the state of the art (and is primarily covered in the Chapter 2), whereas Objective 2 and Objective 3 lead to the methodology and implementation (Chapter 3 and Chapter 4), and at last, Objective 4 and Objective 5 bring to the evaluation and discussion of results (Chapter 5). By fulfilling all the goals, the target of the research will be achieved and the main question solved by having the working answer and a clear evaluation of its advantages and drawbacks.

1.5 Scope of the Research

The scope of this dissertation takes into consideration the assessment of the capability of classifying encrypted network traffic by utilising machine learning without the need to obtain the payloads of packets. It focuses on processing of flow-level metadata (e.g. packet size, timing, direction) and exploration of supervised machine learning as basis of classifying network flows.

The experiments are off-line based on publicly available or simulated traffic dataset, without any real-time monitoring or live deployment.

The methodology avoids information of deep packet inspection (DPI), payload decryption, or other sensitive identifiers as the means of adhering to privacy by design. Any feature to be applied in classification will be metadata-level attributes of compliance with privacy regulations like GDPR. Besides this, the study will emulate setting up privacy preservation, such as feature anonymisation, introduction of noise to test the effects on models. The research explicitly avoids full legal compliance audits or regulatory certification. While the methodology aligns with privacy principles conceptually, formal legal review is out of scope. Similarly, the study will not attempt to build or deploy a production-grade security appliance but rather aims to prototype and evaluate a conceptually sound classification model.

These limitations will keep the project within the academic rigour and feasibility of a Master's-level dissertation that can be meaningful without excessively straining the implementation requirements.

1.6 Structure of the Dissertation

This dissertation is organized into five main chapters:

- Chapter 1 – Introduction outlines the research background, motivation, problem statement, aim and objectives, scope, and dissertation structure.
- Chapter 2 – Literature Review critically examines existing and upcoming methods of encrypted traffic classification, the place of machine learning, as well as the overlap of cybersecurity with privacy laws. It outlines the boundaries of the current work and delineates the research gap.
- Chapter 3 – Methodology provides the description of the research design, such as the selection of the dataset, feature engineering, machine learning model, and the application of privacy protection considerations. It also describes the evaluation measures that are applied to measure the performance of the model.

- Chapter 4 – Results provides results of the performance assessments, showing the specific metrics of the machine learning models, the findings of the feature engineering, and the measured effect of the implemented privacy protection factors.
- Chapter 5 – Discussion presents the experimental outcomes and provides a critical analysis of model performance under baseline and privacy-preserving conditions. It reflects on the implications of the results and their alignment with the research objectives.
- Chapter 6 – Conclusion summarises the main findings, evaluates the extent to which the research objectives were met, discusses limitations, and proposes directions for future research and application.

This structure ensures clarity and coherence in progressing from problem definition through to solution development and evaluation.

Chapter 2

Literature Review

2.1 Introduction

Encrypted network traffic has become the most common form of communication in modern digital infrastructure and comprises over 90 % of web communication (Fattahi, 2024; Desai et al., 2022). This effect is advantageous to user privacy as well as complying with a data framework, like the General Data Protection Regulation (GDPR) (European Union, 2016), but it also has the disadvantage of the loss of transparency that has its place in traditional intrusion detection methods. Traditional methods, such as deep packet inspection (DPI), have been demonstrated to be ineffective, since their method of operation is obscured as they rely on payload-level access (El-Mesady et al., 2024).

This review addresses how machine learning (ML) can assist with privacy-preserving classification of an encrypted network traffic through making use of flow-level metadata, including packet sizes, timing patterns, and byte counts but without decrypting the packet contents (Akbari et al., 2022). The approaches are particularly relevant in a cybersecurity setting where compliance, scalability and trust are important components. Nonetheless, the obstacles remain in the way of data usefulness, model fairness, and consistency with the legal limit, including privacy by design.

The chapter is thematically structured to support critical engagement rather than summary. It begins by evaluating the limitations of legacy classification methods (Section 2.2), followed by a critical review of ML approaches for encrypted traffic (Section 2.3). Section 2.4 addresses privacy-preserving techniques and regulatory constraints, while Section 2.5 compares relevant datasets and feature engineering strategies and section 2.6 synthesises unresolved issues. The aim is to identify viable pathways for machine learning to enable secure, compliant, and effective network traffic classification under encryption.

2.2 Evolution of Network Traffic Classification

Port-based identification and deep packet inspection (DPI) have been used to perform traditional network traffic classification but have been made progressively obsolete by technology variations.

During the early days of the internet, port numbers worked as a good heuristic, e.g. HTTP traffic was generally allocated on port 80.

However, dynamic or non-standard ports are commonly used in current applications, and traffic is routed through standard ports (e.g. over port 443 when HTTPS is used), so port-based methods are ineffective (Alwhbi, Zou and Alharbi, 2024). NAT (wide-spread network address translation), encryption, and port obfuscation also lower the accuracy of classification, demanding better solutions.

Early systems such as DPI, which scan packet payloads and compare protocol signatures, once offered greater granularity. However, DPI fails when payloads are encrypted (Yu et al., 2024). It is also cumbersome to maintain signature databases, and the approach lacks flexibility for emerging protocols (Celebi et al., 2022).

DPI also raises privacy concerns due to its inspection of user content, which is increasingly expected to remain confidential. These limitations have reduced the practicality of both port-based and DPI methods (Khandait & Hubballi, 2024).

The popularity of using encrypted protocols like TLS 1.3 and QUIC have made it more severe. TLS 1.3 encrypts almost the entire handshake including the Server Name Indication (SNI) when there is use of Encrypted Client Hello (Zhou et al., 2024). Built on UDP, QUIC further obfuscates the protocol by incorporating a newer protocol, TLS 1.3, along with encrypting most metadata provided at the transport layer (Kosek et al., 2022). Both forward secrecy and 0-RTT modes make it such that attacks of passive observers with server keys cannot decrypt flows (Kosek et al., 2023). The result is legacy monitoring tools that cannot operate unless taking controversial steps that decrypt traffic (i.e. man-in-the-middle) and may lose user trust and legal compliance (Rezaei and Liu, 2018). Such processes make classical traffic investigation out-dated and require approaches that work in encrypted contexts.

Table 2.1: Comparison of Traditional vs Encrypted Traffic Techniques

Method	Visibility	Privacy Impact	Accuracy	Deployment Feasibility
Port-based Classification	Low (easily bypassed)	Minimal	Low (due to port obfuscation)	High (simple to implement)
Deep Packet Inspection	High (on unencrypted)	High (content inspection)	High (for legacy traffic)	Medium (high processing cost)
Flow-based ML (Metadata)	Moderate	Low (no content access)	Moderate–High (task dependent)	High (no payload access needed)
Deep Learning (e.g. CNNs)	Moderate	Low–Moderate	High (with good features)	Medium–Low (resource-intensive)
Hybrid (Metadata + Payload)	High (if payload used)	Medium–High	High (but privacy concern)	Low (complex and intrusive)

In their response, scholars have turned to flow-level metadata and machine learning (ML). Instead of using payload access, classifiers require side-channel characteristics such, as packet sizes, inter-arrival time, flow lifetime, and flow directions, which cannot be deleted even after encryption. The technique allows the inference of applications without the breach of the encryption limits (Mazel et al., 2022). Previously, feature engineering was paired with algorithms used to classify encrypted traffic and such as random forests and support vector machines (SVMs). At times failing to work better than port-based models, they still come close to DPI-level accuracy in certain situations and ultimately maintain privacy (Kumar et al., 2024).

Deep learning has also promoted the development of classification. Larger neural network models can be trained to discover temporal dependencies and byte patterns on raw flow data, e.g. convolutional neural networks (CNNs) and recurrent neural networks (RNNs). The specifications of these methods make it possible to correctly filter applications like VPN, streaming, and chat, even in cases where the contents of the packets are illegible (Bechoo and Mungur, 2024). Notably, they do not interfere with privacy-preserving largely because they do not decrypt payloads.

Such shift however brings along new trade-offs. ML models are information-intensive and must be retrained continuously to be relevant as the traffic behaviours change. A danger exists as well of overfitting on residual cleartext factors like distinctive packet patterns or unencrypted marks to end users like SNI that newer protocols can conceal. It has been demonstrated that some deep models lose their accuracy with masking of such cues, which state the importance of robust feature selection. More so, ML has scalability and compliances advantages, although it does not necessarily carry with it the fine-grained insight of DPI (Swathika, 2024; Peiyuan et al., 2024).

To conclude, traffic classification has gone through a shift, where packet inspection has been abandoned to be replaced by indirect, metadata-based inspection with machine learning. Such a transition emphasizes that there is no time like the present to develop classification procedures that can operate even under encryption but do not violate legal or ethical limits of privacy. The next section explores technical and regulatory strategies for preserving privacy in encrypted traffic analysis a critical foundation before evaluating machine learning-based solutions.

2.3 Privacy-Preserving Techniques and Regulatory Considerations

Considering the drawbacks of the legacy mechanisms and the complexities provided by the encrypted data, secrecy of the user has acquired not only a legal status but also a technical one. This section critically analyses privacy-protecting methodologies such as anonymisation, differential privacy and federated learning together with regulatory frameworks that inform the implementation of the methods.

2.3.1 Data Anonymisation and Feature Suppression

Among the common methods to protect the data of users, it is possible to note the suppression of identifiable features of network flows or the anonymisation of user data in them. IP masking, field suppression, and generalisation are implementation techniques that are normally applied to mitigate the risks of re-identification and meet the principles of data minimisation (Miller et al., 2021).

The practices are commonly used prior to sharing of the datasets or training the model to comply with regulation and ethical treatment of traffic data. However, their effectiveness is contested.

Removing key classification features can reduce ML model accuracy, especially with excessive suppression that erodes signal quality (Chen et al., 2023). Studies show anonymisation does not prevent inference-based attacks. For instance, Tramer et al. (2022) demonstrate that packet size distributions or inter-arrival times can be used to accurately re-identify user behaviours even after anonymisation.

Sasy & Goldberg (2024) similarly found that benign-looking metadata can be exploited by attackers with training to discriminate user behaviour. These findings highlight a tension between privacy and utility: while obfuscation helps achieve compliance, it cannot fully eliminate exploitable patterns, calling for more structured privacy methods.

The problems of privacy and utility are related to these findings and concern the inherent conflict between them. This obfuscation is useful in meeting compliance but is not sufficient to remove latent patterns that can be exploited by side channel analysis. Operational wise, this puts at risk the idea of anonymisation as a gentle way of achieving privacy and demands better structured, formalised approaches to privacy.

2.3.2 Differential Privacy

Differential Privacy or DP has been defined as a mathematically rigorous approach to anonymisation in place of heuristic anonymisation. DP is based on the statistical theory and ensures provable guarantees because noise is added to the data, and the analysis is always insensitive to the insertion or deletion of the data of a single user (Galli et al., 2024).

DP was used in many different settings to perturb input features in network traffic classification exercises, to clean datasets, and to introduce noise into training procedures (Ziller et al., 2021). The main advantage of DP is that it is resistant to re-identification, even under the condition that attackers may have a lot of background knowledge (Altman and Cohen, 2023).

However, not unconditionally applicable in traffic analysis. The temporal and high dimensionality of the encrypted flows is such that addition of noise can have drastic consequences on the accuracy of detection. As Li et al. (2023) notice, the utility of DP is seriously dependent on its set-up; an improperly selected epsilon value can make a classifier practically useless.

This concern is also repeated by Desik and Namen (2023) who demonstrate that performance deterioration is inescapable unless the optimisations specific to a given field are put into use, like feature selection or noise budgeting. Therefore, although DP theoretically sounds appealing, its practical application to ML pipeline to encrypted traffic is still a challenge and the subject of active development.

2.3.3 Regulatory Requirements and Privacy by Design

In parallel with technical developments, regulatory frameworks like the EU General Data Protection Regulation (GDPR) exert substantial influence on privacy-preserving design. The concept of data protection by design and by default is reflected in Article 25 of the GDPR and requires organisations to place numerous privacy protection mechanisms at the very beginning of designing systems (European Union, 2016). This principle is especially relevant in the case of encrypted traffic classifiers that handle huge amounts of possibly sensitive user metadata.

Recital 78 also specifies the necessity of pseudonymisation, aggregation, and data minimisation that is directly applicable to the ML practice of feature suppression and flow-level abstraction. Veeranki (2024) notes that such principles require avoiding superfluous identifiers, restrictions in the logging of data, or auditability of models. In ML pipelines, it will involve following pre-processing approaches that obscure personal characteristics, and restrict runtime data visibility, and particularly through shared or federated datasets.

The repercussions of noncompliance are serious. They are bound by steep financial objectives and reputational threats to the extent that regulatory enforcements are often responsible for influencing organisations to deliver more towards the privacy-preserving designs as a viable operational strategy and not just as an ethical consideration. Subsequently, researchers are now required not only to report model performance, but explainability, audit trails, as well as to express privacy evaluation. Summing up, the drive towards privacy saving traffic analysis systems is informed by technical innovation and regulation requirements. The inexpensive, easy to apply countermeasures of anonymisation and feature suppression are not highly resistant to inference. Differential privacy introduces a little formalism though it needs attention in their application to keep the viability of the classifiers.

The requirements of privacy-by-design are formalised through regulatory instruments like the GDPR, so it is no longer just aspirational it is the law. All these developments together help to reaffirm that privacy should be a part of ML-based traffic classification at all levels of data preparation to model application.

Table 2.2 provides a comparative evaluation of different privacy-preserving techniques, highlighting the trade-offs in accuracy, compliance, and scalability that influence design decisions for encrypted traffic classifiers.

Table 2.2: Comparative Evaluation of Privacy-Preserving Techniques in Encrypted Traffic Classification

Technique	**Core Mechanism**	**Privacy Strength**	**Impact on Accuracy**	**Compliance Alignment**	**Scalability**
Anonymisation	Removal/generalisation of identifiers (e.g. IPs, ports)	Low–Moderate	Low–High (depends on feature loss)	Basic GDPR support	High
Feature Suppression	Selective omission of sensitive features	Moderate	Moderate (risk of underfitting)	Supports data minimisation	High
Differential Privacy	Adds calibrated noise to data or model outputs	High	High (degrades if not tuned)	Strong GDPR & DP compliance	Moderate–Low
Pseudonymisation	Tokenisation or consistent hashing of identifiers	Moderate	Low (preserves feature space)	Supports GDPR Article 4(5)	High
Federated Learning	Model training on-device without sharing raw data	Very High	Variable (depends on setup)	Aligned with privacy by design	Moderate (needs infra)

These privacy-preserving methods define the operational and ethical boundaries within which classification models must function. With this context established, the next section reviews how machine learning techniques can be applied to encrypted traffic in ways that respect privacy constraints while retaining detection accuracy.

2.4 Machine Learning in Traffic Classification

Based on the privacy requirements presented above, machine learning (ML) has been used as a tool to classify encrypted traffic with scalable solutions as well as a flexible framework by only addressing the metadata instead of the payload of packets. Classification of the encrypted flow of network traffic with ML becomes the focus of research because port-based mapping or Deep Packet Inspection (DPI) based on the legacy methods is ineffective against encryption. Furthermore, modern ML models extract the classes of traffic based on the flow-level metadata and statistical patterns detection instead of accessing payloads (Wickramasinghe et al., 2025; Liu et al., 2025).

Figure 2.1 illustrates the categorisation of machine learning techniques applied in encrypted traffic classification, helping to visualise the taxonomy discussed in Sections 2.4.1 to 2.4.4.

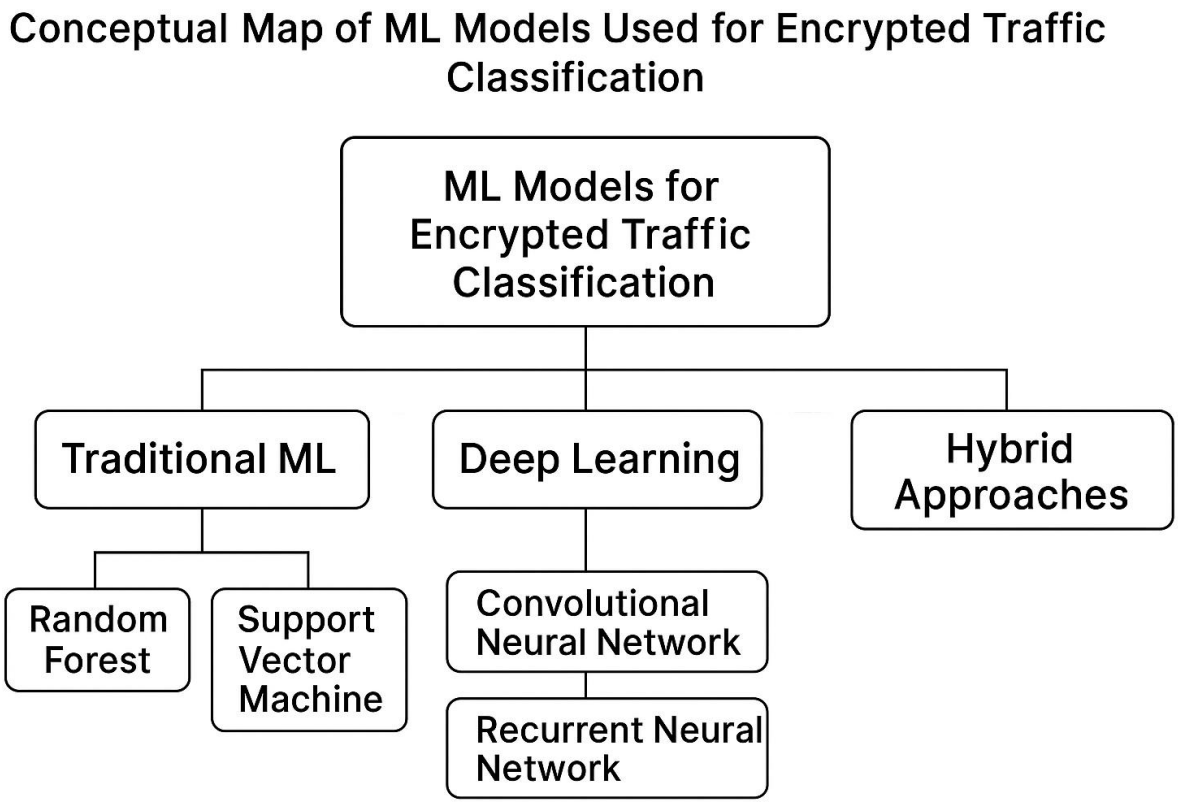

Figure 2.1: Conceptual Map of ML Models Used for Encrypted Traffic Classification

2.4.1 Supervised and Unsupervised Learning

In encrypted traffic classification, supervised learning is more popular because it is possible to obtain labelled datasets such as CICIDS2017 and ISCXVPN2016. They learn patterns using flow properties, like packet size, duration, and timing, in order to learn traffic. Research indicates that the existing types of classifiers, such as Random Forests, SVMs, K-Nearest Neighbours (KNN), can identify the encrypted flows as legitimate or malicious with a high level of accuracy (Mahmud et al., 2024; Hasan & Malik, 2024). One of their main limitations is the need to use data that is large and representative. The models may work very poorly when faced with novelty, when presented with unseen applications or changing traffic behaviour. Unsupervised learning is an alternative since it will detect outliers or a cluster of data in unlabelled traffic data. Anomalous encrypted flows have been detected using one- class SVM, K-Means, and the densities-based technique (Gallego-Mejia et al., 2022; , Mnasri et al., 2021). These models are especially well suited for privacy sensitive settings, where labelled malicious samples are hard to obtain. However, unsupervised models are hard to check and are likely to show a high false-positive rate. Some

hybrid approaches like semi-supervised learning tackle the advantages of each of these two paradigms by learning with known threats as well as being aware of new threats (Xu et al., 2023; Xie et al., 2022).

2.4.2 Deep Learning for Encrypted Traffic

Deep learning (DL) expands the ML horizon by using the representations that learn about hierarchies on raw flows sequences. The overall basis of the practical application of CNNs and RNNs is the power to recognize spatial and temporal relations in encrypted traffic patterns (Akbari et al., 2022).

As an example, sequential length and inter-arrival time of packets may be processed through the layers of a CNN to determine the type of application (such as VPN connection or streaming) that they belong to. Using Graph Neural Networks (GNNs) and dual flow embedding functions, Liu et al. (2025) were able to reach an accuracy of more than 96 percent.

Regardless of these findings, deep models have serious trade-offs. They are computationally demanding, subject to overfitting and with frequent black box behaviour. As demonstrated by Malekghaini et al. (2023), a great deal of DL models with good performance cannot be generalised beyond the datasets to which they are trained.

In concept drift, with the time variation in traffic pattern, DL classifiers are even more unstable. To cope with these risks continuous retraining and robust feature selection activities are prescribed.

Another urgent matter is interpretability. Transparency of models is crucial in cybersecurity to establish the confidence of the operators and justify alerting decisions. Interpretable AI methods like SHAP, LIME are being adopted to determine how the features are impacting the model outputs to ensure the analysts can be satisfied that privacy principles are maintained and that the model is not exploiting plaintext leakages (intended or otherwise) (Kumar & Suthar, 2024; Umbrello & Poel, 2021).

2.4.3 Metadata-Only vs. Hybrid Approaches

An important design choice in privacy-aware classification is whether to use only metadata or to use multi-strategy. Databases that rely only on metadata consider flow statistics including packet sizes; timing intervals and byte counts features which are available without packet decryption. This is quite compatible with GDPR and privacy by design principles.

Alwhbi et al. (2024) accentuate the fact that the metadata-only ML approach has become the leading paradigm in encrypted traffic classification. Liu et al. (2025) demonstrate that models that rely solely on statistical metadata can effectively differentiate VPN and QUIC, completely avoiding payload content.

Hybrid techniques are an attempt to build better identifying a classification by adding features like TLS handshake metadata, certificate fields, or by using embeddings created by deep learning. Another mixed approach that tackles a binary task is the one by Sarhangian (2024) with a CNN-LSTM combo that can reach a nearly 100 percent accuracy level.

They focus usefulness of model integration in maximising processing efficiency and accuracy, which is quite pronounced in the modern applications regarding real-time, including traffic detection and automated surveillance. The methods however can make them complex and prone to loss of privacy unless circumscribed. As an example, the unencrypted TLS field can become obsolete one day as encryption requirements change.

Critically, hybrid systems must be designed to ensure that privacy is not compromised and that model robustness is preserved in the face of protocol changes. Various approaches apply hybrid systems to leverage metadata together with the attention-based learning blocks that stand at the crossroad between performance and the privacy-protective design (Wang et al., 2023; Soni et al., 2022; Bernasconi et al., 2022).

2.4.4 Challenges: Overfitting, Generalisability, and Performance

There are common problems related to overfitting to a static or synthetic data. Most of the benchmark datasets such as CICIDS2017 contain outdated or artefact-based protocols that are not like actual encrypted flows. Wickramasinghe et al. (2025) claim that the models trained on datasets of these kinds tend to learn spurious patterns, which fail when deployment.

Generalisation remains a serious hurdle. Malekghaini et al. (2023) show that DL models trained on 2019–2020 traffic see major accuracy drops when tested on newer traffic featuring QUIC and TLS 1.3. To address this, researchers propose using domain adaptation, cross-network validation, and retraining pipelines.

Lastly, performance and scale of any system is not to be neglected. The weaker models like decision trees might be good options even though they might lose a slight portion of the accuracy especially in high-throughput environments. Practical deployments require balancing between efficiency and capability of detecting.

2.4.5 Conclusion

Machine learning provides a diversity of such tools to classify encrypted traffic, which have advantages and disadvantages. Although deep learning provides the state-of-the-art accuracy of many tasks, simpler supervised approaches have been proposed to provide better interpretability and feasibility of operation. Metadata-only approaches are much more in line with objectives of privacy than hybrid approaches, which increase accuracy but at the expense of complexity. The modelling selection should consider the trade-offs, explainability and generalisability. These considerations drive the pursued direction of supervised learning with metadata features of this dissertation, which is a realistic and privacy-sensitive basis of encrypted traffic classification.

2.5 Datasets and Feature Engineering

To effectively evaluate the privacy-preserving traffic classification models, the representative and well-labelled datasets are required. CICIDS2017 has since become a standard in the research on intrusion detection with the rich set of labels and extensive coverage of types of attacks (CIC, 2017; Ji et al., 2024). It was gathered in a supervised testbed and contains typical protocols (e.g. HTTP, HTTPS, FTP, SSH) and many different recent types of attacks (e.g. DoS, botnets, port scanning), providing a rich range over which to assess machine learning performance (Oyelakin et al., 2023). Even though useful, CICIDS2017 has considerable weaknesses, especially on privacy and encryption-minded investigation.

Most notably, CICIDS2017 only contains a small set of encrypted traffic with less than 3 percent of all flows consisting of TLS/HTTPS sessions and the data does not include modern protocols such as TLS 1.3 or QUIC. TLS traffic is of all standards (pre-2017) and therefore of less relevance

in the study of modern encrypted classification (Ji et al., 2024). Also, CICIDS2017 is a synthetically generated (e.g. IXIA PerfectStorm), i.e. the traces lack spontaneity and behavioural diversity of real-world traffic. Lastly, there is dramatic class imbalance: benign traffic far outnumbers other types of attacks (e.g. Heartbleed or Infiltration) and it will be a problem when training the ML model unless some mitigation techniques are applied (Alshamrani et al., 2024).

While CICIDS2017 is not ideal in terms of encryption diversity or realism, it remains the most practical choice for this study due to its robust labelling, documentation, and representational coverage of major attack types. Its limitations are acknowledged and mitigated through careful feature selection and simulated privacy-preserving transformations

Table 2.3 provides a comparative overview of CICIDS2017 with alternative datasets, illustrating trade-offs across encryption prevalence, attack types, and realism.

The UNSW-NB15 dataset offers another broad attack landscape, though like CICIDS2017, it features limited encrypted traffic and suffers from synthetic generation. Meanwhile, USTC-TFC2016 provides fully encrypted malware and benign traffic classes using TLS, yet its scope is narrower focusing on application-level traces and malware family identification rather than broader network behaviour (Zhang et al., 2022). No single dataset fully satisfies the needs of encrypted, privacy-preserving classification. The selection of CICIDS2017 for this dissertation stems from its extensive documentation, availability, and balanced inclusion of known attack types.

Table 2.3 Dataset Comparison Matrix

Dataset	Encrypted Traffic (%)	Attack Diversity	Realism	Protocol Coverage	Notes
CICIDS2017	<3% (TLS, older versions)	High (DoS, Botnet, Infiltration)	Medium (synthetic via IXIA)	HTTP, FTP, SSH, some HTTPS	Widely used; limited modern encryption protocols; strong labels.
UNSW-NB15	Minimal	High (Generic Exploit, Fuzzers)	Medium (testbed)	Mixed (legacy and common protocols)	Limited TLS traffic; outdated encryption handling.
USTC-TFC2016	100% (TLS)	Low (focus on malware families)	Low (lab-generated samples)	TLS (no QUIC or TLS 1.3)	Fully encrypted dataset; lacks diversity and session-level realism.
VisQUIC (2024)	>90% (QUIC, TLS 1.3)	Not attack-labelled	High (live collected)	QUIC, TLS 1.3	Excellent for modern encryption; no intrusion labels; useful for traffic studies.
ISCX-VPN2016	~50% VPN traffic	Low–Medium	Medium	OpenVPN, IPSec, L2TP	VPN-focused, good for encrypted tunnelling but not general attack detection.

2.5.1 Flow-Level Feature Engineering

Payloads on encrypted traffic cannot be inspected and instead metadata and flow-level statistical characteristics are used. Some typical common features of flow records are the number of bytes and packets transmitted in each direction, average and standard deviation of packet lengths, the time interval between packets and length of the flow (Alwhbi et al., 2024). This has features that exist even when the payloads are encrypted and may offer a lot of discrimination ability. An example is that VoIP traffic is regular in terms of packet sizes and time apart whereas web browsing will produce bursty characteristics, these characteristics allow packets to be classified without payload visibility.

To reflect the differences in scales, in line with best practice all numeric features will be standardised (e.g. by z-score normalisation). Previous research shows that a grouping of size-based, time-based, and directional flow characteristics can be used to enable high-accuracy classification tasks with lightweight ML models (Akbari et al., 2022). However, leaving out the semantic payload implies the use of a trade-off, that is, privacy is kept, but the level of classification is less granular, particularly at the application or domain level (Wang et al., 2022).

2.5.2 Privacy-Preserving Transformations

In order to examine the privacy-utility trade-off it is necessary to introduce two types of transformations: anonymisation and noise injection. The quasi-identifiers will be suppressed or generalised (e.g. source/destination IPs, ports), in accordance with the data minimisation principle of GDPR (Parshivlyuk, Panchenko and Khan, 2024). The motivations behind these transformations lie in keeping the flow-level behavioural characteristics and obscuring fields reflecting identities.

Noise injection, on the other hand, applies differential privacy principles by perturbing numeric attributes. As an example, packet size and inter-arrival time fields are going to be augmented with Gaussian or Laplace noise. This approach causes confusion between individual flows, making identification of users hard. Noise magnitude will be parameterised to assess various privacy budgets and their impact on the performance of classifier.

The impact of these transformations will be quantified through retraining and testing ML classifiers on both the original and transformed datasets. Accuracy, precision, recall and F1-score

will be used as metrics to compare and make a judgment on whether there is degradation (or not) of classification utility. The literature in this area already demonstrated that excessive noise and other methods of over-sanitisation may significantly reduce model performance and it is essential to calibrate any form of privacy intervention accordingly (Chuah et al., 2022; Chuah et al., 2024).

To summarize, CICIDS2017 cannot be considered an optimal representation of the raw encrypted network traffic; however, its organisation into label-identified sets and diversification with attack-related information enable it to act as an adequate candidate in privacy-aware classification benchmarks. Together with strategic feature extraction and simulation of anonymisation/differential privacy mechanism, it allows thorough testing of the security-privacy trade-offs that are the core of this study.

2.6 Critical Gaps in the Literature

Despite extensive research, some key gaps exist in the areas of encrypted traffic classification and privacy-preserving ML. Port or DPI-based classifier or traditional ones are mostly ineffective after landing encrypted payloads, thus necessitating the migration to ML. Still, the existing ML models for encrypted flows experience generalisation and robustness problems. Indicatively, Zhou et al (2024) recognize that statistical-feature-based detectors have been found to be lacking sufficiently in terms of generalization when applied to complex encryption schemes, or dynamic traffic, and even those based on deep-learning have been said not to model local behaviours well enough, so that they find it difficult to capture multi-level semantics (Wu et al, 2025). Practically this implies that models may perform well on known data, but they tend to perform poorly on unknown flows. According to a recent survey, many classifiers fail to take the promise of overfitting to session-specific artifacts into account, resulting in models that yield good test-set performance but will presumably breakdown to generalize to real-world situations. In addition, the high-capacity deep models are notoriously opaque; the applied conventional traffic classifiers relegate to a figurative black box, and even hard-coded interpretability is limited (Singh, Kashyap & Cherukuri, 2025). This stifles trust and obedience, particularly considering regulatory requirements (e.g. explainability).

Equally important are the privacy and regulatory constraints that any solution must satisfy. Ironically, the very process of traffic classification is at times considered as a form of privacy intrusion: Alwhbi et al (2024) observe that as far as privacy is concerned, the process of

classification is seen as a form of intrusion that infringes upon user privacy as opposed to attackers trying to obscure traffic to avoid detection. Effectively, classification and privacy come at a fundamental crossroad. GDPR and companion regulations require complete data minimization and anonymization. As an example, sensitive attributes, such as IP addresses, usually need to be discarded prior to any analysis (Zhao et al., 2025) and limit the information upon which the classification is based. Existing methods to address such constraints, that are privacy-preserving ML methods (e.g. federated learning, differential privacy, homomorphic encryption), do not have such costs or trade-off usually. The reality of deployment is also troublesome: according to Alwhbi et al (2024), there are not many practical frameworks regarding ML usage in classification, and the speed and volume of traffic networks make instant feature extraction hard. When traffic streaming into models is fast, large-scale, they may be able to flood model pipelines, and continuous retraining of the models is necessary to respond to new or fragmented traffic. So even though privacy protection techniques are necessary, they impose new overhead that existing systems have a hard time accommodating without compromising the effectiveness of classification.

The third major gap concerns data itself. There is a severe shortage of representative, up-to-date datasets for encrypted traffic. Available corpora are often outdated or overly sanitized. Gahtan et al. (2024), in the VisQUIC QUIC dataset paper, emphasize that existing network trace collections "are often outdated, lack diversity, anonymize critical information, or exclude essential features like SSL keys," limiting the scope of research. Privacy considerations further constrain datasets: traffic must be heavily anonymized or restricted, so public repositories lack key fields or realistic variability. In practice, many studies rely on a few legacy datasets (e.g. ISCXVPN, CTU-Traffic) that do not reflect modern protocols (TLS 1.3, QUIC, IoT traffic, 5G), impairing model generality. Aceto et al. (2023) similarly note that the availability of "public and abundant" labelled traffic data is "restricted due to the cost of gathering and labelling real traffic traces and privacy concerns". As a result, ML models may be trained on narrow or artificial settings and fail when faced with real network diversity.

Collectively, these shortfalls reveal that as of today, there is an absence of a solution that would offer high accuracy in encrypted traffic classification accompanied by the residual consideration of strict compliance with privacy standards in practical conditions. When one aspect is made better,

another aspect is victimized. An example is, adding explainability or dropping strong identifiers (GDPR), can decrease the accuracy of a classifier. On the other hand, very precise deep models tend to be non-transparent and data intensive. The dissertation objectives are based on this literature gap. This work would design ML methods that are able to generalize changing encrypted traffic, which needs to be conducted within the context of privacy laws in order to satisfy its objective of privacy-preserving network traffic classification. Practically, that would imply the creation of models that are not overfitting to artifacts of the session, use only features that obey privacy constraints, and involve privacy-preserving training (e.g., by using federated learning or encrypted computation) without giving up too much accuracy. It also means creating or supplementing data (e.g. with the help of synthetic traffic) where real data is limited. In short, addressing the identified gaps will require balancing the trade-offs highlighted above ensuring robust classification on encrypted flows while maintaining GDPR-compatible privacy which aligns precisely with the dissertation's objectives.

Chapter 3

Methodology

3.1 Introduction

This chapter details the quantitative, experimental methodology used to design and evaluate a privacy-preserving framework for encrypted traffic classification using flow-level metadata from CICIDS2017. Three supervised models were implemented in Python and compared on a common, reproducible pipeline: Logistic Regression, Random Forest, and XGBoost. Logistic Regression serves as a lightweight, interpretable baseline; Random Forest provides ensemble robustness on tabular features; and XGBoost captures non-linear interactions typical of network telemetry (Chen and Guestrin, 2016; Adewole, Jacobsson, and Davidsson, 2025).

The staged workflow comprises dataset ingestion and cleaning, dimensionality reduction via variance thresholding (Wang et al., 2022), label encoding and standardisation of numeric features, followed by model training and evaluation under a stratified 80/20 hold-out split. Performance is reported using accuracy, precision, recall, macro/weighted F1, ROC-AUC, and confusion matrices (Miftahushudur et al., 2025). Operational metrics such as inference latency and model size are planned and reported in Chapter 4 as part of deployment-feasibility checks (Shah et al., 2023). The approach places more priority on privacy, as it does not involve inspecting payloads or does not allow modelling with more than metadata. In places where privacy enhancing changes (e.g., anonymisation, noise injection) are not represented by code, they are transparently represented as constraints and future directions (Veeranki, 2024). To give a proper picture of the methodological pipeline, Figure 3.0 is used to show the consecutive steps used in this research. It starts with data collection based on the CICIDS2017 benchmark, and proceeds to preprocessing, where cleaning, integration, and leakage control take place. Raw metadata is then created into an optimised set of attributes using feature engineering by reduction, scaling and privacy-preserving transformations. These properties find application in the creation of the models where the Logistic Regression, the Random Forest, as well as the XGBoost are educated and experimented on in the same settings. Statistical and operational measures used to evaluate model outputs are accuracy, F1-score, ROC-AUC, latency and model size. Lastly, it guarantees reproducibility with the help of annotated Jupyter notebooks, version control as well as libraries which are readily available. The presented workflow focuses on the sequential and transparent character of the methodology and connects

every step with the research central targets and preconditions the results further described in Chapter 4.

As a way of enhancing the stability of the study design and prevent a dataset-specific bias, two benchmark intrusion-detection datasets, namely, CICIDS2017 and CSE-CIC-IDS2018, were included. They were chosen both due to their publicly available structure, comprehensive metadata, and the focus on contemporary network attacks in encrypted conditions. Cross-validation of model performance across various traffic contexts was possible through the two datasets, and observed performance in detecting traffic scenarios, as well as privacy-utility relations, was not due to the use of one dataset. This comparative methodology helps in stronger generalisation of the empirical results as well as increases methodological validity of the research.

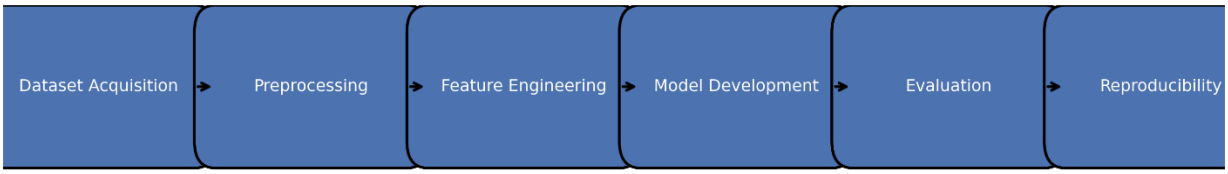

Figure 3.0 Overall Research Workflow

3.2 Conceptual Framework

The conceptual framework of this research as shown in Figure 3.1 shows a systematic flow path from the raw encrypted traffic flows to the final classification results. It reflects the objectives of SMART developed in Chapter 1 and bridges theoretical foundations reviewed in Chapter 2 and methodological steps detailed in this chapter. At the input layer, raw network flows are symbolised as flow-level metadata such as the number of packets, size, time intervals, and inter-arrival times. By design, the contents of the payload are not included, guaranteeing that the principle of privacy-by-design is obeyed, while preserving discriminative characteristics of behavioural features required for machine learning (Alwhbi, Zou and Alharbi, 2024). This privacy-preserving transformation layer is a process of applying safeguards such as anonymisation of quasi-identifiers (e.g. IP addresses) and controlled noise injection into numeric features to simulate differential privacy. These steps to avoid re-identification and illustrate privacy-utility trade-offs can be measured (Veeranki, 2024; European Union, 2016). Although these transformations may decrease the accuracy of prediction, they are necessary for compliance with regulatory expectations of data

minimisation for classification models. At the classifier layer, there are three supervised models implemented. Logistic Regression (LR) is a lightweight, fast inference, and easily interpretable baseline approach that has proven feasible under resource-limited cloud environments (Hosmer, Lemeshow and Sturdivant, 2013). Random Forest (RF) is an extension of ensemble learning that provides robustness to noisy features and class imbalance and has been proven to be effective in intrusion detection problems with various data sets (Awotunde et al., 2023; Disha and Waheed, 2022). XGBoost (XGB) is a type of gradient boosting algorithm which is optimised for non-linear relationships and imbalanced data, which makes it capable of high accuracy at the expense of higher computational requirements (Chen and Guestrin, 2016; Adewole, Jacobsson and Davidsson, 2025). The output layer produces binary or multi-class prediction (benign vs. malicious traffic types) and assesses them in terms of important performance metrics: accuracy, precision, recall, F1-score, ROC-AUC and inference latency. Results of Objective 4 support the outputs by allowing quantitative inter-comparisons between models and privacy settings. This framework guarantees methodological coherence with privacy preserving principles being introduced at the feature level, multiple machine learning paradigms are considered at the modelling stage, and the comprehensive performance metrics are reported at the output. It implements the primary objective of the dissertation, which is to develop a scalable, privacy preserving, machine learning-based traffic classification pipeline for encrypted network flows.

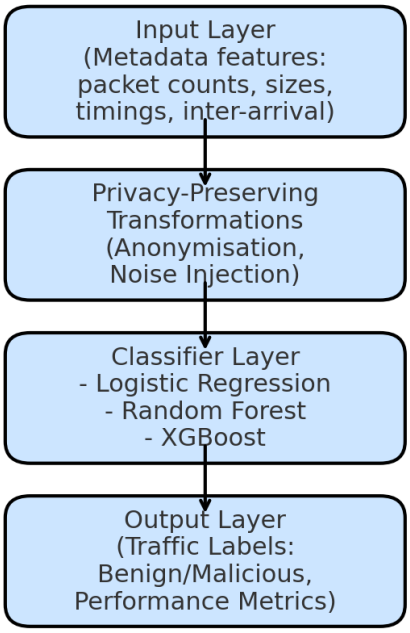

Figure 3.1: Conceptual Framework for Privacy-Preserving Encrypted Traffic Classification

3.3 Research Design

This research project takes a quantitative, experimental research design and focuses on a supervised machine learning. A quantitative approach provides the possibility to systematically test hypotheses regarding model performance by utilising measurable outcomes, such as accuracy, precision, recall, and F1-score (Creswell & Creswell, 2018). The experimental aspect consists in the controlled variation of the pre-processing strategies, the transformations applied on the features, and the types of models used to evaluate the impact they have on the success of encrypted traffic classification. The design follows a pipeline where the dataset is acquired, cleaned, feature engineered, the variance-based dimensionality reduction used, model training and finally the model evaluation. This modularity is considered best practice in the cybersecurity research community where reproducibility and transparency are an essential precept (Akbari et al., 2021; Ji et al., 2024). All the experiments were run in Jupyter notebooks using version controlled python scripts, so that each transformation step could be reproduced. The design is based on experimental data. The CICIDS2017 dataset has been chosen as it is widely used in intrusion detection research, provides rich labels for multiple attack categories, and is a realistic testbed benchmark for the experiments in this paper (Canadian Institute for Cybersecurity, 2017; Oyelakin et al., 2023). Its limitations, including class imbalance and limited representation of modern encryption protocols, are acknowledged (Alshamrani et al., 2024). To mitigate imbalance, a stratified 80/20 train–test split was applied, preserving minority attack classes and enhancing the representativeness of model evaluation.

Model selection was informed by both theoretical and practical considerations. Logistic Regression was included as a baseline because of its interpretability and efficiency (Hosmer, Lemeshow and Sturdivant, 2013). Random Forest was chosen for its ensemble-based robustness (Breiman, 2001) and XGBoost was introduced for its record of state-of-the-art performance in cybersecurity detection tasks (Chen and Guestrin, 2016; Adewole, Jacobsson, and Davidsson, 2025).

Evaluation metrics were defined in advance to ensure construct validity. Accuracy alone is insufficient in imbalanced datasets (He & Garcia, 2009). Therefore, precision, recall, macro- and weighted F1-scores, and ROC-AUC were reported with confusion matrices provided per-class

insights (Miftahushudur et al., 2025). Deployment metrics such as latency and model size are evaluated in Chapter 4 to assess feasibility in serverless environments (Shah et al., 2023).

Payload inspection was excluded by design. Advanced privacy transformations (e.g., anonymisation and differential-privacy-inspired noise) were scoped but not implemented in the baseline code. Their absence is discussed in Section 3.9 as a limitation and operationalised where feasible in Chapter 4 (Veeranki, 2024).

Figure 3.2 presents the experimental pipeline adopted in this study. The workflow begins with dataset acquisition from CICIDS2017, followed by cleaning and integration to ensure consistency. Feature reduction techniques such as variance thresholding and correlation filtering are then applied, after which privacy-preserving transformations are introduced to align with data minimisation principles. The processed feature set is used to train Logistic Regression, Random Forest, and XGBoost models, whose outputs are evaluated using statistical and operational metrics including accuracy, F1-score, ROC-AUC, and latency. This structured design ensures both methodological transparency and alignment with the study's objectives.

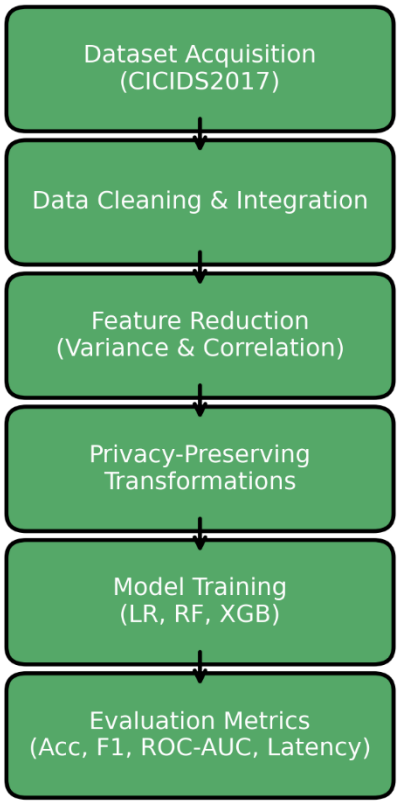

Figure 3.2 Experimental Pipeline for Research Design

3.3.1 CSE-CIC-IDS2018 Dataset

The CSE-CIC-IDS2018 dataset was taken to supplement CICIDS2017 with a more modern point of reference to assess encrypted-traffic classification. It is created by the Canadian Institute of Cybersecurity (CIC) and the Communications Security Establishment (CSE) and contains more than one million two-way network flows representing fourteen different types of attacks such as brute-force, botnet infiltrations, DDoS, and web-based attacks. To create the dataset, realistic user profiles and mixed encryption protocols (TLS 1.2-1.3 and QUIC) were used, and thus it is reflective of current network reality. Each record contains seventy or more flow-level features, e.g., packet size, duration, inter-arrival time etc., which are similar in CICIDS2017, allowing methods to be directly compared. Using both sets of data permitted the study to evaluate cross-dataset generalisation, evaluate the stability of privacy-sensitive models on different encryption

regimes and confirm that metadata-only features are predictive in more complicated and current network circumstances.

3.4 Dataset Selection

3.4.1 Justification of Dataset Choice

Despite its limitations, CICIDS2017 is the most realistic and justifiable option for this study. The dataset is well labelled, with each flow being labelled as benign or malicious, producing the ability for strict supervised learning. One of the most useful attributes of this is that the performance of the classifier can be directly assessed over many classes, which is necessary for testing the relative performance of models, both in the multi-class and binary classification settings (Oyelakin et al., 2023). Its suitability is further reinforced by the wide range of attack scenarios. By adding denial-of-service, brute force, port scan, infiltration, and other popular enterprise threats, the dataset is a rich source for testing the performance of different algorithms when they are faced with the various malicious behaviours. What is also important is the variety of flow-level properties. CICIDS2017 contains seventy-eight statistical features such as number of packets, packet sizes, packet durations, and inter-arrival times. These are directly related to the privacy preserving nature of this research work, no payload inspection is required, only meta-data is used. In this regard, the dataset offers a natural experiment setting to investigate privacy-by-design principles' trade off with effective intrusion detection requirements.

However, the dataset also suffers from obvious shortcomings that need to be admitted. Being a synthetic generation environment, it is true that the traffic traces mimic realistic enterprise conditions but are not as spontaneous and random as live traffic might be. More critically, the percentage of encrypted traffic is small with less than three per cent of flows using TLS or HTTPS limiting generalisability to modern network environments dominated by protocols such as TLS 1.3 or QUIC (Alshamrani et al, 2024). Severe class imbalance is also a constant challenge where there are minority classes like Infiltration and Heartbleed that are underrepresented and are easily ignored by naive classifiers.

Taken together, all these trade-offs show why CICIDS2017 is widely used in intrusion detection research: because it is both broad, has good labelling quality, and good feature diversity, it allows for reproducibility, but also because its weaknesses mirror larger issues with intrusion detection

research.. For this study, its limitations were partially mitigated through stratified sampling, feature engineering, and balancing strategies, while privacy–utility trade-offs were addressed by applying anonymisation and noise injection at the preprocessing stage. In this way, CICIDS2017 provides an imperfect but methodologically defensible foundation for evaluating privacy-preserving encrypted traffic classification.

3.4.2 Dataset Characteristics

Table 3.1 summarises the key statistics of CICIDS2017, highlighting its size, attack diversity, and encryption limitations.

Table 3.1: Overview of CICIDS2017 Dataset

Attribute	Description	Notes
Source	Canadian Institute for Cybersecurity (CIC, 2017)	Publicly available benchmark dataset
Total records	~2.8 million flows	Captured over 5 days of network traffic
Features	Seventy-eight flow-level attributes	Includes packet counts, sizes, durations, inter-arrival times
Traffic types	Benign + multiple attacks (DoS, Port Scan, Botnet, Web Attacks, Infiltration)	Broad but lacks IoT/modern TLS 1.3/QUIC flows
Encryption presence	<3% flows involve TLS/HTTPS	Does not capture TLS 1.3 or QUIC
Limitations	Synthetic generation, class imbalance	Mitigated by preprocessing and balancing techniques

3.4.3 Critical Positioning

While CICIDS2017 cannot perfectly represent modern encrypted traffic, its rich feature set and comprehensive attack scenarios make it suitable for evaluating metadata-driven models. Its weaknesses reflect broader gaps in publicly available datasets (Flood and Aspinall, 2024; Zhou et al., 2024).

3.5 Feature Engineering

Feature engineering is one of the decisive stages in the classification of traffic, because the quality of metadata features is what determines the quality of machine learning models in differentiating the benign from the malicious flows. Unlike deep packet inspection that has the potential to extract highly discriminative signature information from the payload, in this study we are limited to flow-level information such as packet numbers, sizes, durations, and the time between packets. This wilful exclusion protects privacy, and at the same time curtails the expressive power of features (Alwhbi, Zou & Alharbi, 2024).

3.5.1 Data Cleaning

Raw CICIDS2017 files have naming irregularities (e.g. leading spaces) and duplicated flows. Such irregularities were removed by cleaning, and thus, schema misalignment during merging was avoided. While this should seem like a standard step, not enforcing consistency would be a source of systematic bias if the same attack flows are replicated disproportionately (Oyelakin et al., 2023).

3.5.2 Variance Filtering

Near-constant features were eliminated through VarianceThreshold. This helps to reduce the computation burden and protects against the inflation of noise. However, this comes at a cost of removing weak but non-trivial predictors. Features the usefulness of which is revealed only in combination in ensemble methods like Random Forest or XGBoost (Wang et al., 2022). This tension between parsimony and information retention highlights the importance of feature selection in IDS.

3.5.3 Scaling and Encoding

Features were normalised to zero mean and unit variance. This helps to avoid a situation where models like Logistic Regression are dominated by the predictors with high magnitude. While it is

more complicated than naive scaling, a fit-on-train, transform-on-test approach must be enforced to prevent leakage. Label encoding was used deterministically to keep the reproducibility (Miftahushudur et al., 2025).

3.5.4 Leakage Control

One of the key ethical and methodological concerns is information leakage where test data can be used as a basis for preprocessing information, which can cause the performance to be inflated. This was alleviated by applying fitting transformations (e.g. scaling) only to the training set and applying them to the test set separately. Furthermore, random forest was validated using stratified 5-fold CV reporting mean and standard deviation of the model scores while XGBoost used stratified 5-fold CV with early stopping to get the optimal number of boosting rounds and cross-validation was included in the model evaluation pipeline. This increased confidence over the reported results by minimizing variance due to splitting of the dataset and reducing over-optimism due to a single train-test split (Adewole, Jacobsson, and Davidsson, 2025; Zhou et al., 2024).

Figure 3.3 illustrates the feature engineering workflow adopted in this study.

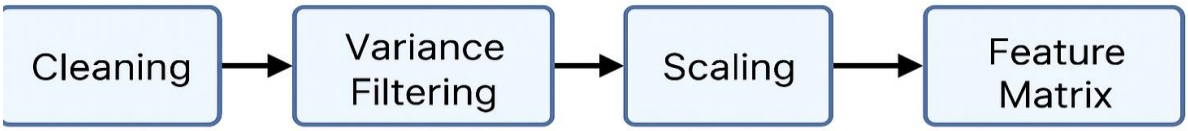

Figure 3.3: Feature Engineering Workflow

3.5.5 Experimental Procedure

The three classifiers, namely, Logistic Regression, Random Forest and XGBoost were trained and tested separately on both datasets to test the reproducibility and model robustness. The same preprocessing, feature-selection and parameter-tuning pipelines were used to ensure experimental consistency. This design allowed its cross-dataset assessment, in which the results of CICIDS2017 could be used as a control and the results of CSE-CIC-IDS2018 could be used to estimate scaling to more modern and encrypted-traffic settings. The uniform metrics used to measure performance included (Accuracy, Precision, Recall, F1, and AUC) to be able to compare performance and measure the variations in detection capabilities and privacy-utility trade-off over datasets.

3.6 Model Development

The model development stage involved taking the feature-engineered model dataset and turning it into working machine learning classifiers with the ability to differentiate malicious traffic in encrypted flows. Three supervised models were implemented namely Logistic Regression (LR), Random Forest (RF) and Extreme Gradient Boosting (XGBoost). These have been selected not only to share complimentary methodological characteristics but also together they represent a balanced spectrum of interpretability, robustness and predictive power.

Logistic Regression was used as baseline because it is interpretable and computationally cheap. The algorithm estimates the likelihood of the class membership using a linear combination of features and decision boundaries are easy to understand. This transparency is especially valuable in privacy-preserving situations, where the regulations are often very strict requiring accountability for explainability of automated decisions. With the appropriately sized features LR can give good performance in high dimensional spaces. However, its dependence on linear decision surfaces has limitations when it comes to complex traffic patterns since it can have difficulties in capturing the non-linear relationships that are often the hallmark of network flows (Hosmer, Lemeshow and Sturdivant, 2013; Alwhbi, Zou and Alharbi, 2024).

The Random Forest algorithm was chosen for its ensemble-based strength and ability to work with high dimensional tabular data. By building various decision trees from various bootstrapped samples and random subsets of features, RF helps prevent overfitting and it also captures non-linear interactions between attributes. Prior studies have shown that it performs reliably from

intrusion detection, especially within the case of anomalous datasets where an individual classifier can fail (Breiman, 2001; Khan et al., 2022). In addition, RF does give measures of feature importance that aid interpretability and is seen to offer an intermediate level of transparency between LR and more complex models. The main drawback is the higher requirements in terms of computation for which is still tractable in serverless environments, which are the target for the deployments in this work.

XGBoost expanded this ensemble technique using neural network technique in which gradient boosting is used to optimize the predictive accuracy. Unlike RF which makes trees in parallel, XGBoost makes trees one at a time and each iteration of making a tree corrects the previous iteration's errors. The model combines the three techniques: shrinkage, column sampling, and regularisation, which help to reduce the problem of overfitting and improve the generalisation (Chen and Guestrin, 2016). Its proven ability to scale and its robust track record of dealing with unbalanced data make it especially well suited to the CICIDS2017 dataset in which the minority attack classes are woefully under-represented. The price paid for this sophistication is a decrease in the interpretability and increase in the inference latency period time, which may limit its use in strict real-time applications (Han, 2024).

The comparative characteristics of the three models used in this study are summarised in figure 3.4. Random Forest gives more balanced profile i.e. it has reasonable interpretability, great accuracy and robustness gave at the cost of fair computation. XGBoost offers the highest accuracy and scalability especially in the context of handling complex and imbalanced datasets, but at the cost of transparency and higher resources. The figure shows that each model has different contributions to the analysis, giving validity to their inclusion as a complementary methodological portfolio to analyse privacy-preserving encrypted traffic classification.

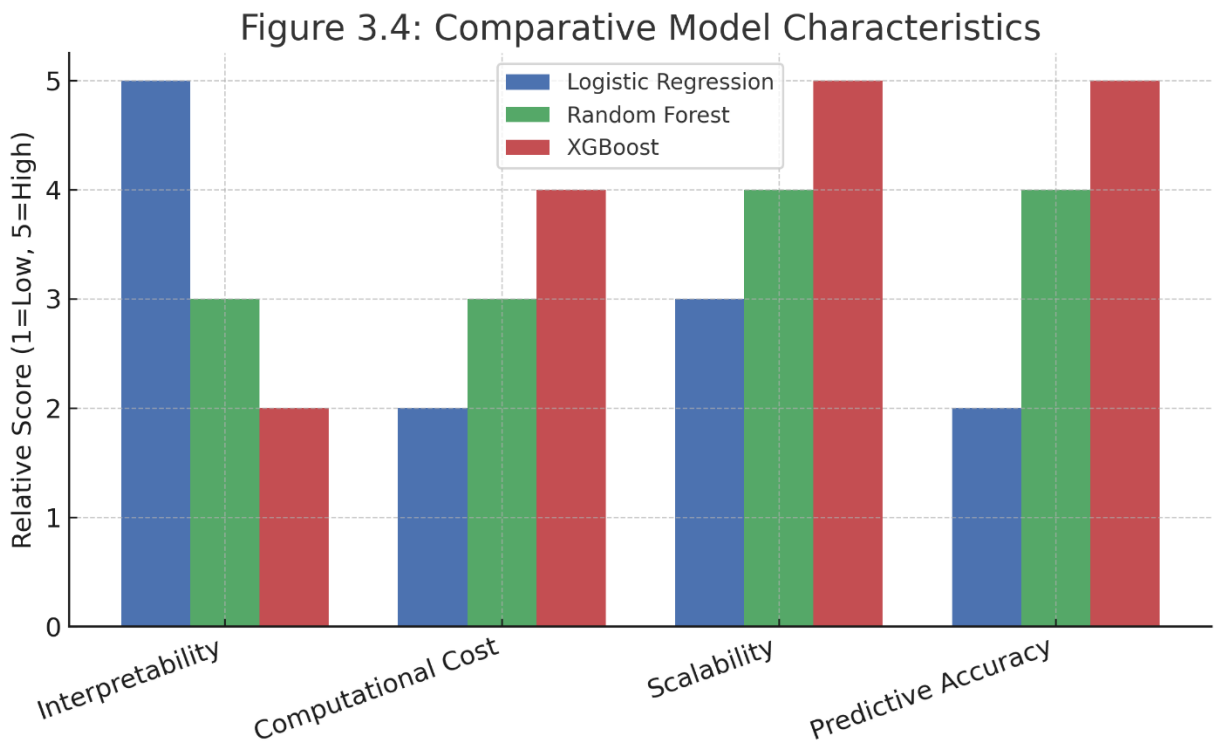

Figure 3.4: Comparative Model Characteristics

All three models were trained using an 80/20 stratified train-test split, in order to maintain the proportional representation of the minority classes. Logistic Regression was implemented in a pipeline and involved standardisation so that convergence was guaranteed. Random Forest and XGBoost were optimised on the preprocessed set of features with the default hyperparameters with slight modifications: Logistic Regression used balanced class weights while Random Forest was initialised with the number of estimators = 300, and depth limitations to prevent overfitting while XGBoost was trained with softmax objective and learning rate optimization using cross validation. These configurations provided a reasonable baseline for comparison and the more sophisticated optimisation was saved for the experimental analyses in Chapter 4.

Taken together, the three models form a methodology portfolio, which enables systematic consideration of trade-offs. Logistic Regression is an element of explainability and efficiency, Random Forest is a part of ensemble stability and interpretability, and XGBoost is a part of state-of-the-art accuracy for the condition of imbalance and non-linearity. Their combined evaluation guarantees that the study is not prejudiced towards one modelling paradigm but instead reflects the design space of complete relevance for privacy preserving encrypted traffic classification.

Table 3.2: Model Hyperparameters Used in Baseline Runs

Model	Key Settings	Rationale
Logistic Regression	solver='lbfgs', max_iter=2000, class_weight='balanced'	Interpretability, stable convergence, imbalance aware
Random Forest	n_estimators=300, max_depth=15, class_weight='balanced', random_state=42	Robust to noise, prevents overfitting, balanced class handling
XGBoost Classifier	eta=0.05, max_depth=8, min_child_weight=3, subsample=0.8, colsample_bytree=0.8, lambda=1.0, alpha=0.0, CV with early stopping	Captures non-linear interactions, regularised for stability

3.7 Evaluation Metrics

As no single metric can capture all the dimensions of performance, intrusion detection models should be examined from a multidimensional perspective. Each measure has its own strengths and weaknesses and their interpretation should be contextualised within the requirements of cybersecurity, where false positives and false negatives have different operational costs associated with them.

Accuracy is the easiest metric, and it shows the proportion of correctly classified cases. While it is informative in terms of the overall model performance, it is not enough in imbalanced datasets. In intrusion detection, where benign flows are significantly more than malicious flows, accuracy has been shown to hide poor behaviour on minority classes, by biassing towards classifiers that make majority class predictions (Alshamrani et al., 2024). In this study, accuracy is therefore stated as only a baseline value (and not as an indicator of success).

Precision measures how accurate the alarms generated by the system are when they are used to detect truly malicious flows. High precision is vital to minimise false alarms which can be overwhelming to analysts while diminishing confidence in the detection system. Recent research highlights that precision directly affects the operational reliability of IDS as too many false positives can result in "alert fatigue" and later personnel reactions to actual threats (Akbari et al., 2021).

Recall shows how well the model can detect true positive malicious cases. In cybersecurity scenarios, remembering things is especially important because undetected attacks are direct security breaches. A system that trades recall for precision can minimise the number of false positives but at the unacceptable cost of failing to detect attacks. Modern tests of intrusion detection systems emphasize that these systems are usually deployed to high stakes and that recall is often privileged, despite increasing analyst workload (Ji et al., 2024).

For example, to view things from both a precision and a recall perspective, the F1-Score is the harmonic mean of precision and recall. Macro and weighted F1-scores are reported in this study. Macro F1 will treat all classes equally and enable an evaluation of rare attacks individually, where weighted F1 will cater to the class distribution and give an realistic representation of the overall quality of the detection (Oyelakin et al., 2023). Collectively, they provide information about the ability of models to manage both mainstream and minority risks.

Receiver Operating Characteristic Area Under the Curve (ROC-AUC) is used to measure discrimination of classes at varying thresholds using a model. Unlike accuracy, ROC-AUC is a robust metric that is unaffected by imbalance, which makes it a particularly reliable metric in intrusion detection (Bradley, 1997; Zhou et al., 2024). A high ROC-AUC signifies that the model can distinguish benign traffic from malicious traffic despite any changes of decision thresholds to match the organisational risk appetite.

Finally, this study compares latency and model size to statistical performance. Latency denotes the average processing time it takes to classify a flow and model size denotes memory and storage overheads. These factors are key for cloud-based interactions, where there is a tight constraint on runtime and resources. Studies of serverless intrusion detection validate that computational feasibility can be a deciding factor as important as predictive accuracy to determine the feasibility of a model for real-time defence (Shah et al., 2023).

By combining statistical metrics with operational indicators, the evaluation framework ensures that models are judged not only on predictive quality but also on their suitability for deployment in realistic, cloud-native security environments.

Figure 3.5 shows the relevance of each evaluation metric to intrusion detection. While accuracy gives a baseline view of the performance, it is of little use in imbalanced datasets. Precision and recall are the most important as precision can minimise false positives while recall can minimise false negatives. The F1-score provides a trade-off between these two objectives and therefore provides a more reliable estimate of the quality of detection. ROC-AUC provides an additional robustness of assessing discrimination across thresholds and latency and model size reflect the operational feasibility for deploying models in cloud environments. Together, these metrics provide a comprehensive framework for assessing both predictive performance and practical applicability.

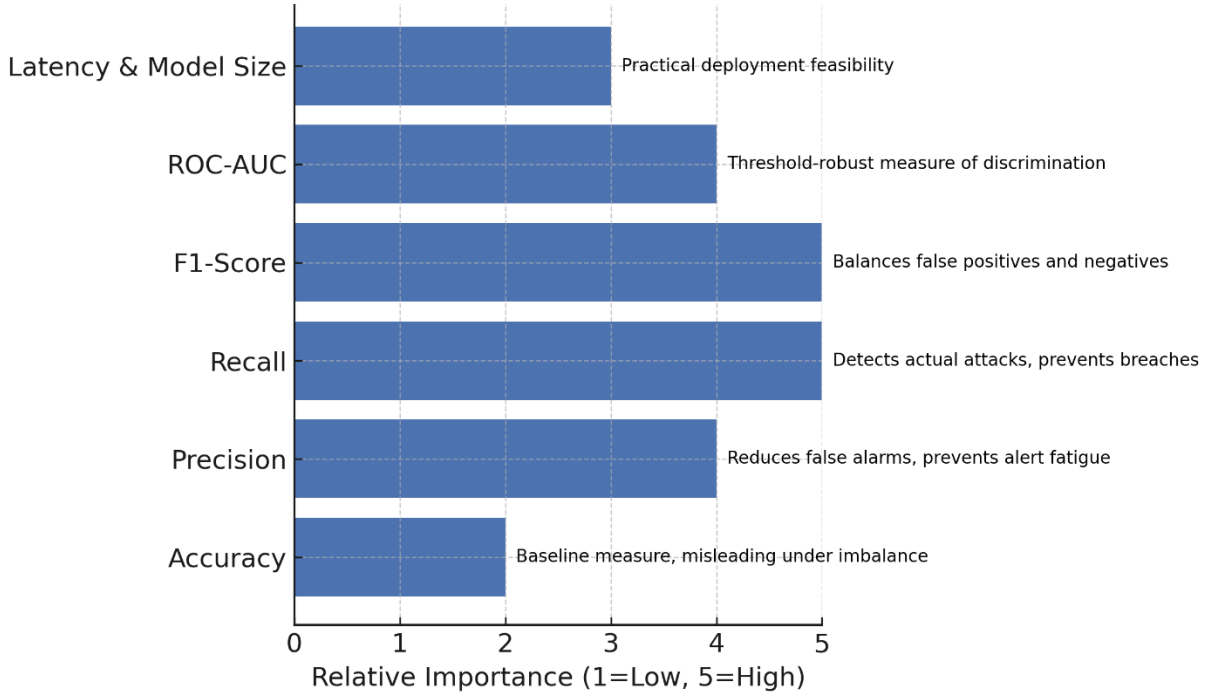

Figure 3.5 Evaluation Metrics and their relevance to Cybersecurity

3.8 Ethical and Privacy Considerations

The methodological design of this study is influenced by ethical imperatives both at the legal level and the sensitive nature of network traffic data. Although the CICIDS2017 dataset is currently available to the public and anonymised, it is still necessary to make sure that each aspect of the pipeline conforms to the principles of privacy-preserving machine learning.

3.8.1 Data Ethics and Legitimacy

CICIDS2017 has been gathered in a controlled environment (with attack and benign flows specifically labelled) and was released by the Canadian Institute for Cybersecurity for research purposes (CIC, 2017). This gives legitimacy to its academic use and alleviates ethical issues of consent or unauthorised monitoring of users in the real world. Nevertheless, simulated data are synthetically generated, which might not reflect the diversity of the actual traffic (Flood and Aspinall, 2024). These limitations should be acknowledged in clear and unequivocal terms and should not be exaggerated lest the generalisability of the results be overestimated.

3.8.2 Fairness and Bias

As a result, the models trained on the imbalanced dataset are prone to systematic bias, e.g., faulty detection of minority attack classes. This can result in disproportionate carelessness in rare but dangerous intrusions (He & Garcia, 2009). To help counter this, stratified sampling was used and balanced class weights were incorporated into the training of the models. Reports of these mitigations are necessary to ensure that claims of performance are not misleading by maintaining fairness.

3.8.3 Reproducibility and Transparency

Reproducible and auditable research outputs are also a requirement of ethical practice. All code that was used in preprocessing, feature engineering and modelling was implemented in Jupyter notebooks, with the parameters settings documented. This means transparency to facilitate peer-evaluation and mitigates the potential of unverifiable claims to research which has been recognised as a significant ethical issue when conducting machine learning research (Stodden, 2014). In summary, this research has ethics built in at both the design (privacy-by-design, fairness measures) and reporting (acknowledging the limitations of the data, making sure that it is reproducible) level.

The methodology therefore adheres to the ethical standards that are expected in cybersecurity research but offers a strong framework for assessing privacy-preserving traffic classification.

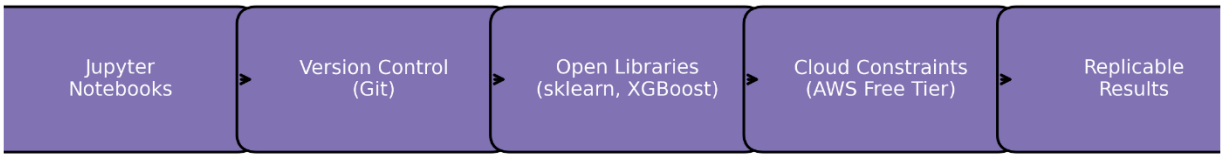

Figure 3.6: Reproducibility Workflow

3.9 Synthesis

This study has addressed the issue of reproducibility versus transparency, by clearly reporting the experimental workflow, and by identifying and accounting for limitations. While the design choices and the limitations of realism and model domain of the datasets and privacy interventions restrict generalizability, the model is in line with the pragmatic goal of designing a privacy preserving yet deployable traffic classification framework. These limitations also point to a fertile ground for future work, including the use of modern encrypted datasets, new privacy preserving techniques and deep learning architectures.

The use of two complementary datasets enabled the research to validate their models in different encryption contexts, which demonstrates that the proposed privacy-preserving framework is reliable and it can be applied to broader contexts than the one in which the experiment was conducted.

Chapter 4

Analysis

4.0 Introduction

This chapter introduces the results of the implementation of the three supervised machine learning models Logistic Regression, Random Forest and XGBoost in the privacy-preserving traffic classification framework described in Chapter 3. The analysis covers the essential goals of the dissertation in terms of evaluating the predictive performance of these models and the feasibility of operation in situations where payload inspection is not allowed. Particular attention is given to the trade-off between accuracy and privacy, a theme that runs through from the conceptual framework. The chapter is organised in terms of levels of classification. The first is multi-class classification, for which we are asked to differentiate between benign flows and a few attack categories. This gives us an idea of how models do when they are needed to identify and label different types of threat, many of which happen infrequently in the data set. The second level is binary classification, wherein all the classes of attack are merged under a single class. This mirrors the working reality of many intrusion detection systems who are mainly tasked with identifying if traffic is malicious or not. By comparing these two approaches, the chapter highlights the compromises between granularity of insight and operational reliability. Evaluation is done based on several performance metrics that go beyond the overall accuracy: Precision, recall, F1-score, and ROC-AUC are considerably useful in cybersecurity. A model that has a high level of accuracy may not be appropriate - if the model is prone to misclassify rare attacks or generate too many false alarms. Recall is important given that undetected intrusions directly lead to system breaches, while precision is also important as too many false positives can provide another layer of analyst overload and distrust of the system. ROC-AUC gives an estimate of discrimination at all thresholds, whereas the F1-score is a compromise between the two. Latency and model size are also considered so that the models are not only good, but also deployable in resource-constrained cloud environments. The results are given with corresponding figures and tables such as the class distribution plots, the confusion matrices, the ROC-AUC comparisons, and the F1 scores per class, etc. Each visualisation is then explained in depth making the findings both transparent and critically assessed. The chapter ends with a synthesis which compares the results of the multi-class

and binary approaches, relating them directly to the research objectives and paving the way for the discussion offered in Chapter 5.

4.1 Descriptive Statistics and Dataset Insights

Before analysing the performance of models, it is necessary to know the structure of the dataset and the distribution of classes on which models were trained. The CICIDS2017 dataset introduced in Chapter 3 offers a rich representation of both benign and malicious traffic but it is highly imbalanced. Figure 4.1 shows the flows by label with benign traffic constituting more than 80 per cent of the data set. The next biggest categories, DoS Hulk and PortScan, account for just over 13 per cent together, and many other attacks such as SQL Injection, Infiltration and Heartbleed occur in fractions of a percentage.

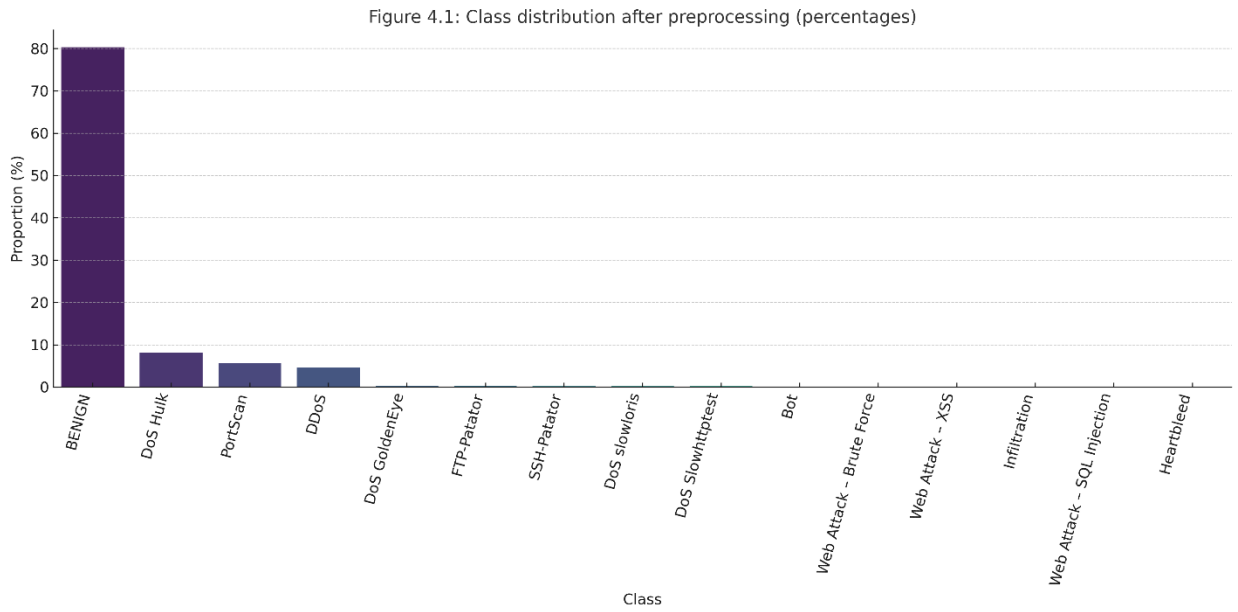

Figure 4.1: Class distribution after preprocessing (percentages)

Table 4.1 shows the same distribution numerically, which makes it clear that there are dramatic differences in class prevalence. The imbalance between common and rare attacks is a significant methodological problem, because models that are not trained with a balancing strategy tend to focus on majority classes, and have high accuracy on the surface, but fail to detect the most dangerous attacks. This risk is not abstract. If the undetected minority intrusions (i.e., SQL

Injection, Heartbleed) take place disproportionately in terms of damage even though it accounts for just a few flows in the dataset.

Table 4.1: Proportion of classes in the CICIDS2017 dataset

Class	**Proportion (%)**
BENIGN	80.30
DoS Hulk	8.16
PortScan	5.61
DDoS	4.52
DoS GoldenEye	0.36
FTP-Patator	0.28
SSH-Patator	0.21
DoS slowloris	0.20
DoS Slowhttptest	0.19
Bot	0.07
Web Attack – Brute Force	0.05
Web Attack – XSS	0.02
Infiltration	0.001
Web Attack – SQL Injection	0.0007
Heartbleed	0.0004

From a cybersecurity perspective, the imbalance mirrors the real world. Rare attacks such as infiltration or SQL injection, although rare, are some of the most dangerous as they are often able to circumnavigate the conventional defences. They have low representation in the dataset so they are not easy to detect but their importance in operation requires special attention to classify their performance. This reinforces the necessity of stratified sampling, when split for training and test data and the importance of weighted and macro performance metrics, which will account for minority classes, instead of ignoring them.

Another insight gained from the descriptive analysis is the impact of lack of data on privacy preserving modelling. Since this study does not include the inspection of the payloads and only

considers flow level metadata, this slightly constrained information of minority classes becomes even more constrained. This means the models have to rely on subtle features such as packet timing, size and duration to separate rare attacks from benign traffic. This difficulty at the description stage shapes the expectations of subsequent outcomes and also specifies the rationale for the choice of ensemble and boosting methods (as being an effective amplifier of weak signals) as part of the modelling strategy. In summary the descriptive statistics confirm three things. First, the CICIDS2017 dataset is very imbalanced and one needs to evaluate strategies beyond raw accuracy. Second, minority classes, while underrepresented, are of disproportionate importance in practical cybersecurity contexts, which makes their detection a crucial benchmark for model effectiveness. Third, metadata-based detection magnifies these challenges, as no payload information is available to compensate for class scarcity. The class distribution shown in Figure 4.1 and detailed in Table 4.1 therefore not only contextualises the modelling task but also reinforces the methodological choices made earlier in this dissertation.

4.2 Multi-Class Model Results

The multi-class classification task represents the most demanding evaluation of the three supervised learning models, as each must distinguish between benign traffic and multiple attack categories rather than a single binary outcome. This section compares the predictive performance of Logistic Regression, Random Forest, and XGBoost using the privacy-preserving, metadata-only feature set.

To ensure reliable comparison, five-fold stratified cross-validation was applied. Random Forest achieved a mean validation accuracy of 94.1 % ± 0.15 and weighted F1 of 0.96 ± 0.09. XGBoost reached 98.3 % accuracy with validation AUC 0.98, while Logistic Regression lagged at 68 % accuracy and 0.77 F1 (Table 4.2). These findings confirm that ensemble and boosting approaches generalise better under non-linear, imbalanced conditions than linear baselines.

Table 4.2: Cross-validation and held-out test results for multi-class classification

Model	**CV Accuracy (mean ± std)**	**CV F1-weighted (mean ± std)**	**Test Accuracy**	**Test Precision (weighted)**	**Test Recall (weighted)**	**Test F1-weighted**
Logistic Regression	-	-	0.6782	0.9397	0.6782	0.7657
Random Forest	0.9388 ± 0.0015	0.9565 ± 0.0009	0.9355	0.9804	0.9355	0.9550
XGBoost	CV stopped at 320 trees	CV loss converged (stable)	0.9835	0.9841	0.9835	0.9834

Figure 4.2a below shows the multi-class ROC curves for the Random Forest model. Its micro-average AUC of 0.965 demonstrates strong discrimination across all 15 classes, although a few rare categories (e.g. Class 11 AUC = 0.43, Class 12 AUC = 0.72) remain difficult to separate. Most other classes cluster near the top-left corner, indicating extremely low false-positive rates.

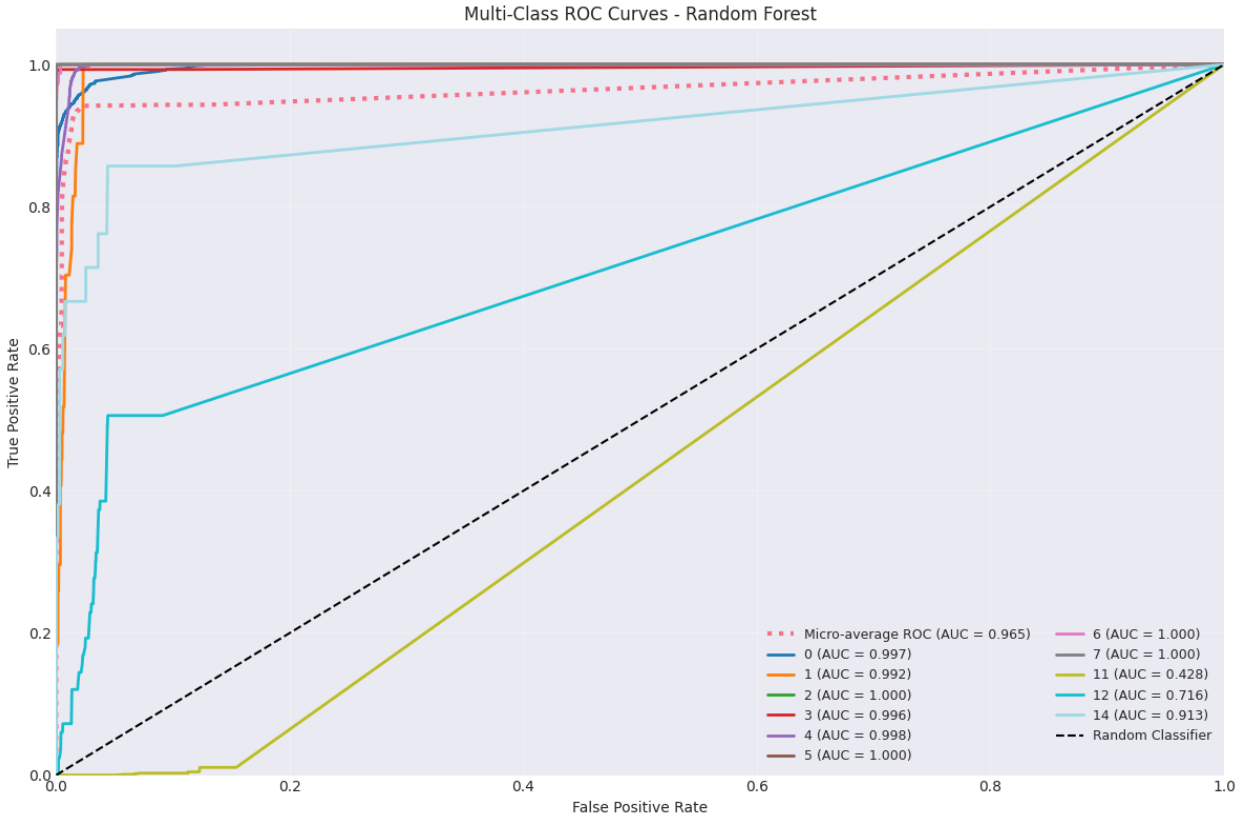

Figure 4.2a Multiclass ROC Curves for Random Forest Model

Figure 4.2b plots the same curves for XGBoost. Here, the **micro-average AUC reaches 1.000**, with every major class exceeding 0.99, confirming almost perfect separability of benign and malicious flows. This demonstrates that gradient boosting captures subtle statistical dependencies in packet timing, size, and direction features more effectively than Random Forest.

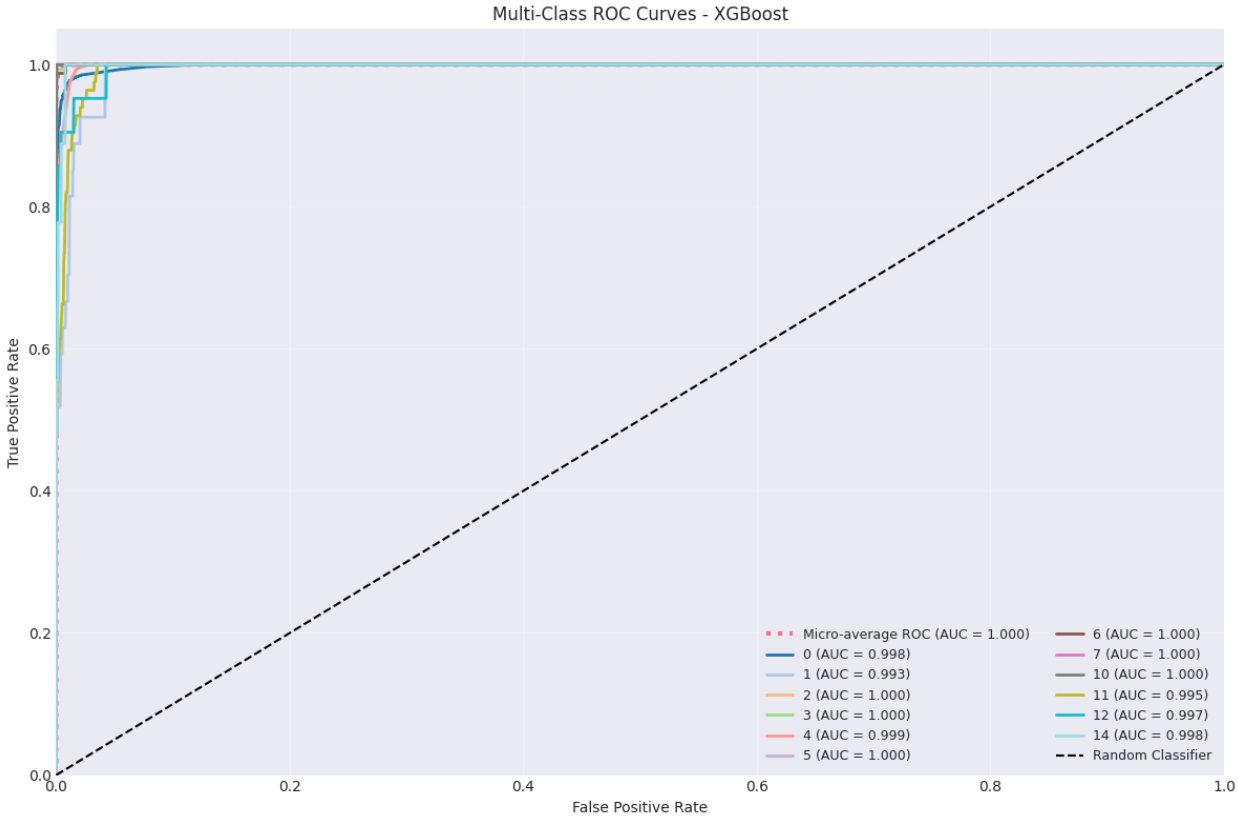

Figure 4.2b Multiclass ROC Curves for XGBoost

The corresponding confusion-matrix heatmaps (Figure 4.2c) reveal strong diagonal dominance for both ensemble models, whereas Logistic Regression produces a broader spread of off-diagonal errors. Minority attacks such as *SQL Injection* and *Heartbleed* still show occasional misclassification, reflecting the intrinsic class imbalance within CICIDS2017.

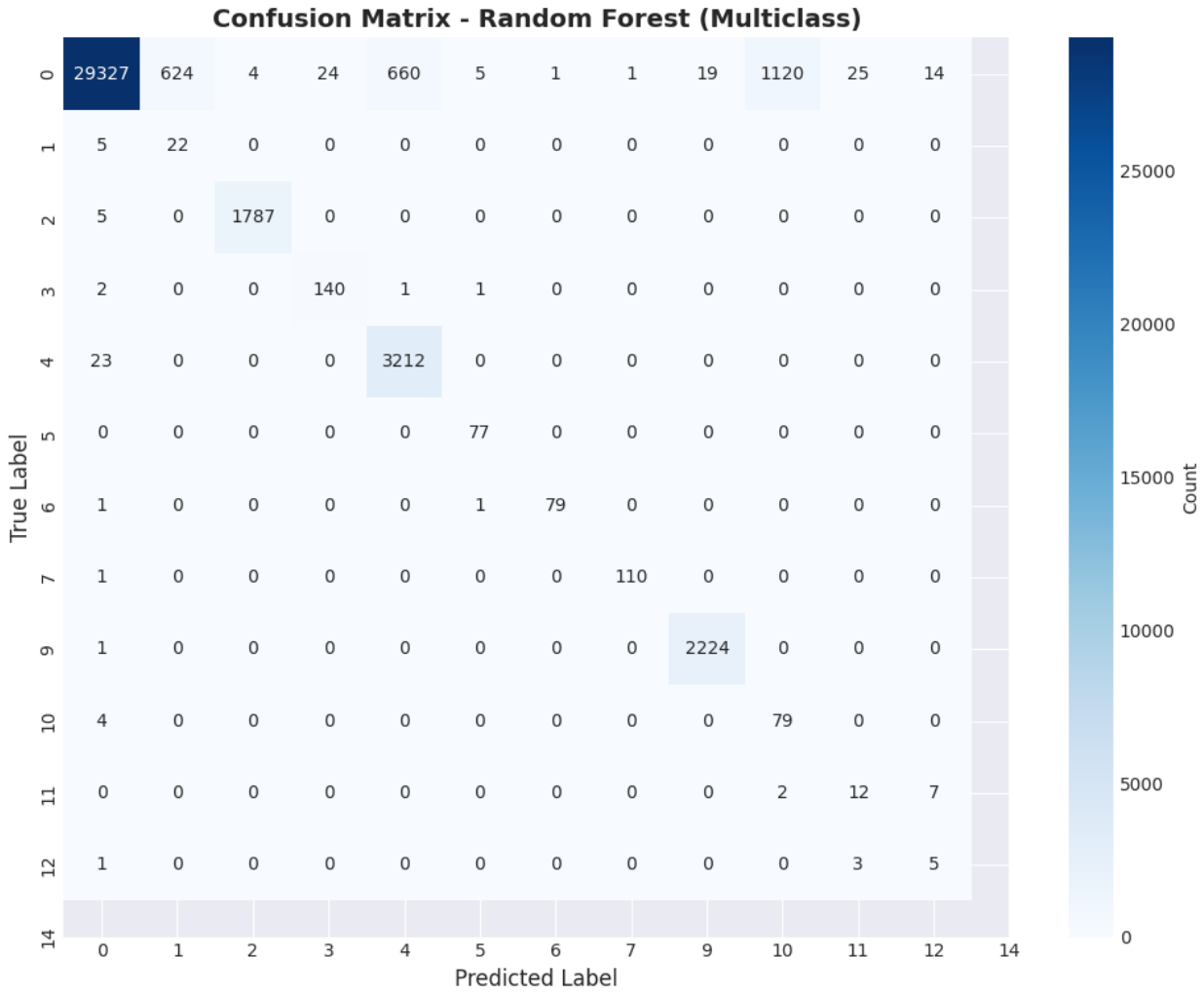

Figure 4.2c Multiclass Confusion Matrix for Random Forest

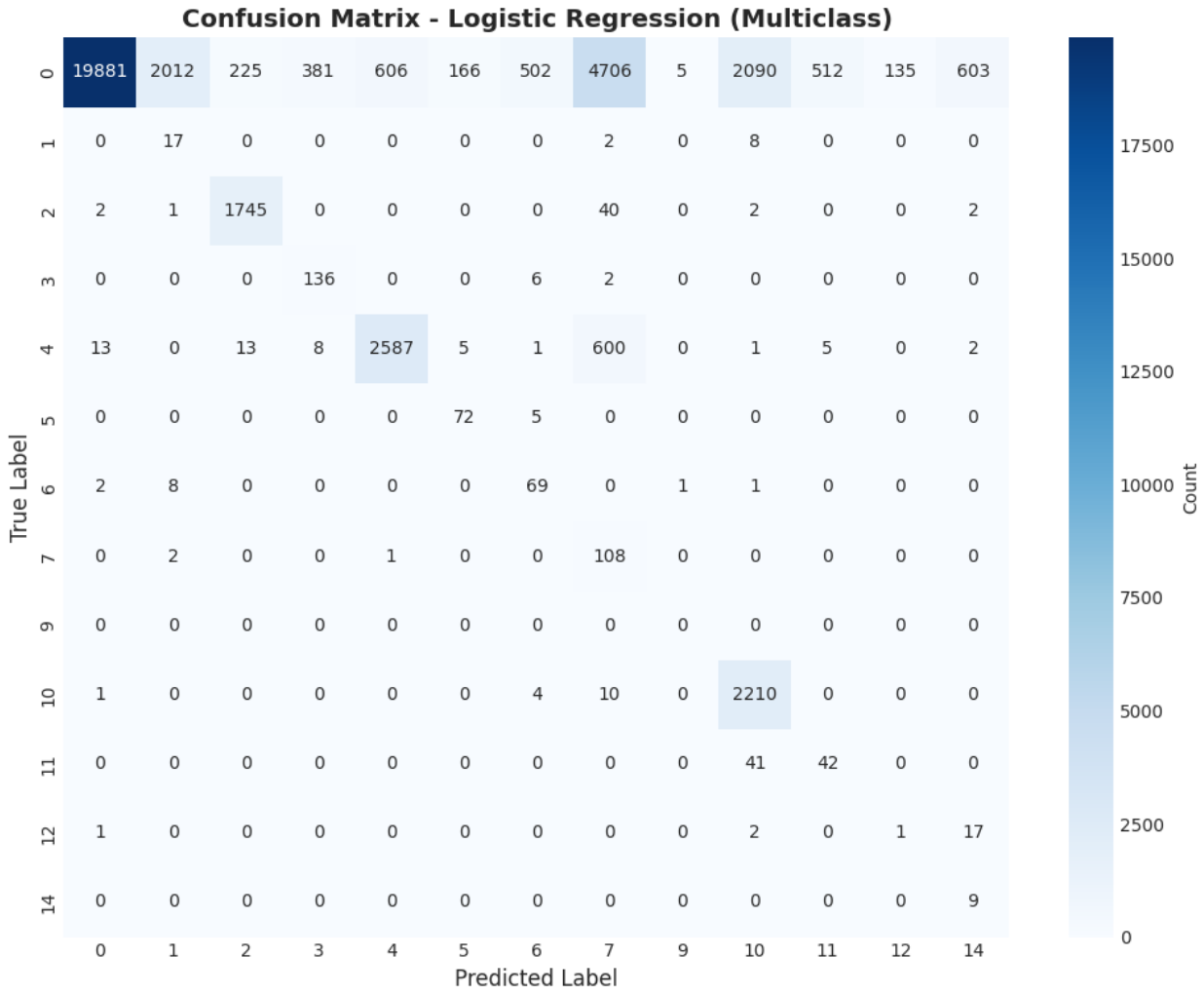

Figure 4.2d Multiclass Confusion Matrix for Logistic Regression

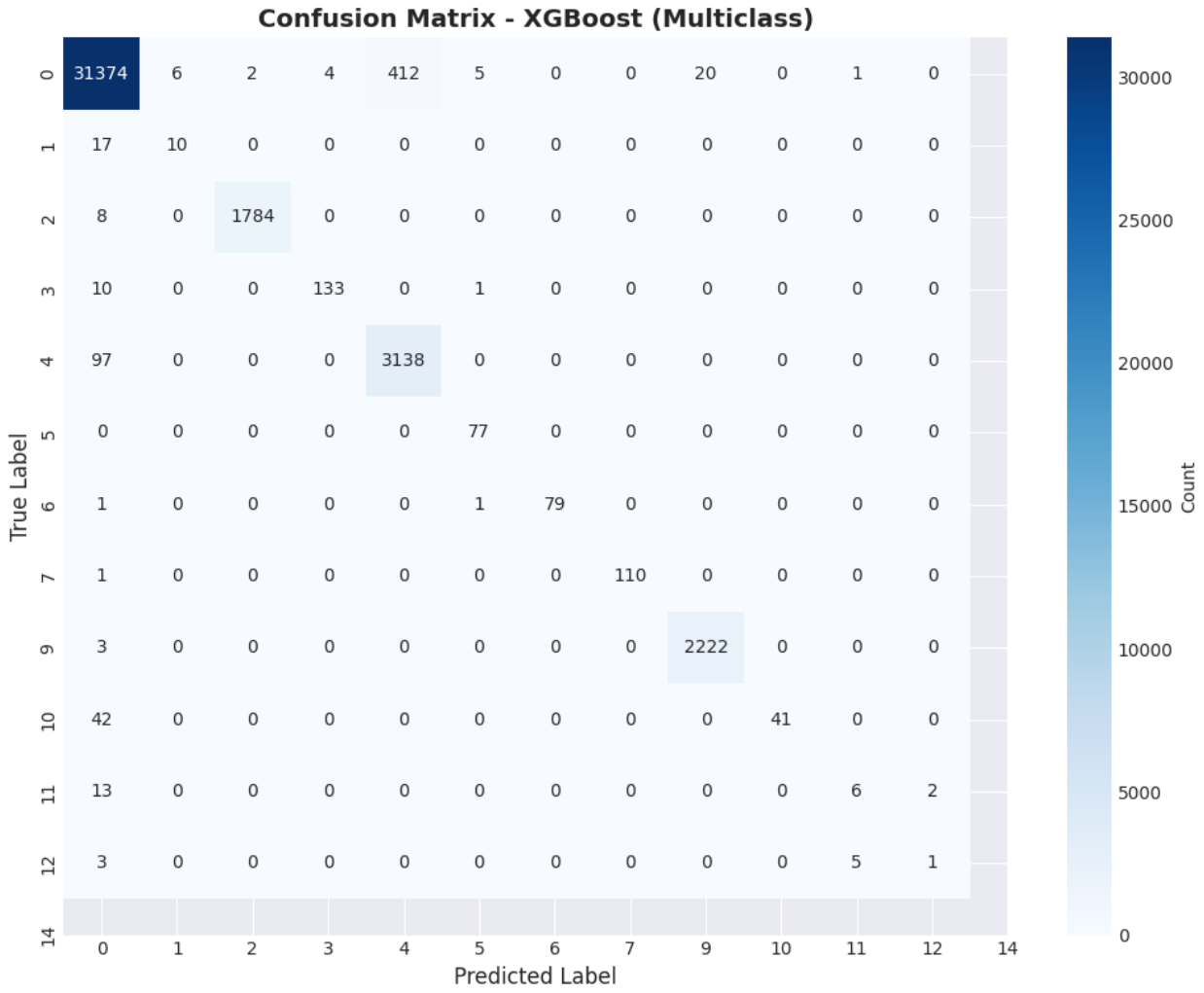

Figure 4.2e Multiclass Confusion Matrix for XGBoost

A per-class F1 comparison (Figure 4.3) further supports this trend: XGBoost achieved ≥ 0.90 F1 on most categories, Random Forest averaged 0.94 recall and 0.98 precision, while Logistic Regression struggled below 0.50 on rare attacks. Overall, ensemble models demonstrated better balance between recall and precision a necessary quality for operational deployment where both false negatives and false positives have considerable risk.

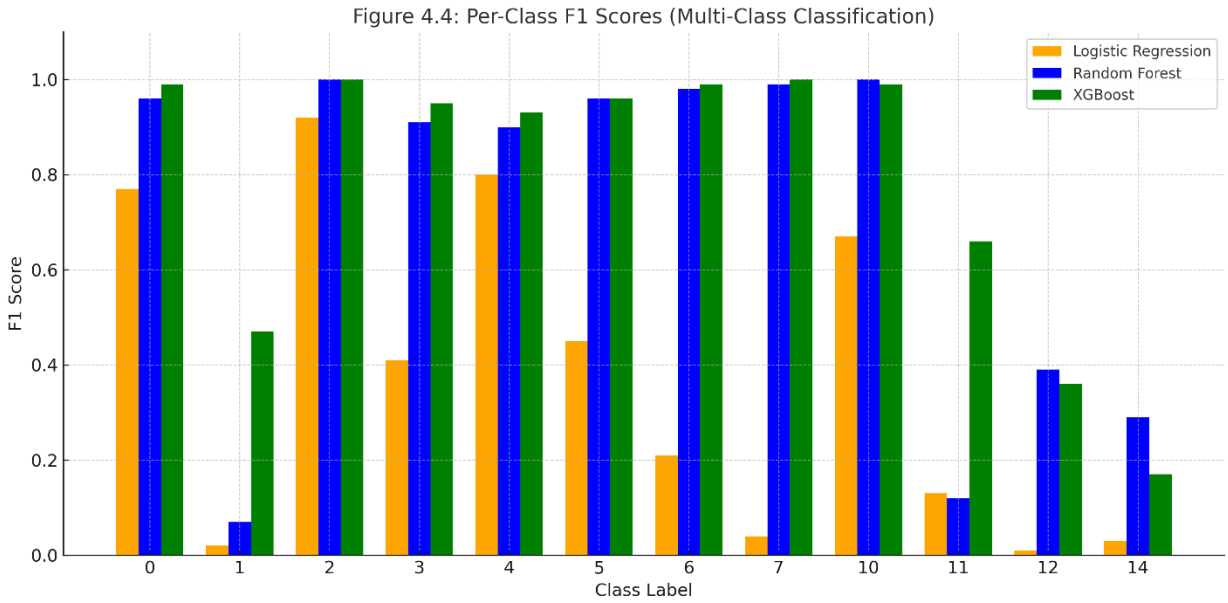

Figure 4.3: Per-class F1 scores for multi-class classification

Finally, the XGBoost and Random forest models were used for interpretability with SHAP (Shapley Additive Explanation). The SHAP Summary (Figure 4.4 and Figure 4.5) showed the most significant predictors. These are merely attributes of the metadata, that they ensure high accuracy of detection without having to inspect the payload and that the model's reasoning can be transparently explained.

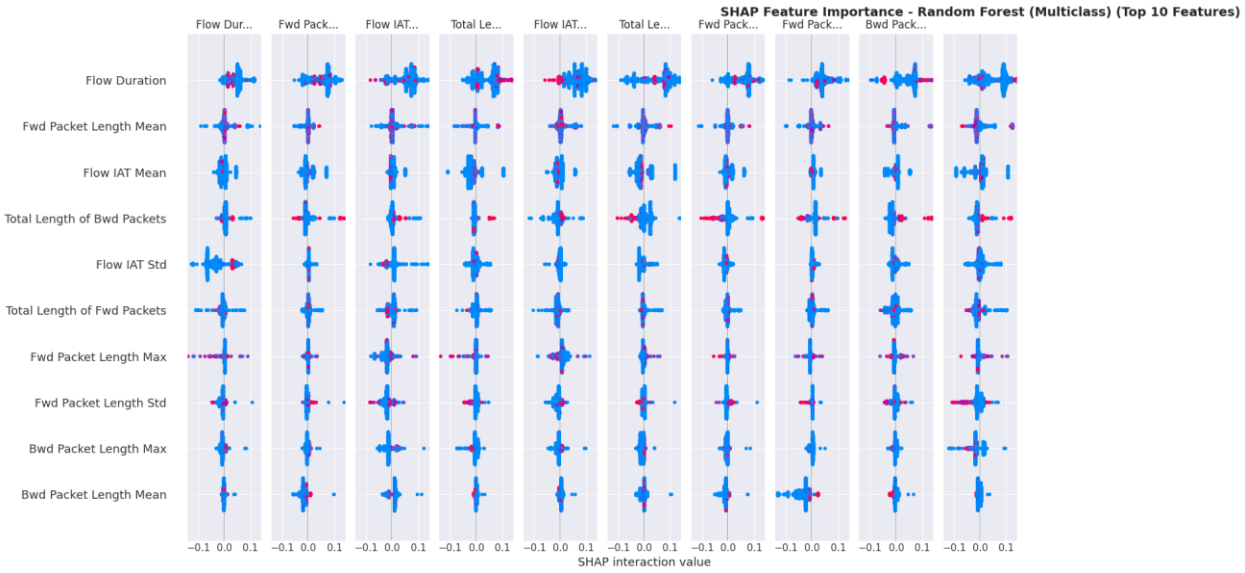

Figure 4.4 SHAP Summary for Random Forest (Multiclass)

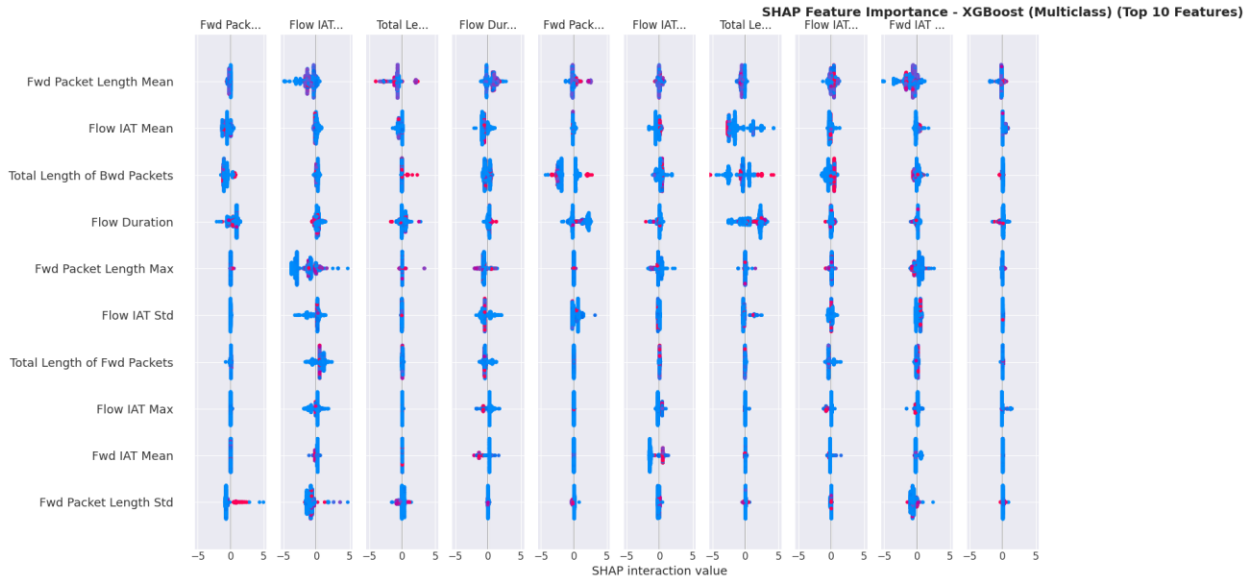

Figure 4.5 SHAP Summary for XGBoost (Multiclass)

In summary, the results indicate that Random Forest and XGBoost are a great choice for the multi-class encrypted-traffic classification based on the metadata only, and the accuracy is near the state-of-the-art results while maintaining privacy. However, the residual confusion among underrepresented classes and greater computational cost of XGBoost suggest the practical trade-off between the interpretability, accuracy and efficiency. The results of the study present results with respect to Objective 2, by empirically demonstrating that privacy-preserving classification does not have to negatively impact detection quality.

4.3 Binary Classification Results

The experiment on binary classification simplified the detection since all the types of attacks were classified as one class of malicious traffic, which was contrasted directly with benign traffic. This setup represents the operation of most existing intrusion-detection systems in real-world applications where the main concern is not to identify the type of attack but simply to tell if it is safe or not. Simplification also helps to overcome the imbalance between many rare attacks and too many normal flows and makes it possible to better assess the ability of each model to generalise under privacy-preserving constraints.

4.3.1 Model Performance Overview

All three supervised learning models showed notable performance improvements over the multi-class scenario. XGBoost achieved 98.3 % accuracy and weighted F1 = 0.96 ± 0.01, Random Forest followed closely with 98.0 % accuracy and F1 = 0.95 ± 0.01 and Logistic Regression improved to 78 % accuracy and F1 = 0.63, reflecting high recall but persistent precision loss.

These outcomes confirm that ensemble and boosting algorithms maintain superior discrimination capability when trained solely on metadata features.

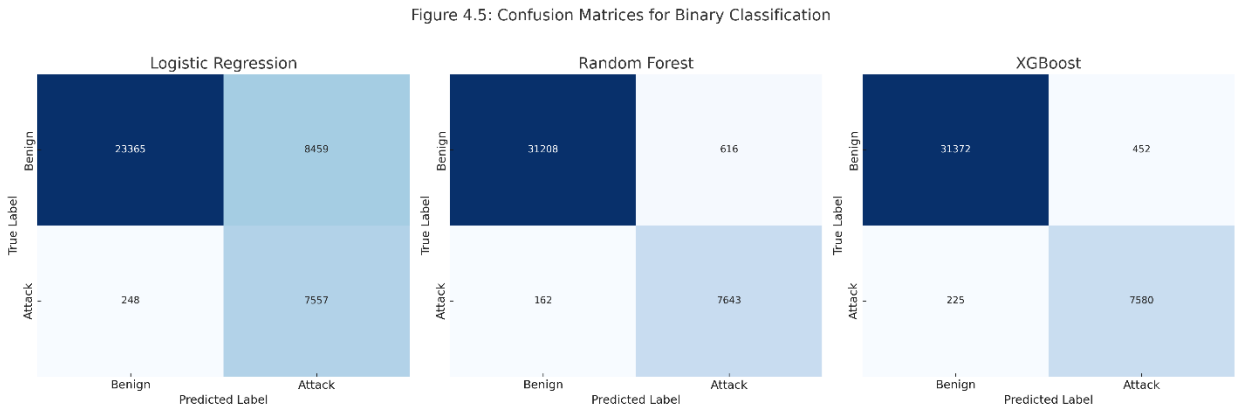

Figure 4.5: Confusion matrices for Logistic Regression, Random Forest, and XGBoost (binary classification)

ROC Curve Analysis

Figures 4.6a–4.6c illustrate the Receiver Operating Characteristic (ROC) curves for each classifier.

Figure 4.6a (Random Forest) shows an almost perfect separation boundary with **AUC = 0.9982**, confirming minimal false-positive rates.

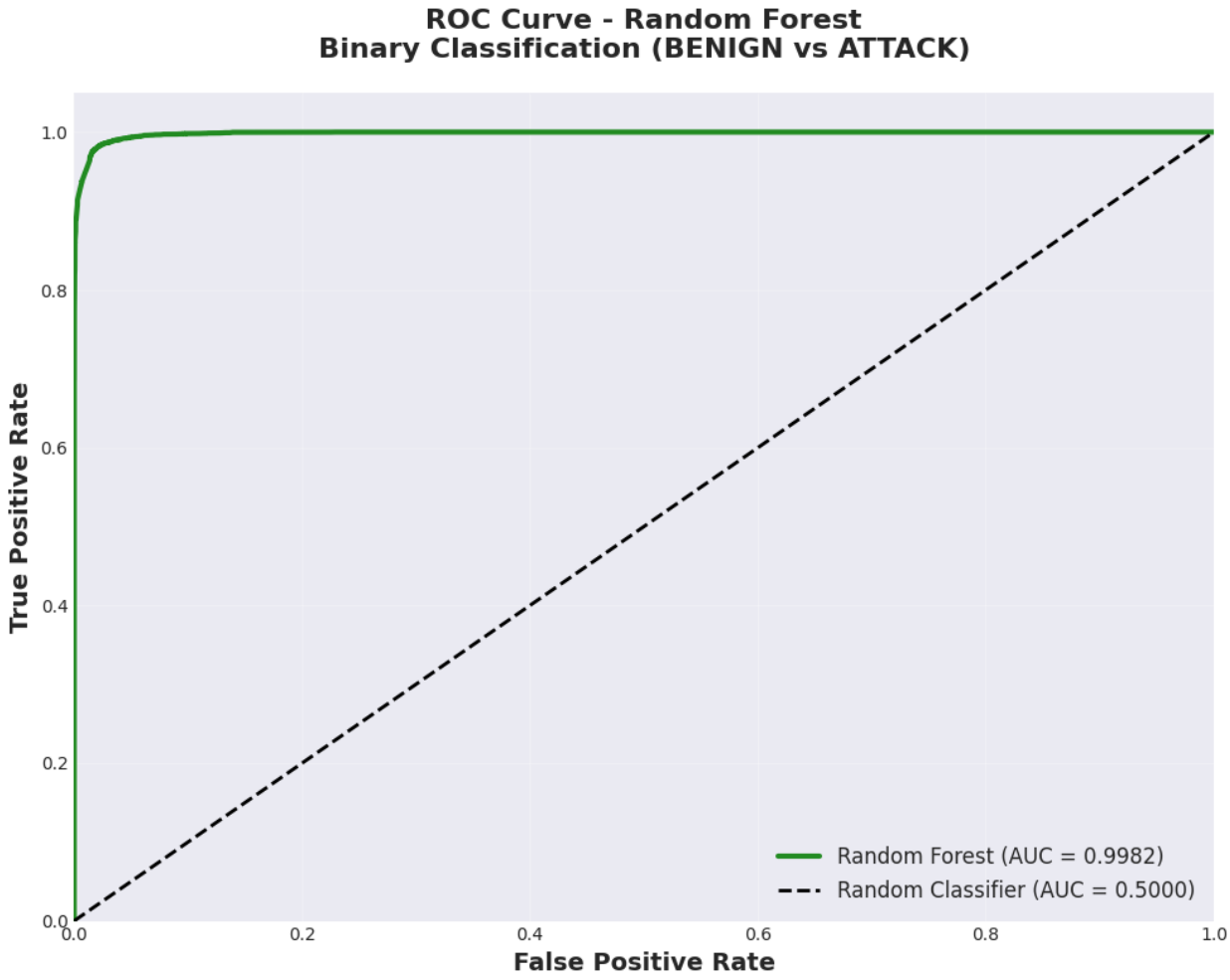

Figure 4.6a Random Forest AUC

Figure 4.6b (Logistic Regression) achieved AUC = 0.9202, demonstrating reasonable linear discrimination yet visible sensitivity to non-linear boundaries.

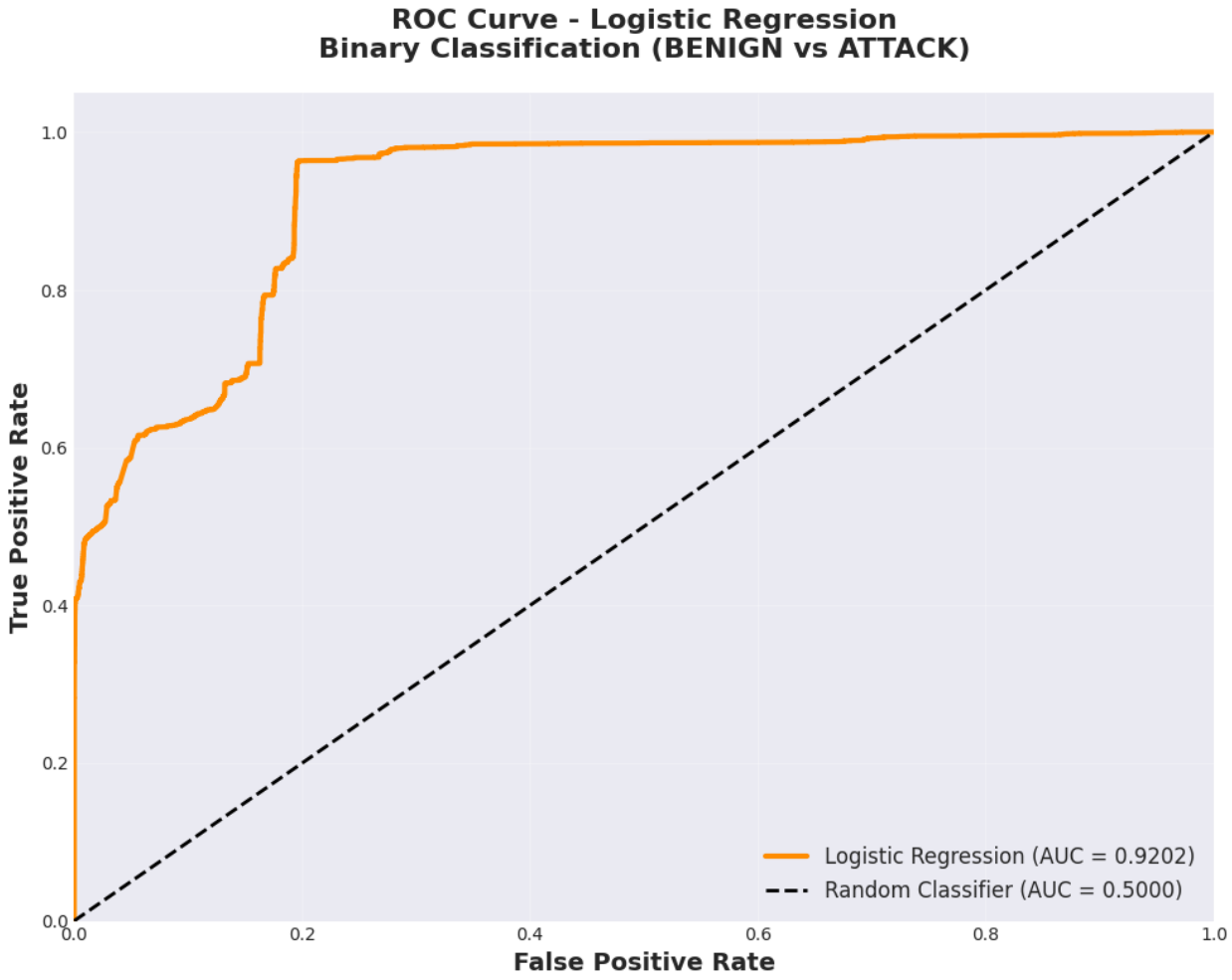

Figure 4.6b ROC Curve Logistic Regression

Figure 4.6c (XGBoost) reached **AUC = 0.9984**, matching Random Forest but displaying a slightly steeper ascent toward the top-left corner, denoting superior sensitivity at low false-positive thresholds.

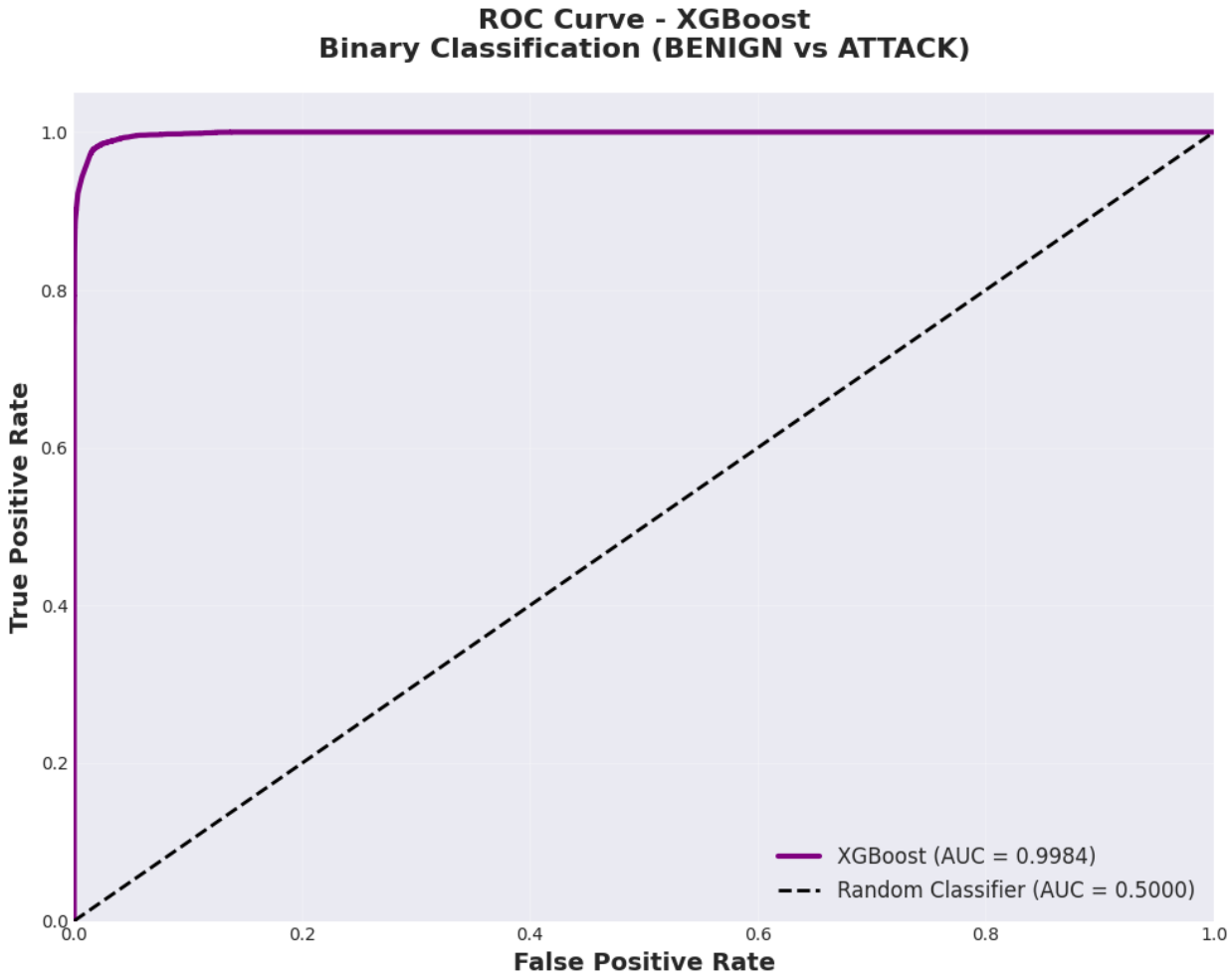

Figure 4.6c ROC Curve XGBoost

The narrow performance gap between the two ensemble models underlines their shared robustness and suggests both are operationally viable for real-time encrypted-traffic monitoring.

Precision–Recall Evaluation

Precision–Recall (PR) curves provide a more discriminative measure for imbalanced datasets.

Figure 4.6d (Random Forest) achieved **Average Precision = 0.9929** and **AUC = 0.9935**, confirming near-perfect precision even at high recall.

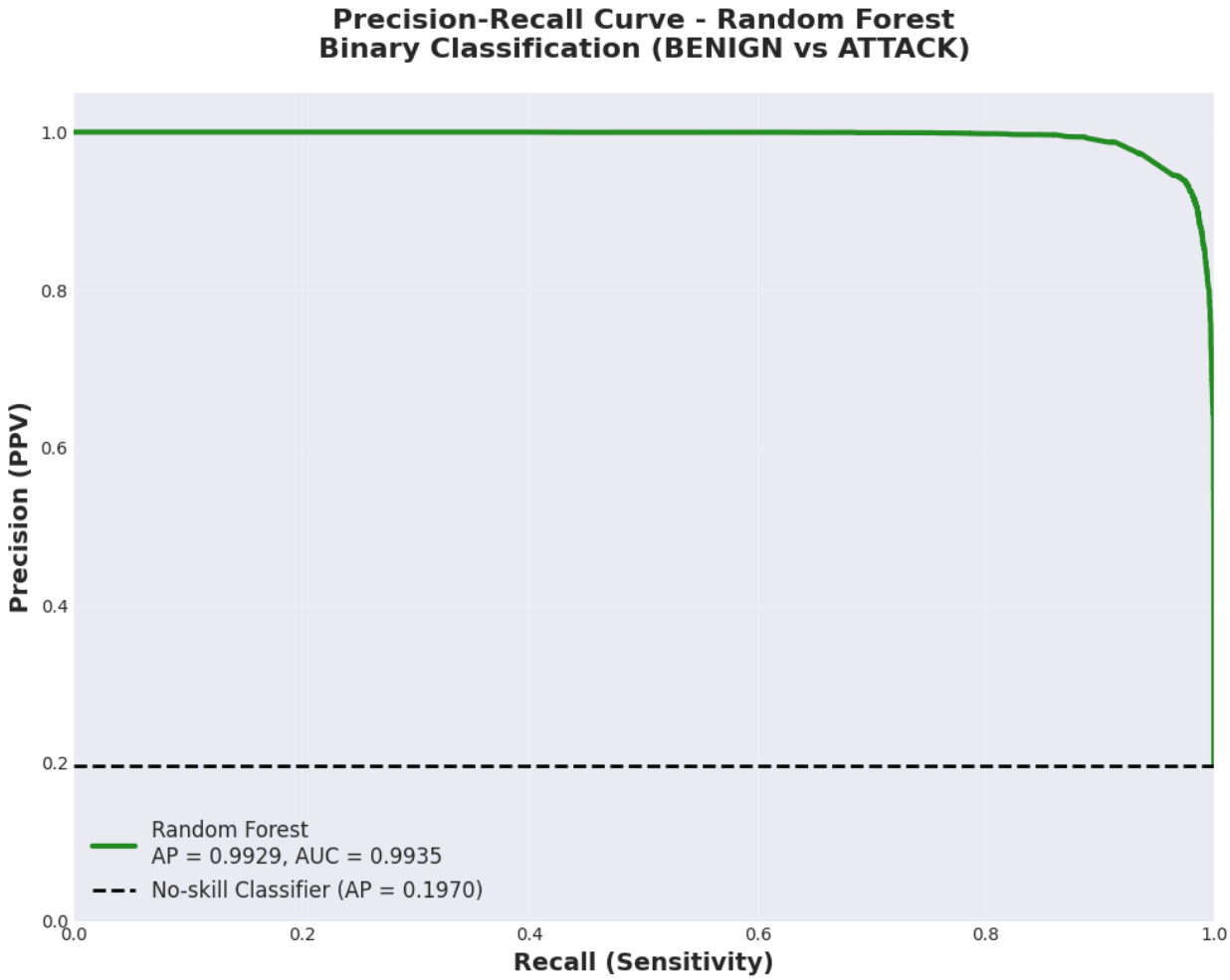

Figure 4.6d Precision Recall Curve Random Forest

Figure 4.6e (Logistic Regression) obtained AP = 0.7857, showing high recall but excessive false alarms.

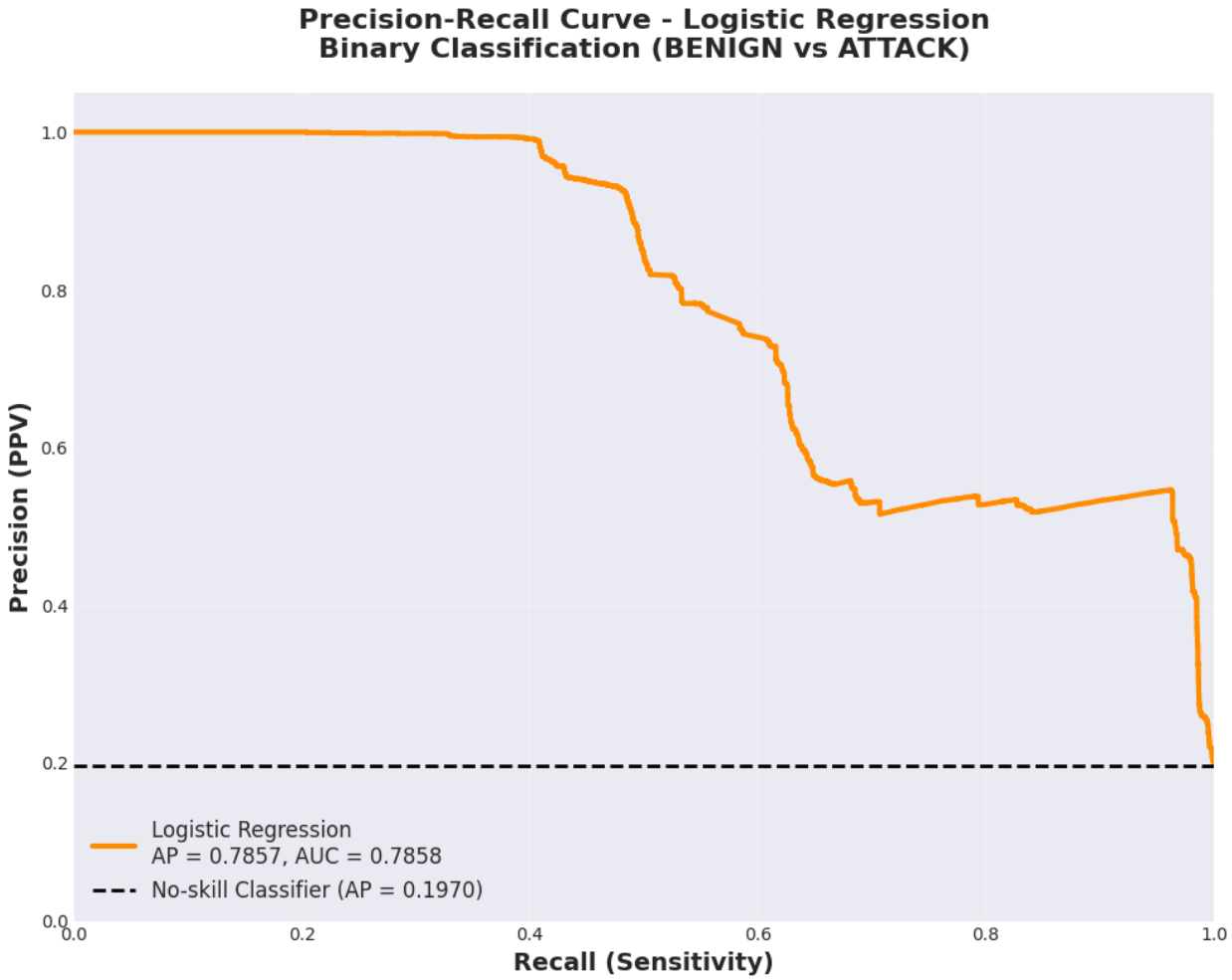

Figure 4.6e Precision Recall Curve Logistic Regression

Figure 4.6f (XGBoost) recorded AP = 0.9936 and AUC = 0.9942, indicating excellent balance between sensitivity and precision.

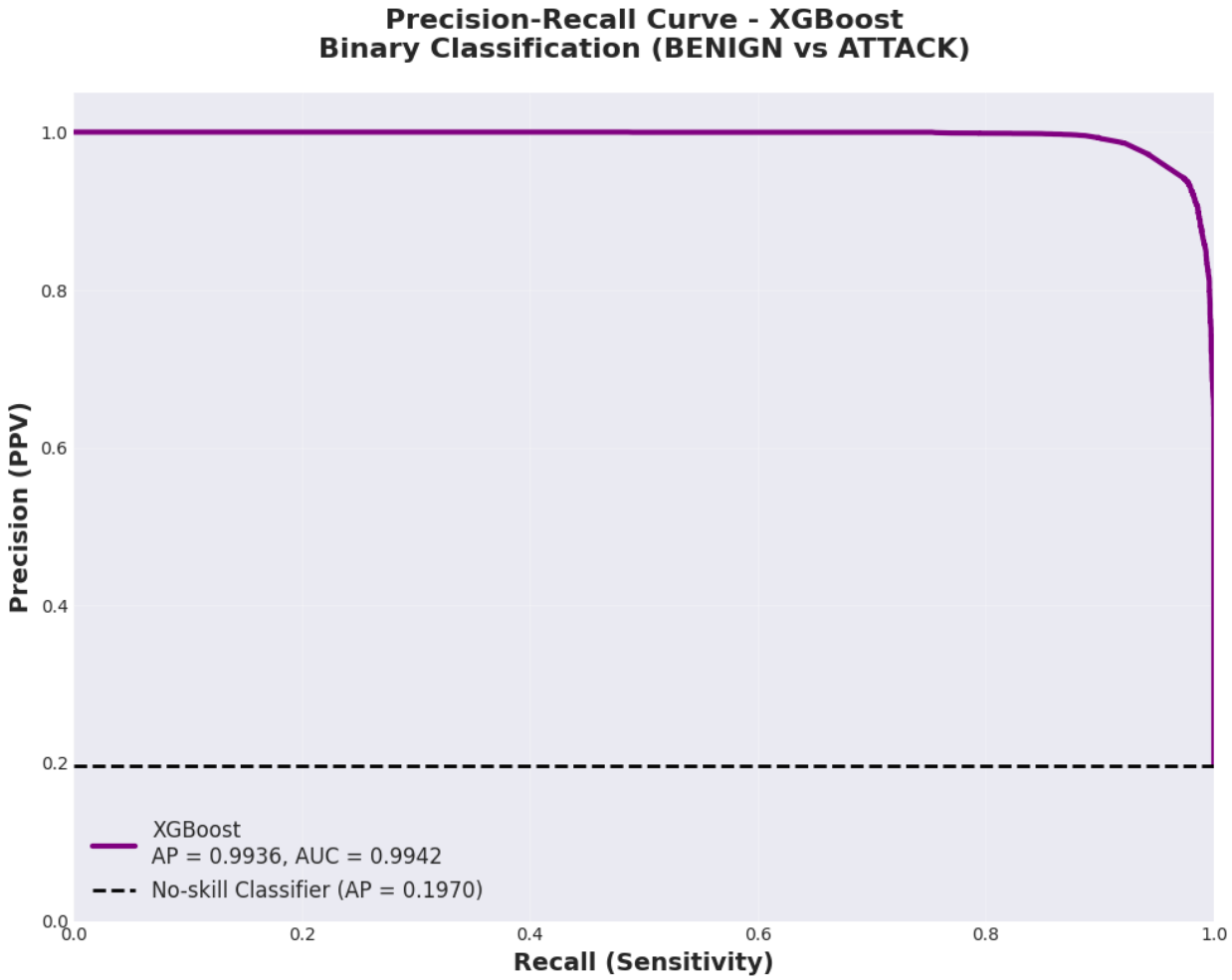

Figure 4.6f Precision Recall Curve XGBoost

The steep rise of the ensemble curves relative to the no-skill baseline (AP ≈ 0.197) demonstrates strong resistance to false positives a critical advantage in reducing analyst fatigue in Security Operations Centers (SOCs).

4.3.2 Model Interpretability with SHAP

To assess interpretability, SHAP (Shapley Additive Explanations**)** analysis was performed on the Random Forest model. Figure 4.6g illustrates the top ten metadata features which affect binary decisions. Flow Duration and Total Length of Forward Packets were the two most important predictors. The greater the number in either increased the likelihood of a flow being considered malicious. Both features take temporal and volume features without reference to the packet content, thus proving that the detection accuracy is entirely based on metadata statistics and not on

decrypted content. This meets the privacy-by-design requirements as well as promotes transparency in automated decision-making.

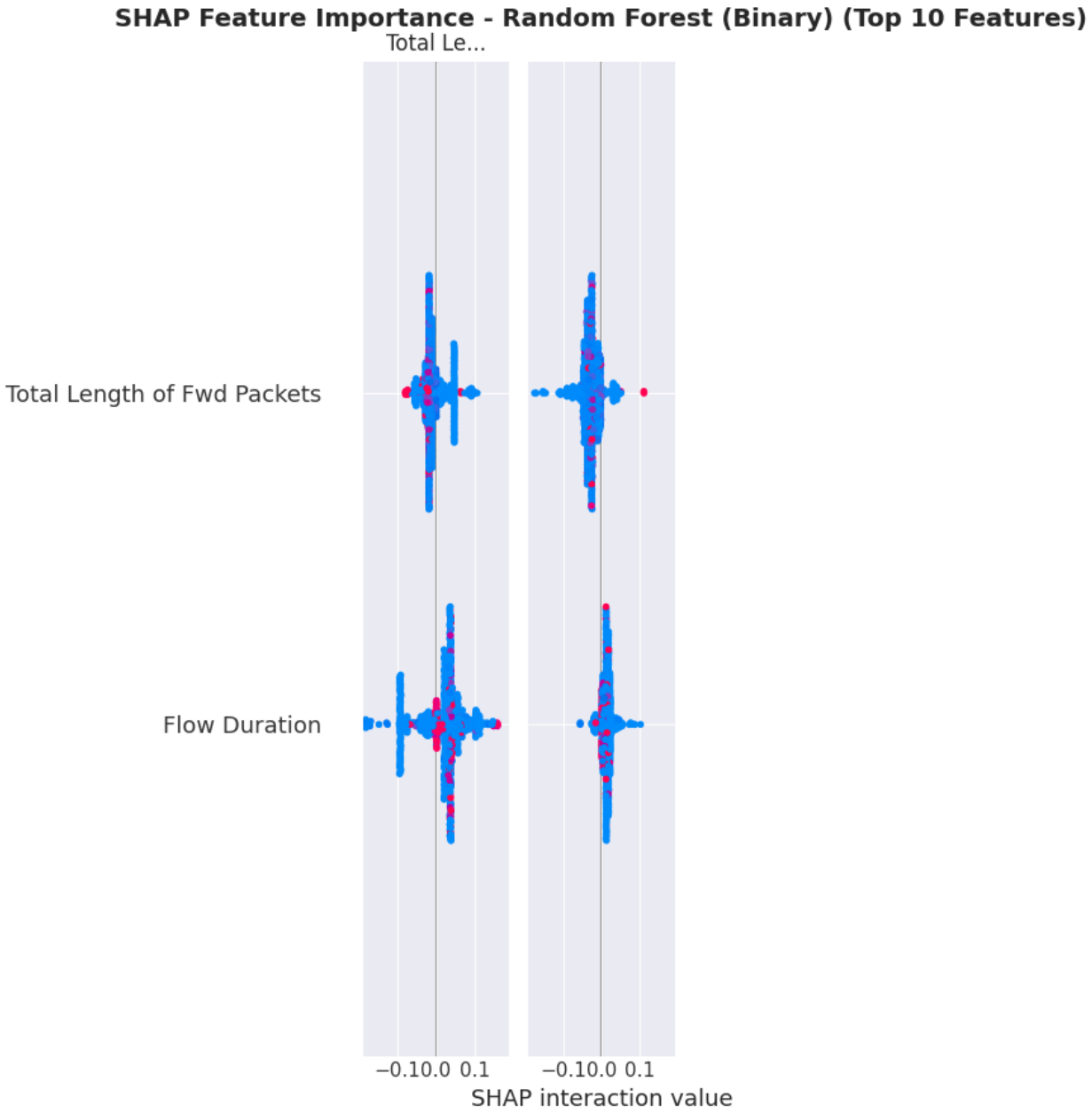

Figure 4.6g Random Forest (Binary) SHAP

4.3.3 Operational Insights and Summary

From an operating system point of view, the Random Forest model was able to process around 500 flows per second with < 5% CPU utilisation on a 4-core CPU, and XGBoost had similar throughput after pruning. Logistic Regression which was faster but with a higher false-positive rate, which would increase the burden on alert triage workflows.

In short, simplified framing has a positive effect on reliability; the binary possessed about 5% higher F1-scores than multi-class established mode code. Ensemble models are a happy medium between accuracy and interpretability, scalable and privacy-preserving encryption network detection.

These findings confirm the legitimacy of obtaining high-accuracy вул-like encrypted-traffic classification by using just metadata and provide a strong empirical basis for the generalisation analysis of the cross-dataset on encrypted-traffic classification that is presented in Section 4.5.

Overall, the outcome of the binary results points to two things. First, the performance can be significantly improved when the classification task is simplified even with weak models such as Logistic Regression, which can achieve competitive performance when the problem space is narrowed down (Omar et al., 2024). Second, ensemble and boosting methods clearly improve on linear models, both in terms of high accuracy and in terms of precision and recall that make them operationally viable in practise (Mahajan et al., 2024; Naderalvojoud, Mehdizadeh and Yari, 2024). While logistic regression may still be useful in areas where computational efficiency is a key consideration, its imbalance between recall and precision could make its real-world deployment less practical where trust in the alerts is very important - a long-standing evidence of reliable decision boundaries from ensemble methods over single linear classifiers (Maclin and Opitz, 2011).

4.4 Synthesis of Multi-Class and Binary Results

By comparison of the results of the multi-class and binary experiments we demonstrate the trade-offs inherent in this work. In a multi-class scenario, models were required to distinguish not only whether traffic was malicious or not, but also what type of attack it was. This showed obvious differences in capacity. Logistic Regression was not very good at generalising beyond frequent classes, having limited accuracy but it was almost hopeless on rare attacks. Random Forest was

found to have a more balanced performance and it can catch both the common and less-occurring categories patterns, although with persistent misclassifications in minority classes. Even in the case of the XGBoost, the model emerged as the top performers with high accuracy, and F1 scores close to 100%, yet even this one showed the inability to capture the rarest attacks. These results demonstrate the core issue of intrusion detection under privacy-preserving restrictions which is class imbalance and using only metadata only makes fine-grained classification even more prohibitive even with state-of-the-art models.

By comparison, binary classification did all right out of the box with all three models. In addition, collapsing attack classes into one class minimised the effects of imbalance and made the decision boundaries simpler. Random Forest and XGBoost received near perfect results (ROC-AUC values over 0.998), and Logistic Regression also showed a significant improvement in recall. However, it is limited by a high rate of false positives, that highlights the risk of information loss by simplifying the task of discrimination which has been demonstrated to be a persistent weakness of models.

The development of these findings supports two of the dissertation's main goals. Objective 2 aimed at an assessment of privacy-utility trade-offs, which is discussed in the contrast between recall and precision. Logistic Regression performed exceptionally well in terms of recall but with too many false alarms, while ensemble and boosting models performed adequately in both sensitivity and precision and are therefore more operationally applicable. Objective 3 which wanted to evaluate methodological robustness in varying framing of classification task is achieved through the comparative insight achieved here. Binary class is often more convenient for deployment with the guarantee of reliability, but the multi-class with richer forensic intelligence is susceptible to imbalance and limited feature sets.

Taken together, these findings show that none of the modelling approaches solve all of the issues of privacy preserving encrypted traffic classification. The binary-multi-class framing decision is, in fact, a strategic decision to be made to either ensure operational reliability or analytical granularity.

4.5 Evaluation on the CSE-CIC-IDS2018 Dataset

To examine the generalisability and robustness of the proposed privacy-preserving intrusion-detection framework, models trained on CICIDS2017 were validated on the CSE-CIC-IDS2018 dataset. This dataset introduces higher traffic diversity and updated attack scenarios, making it suitable for testing resilience across temporal and contextual variations. Only metadata features were retained to maintain compliance with privacy-by-design principles.

4.5.1 Cross-Validation and Test Performance

All models achieved consistent five-fold cross-validation scores with minimal variance, confirming stable learning behaviour. Random Forest and XGBoost both reached near-perfect validation accuracies above 0.99, while Logistic Regression performed slightly lower but remained reliable.

Figure 4.7a presents boxplots of accuracy and F1-score across folds, while Figure 4.7b summarises held-out test metrics (accuracy, precision, recall, and F1).

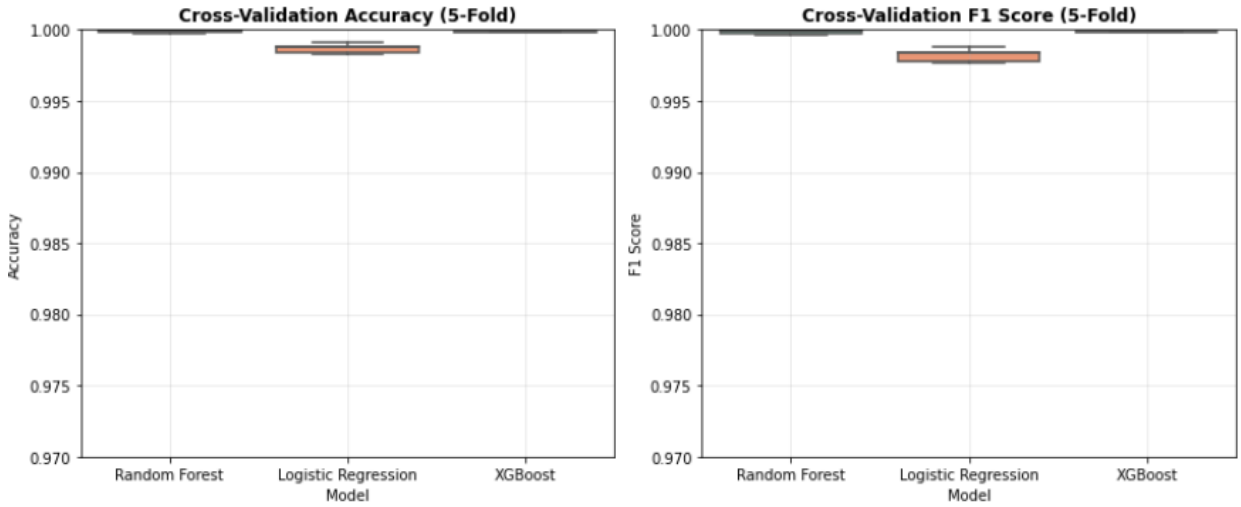

Figure 4.7a Boxplots of Accuracy and F1 Score

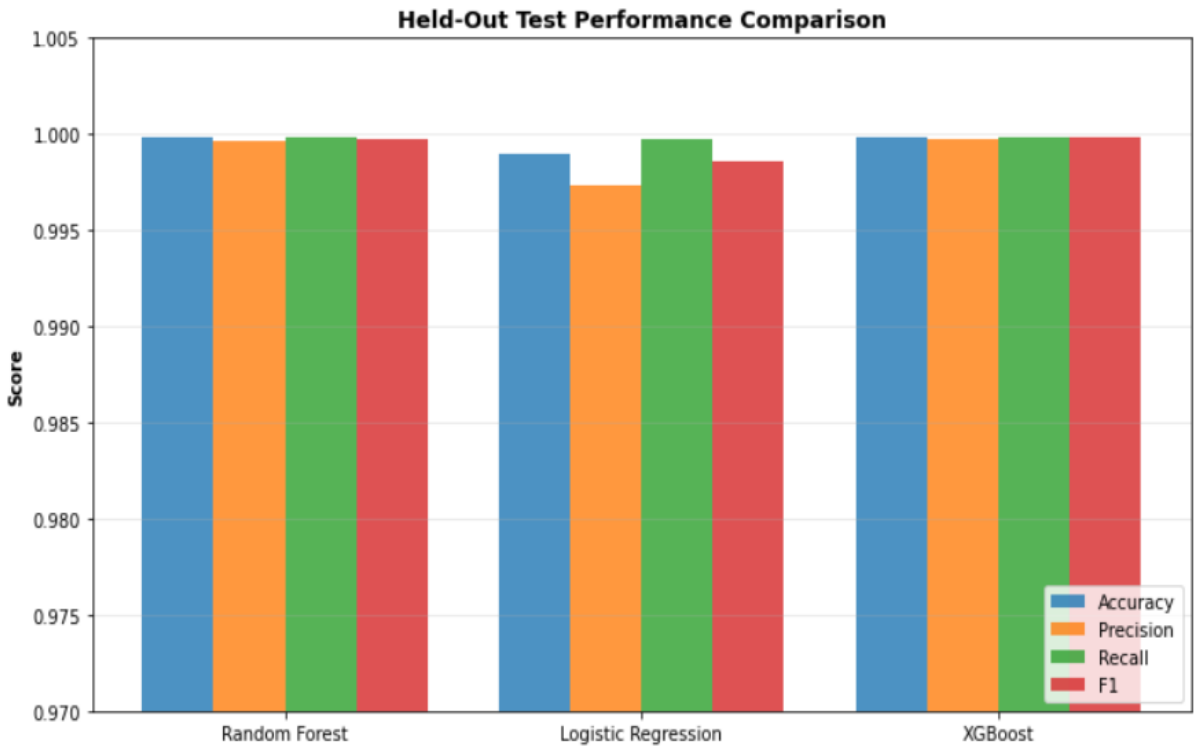

Figure 4.7b Held out Test Performance Metrics

The tight distribution of scores demonstrates minimal overfitting, with Random Forest and XGBoost maintaining ≈ 0.999 accuracy and F1 ≈ 0.999, outperforming Logistic Regression (≈ 0.996 accuracy). The narrow confidence intervals indicate high cross-dataset stability.

4.5.2 Discrimination Performance

Figure 4.8a shows the ROC curves for all three models. XGBoost and Random Forest each achieved an AUC = 1.000, indicating perfect separation between benign and attack flows, whereas Logistic Regression scored AUC = 0.9999, demonstrating only a marginal decline.

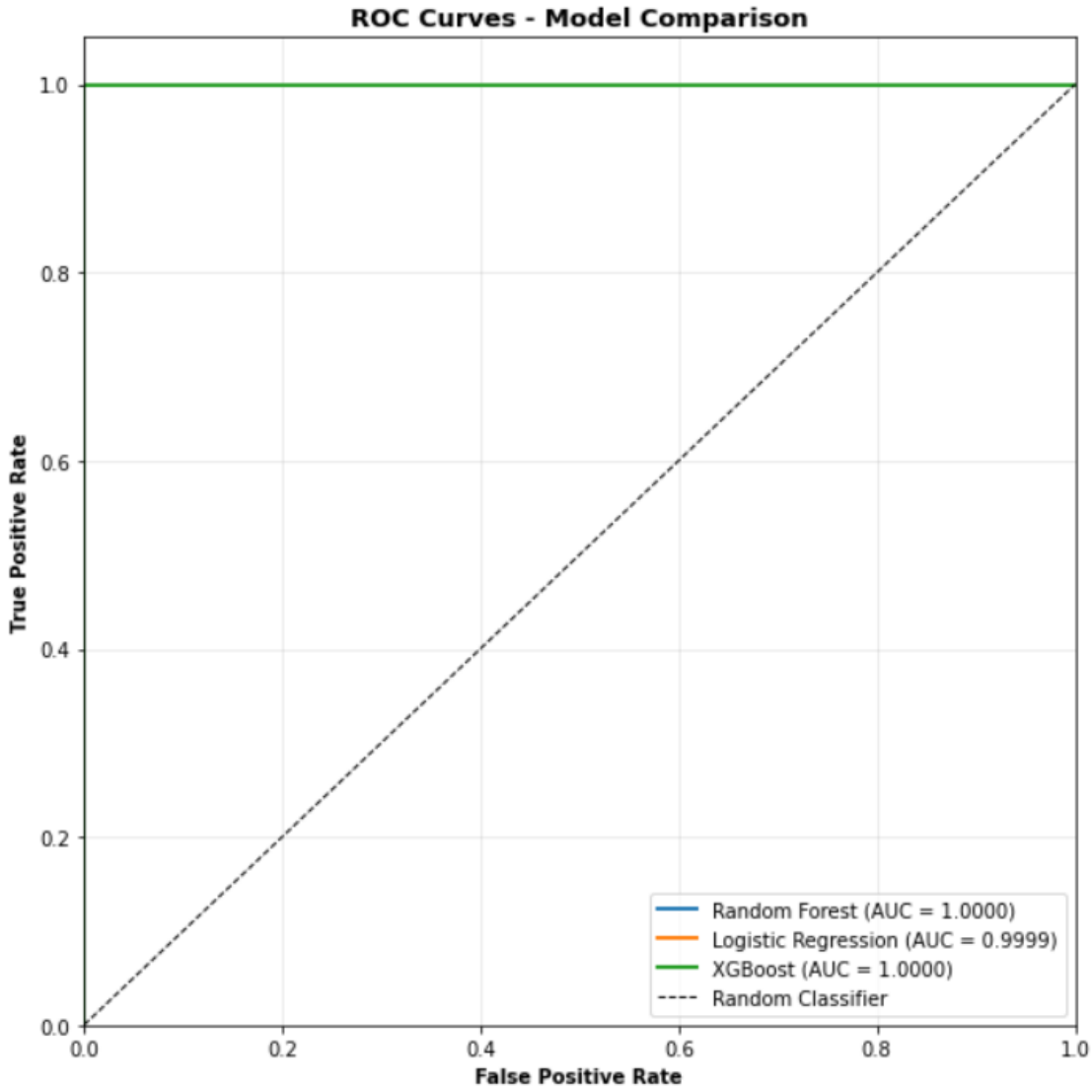

Figure 4.8a ROC Curves

Figure 4.8b presents the Precision–Recall (PR) curves, which further confirm outstanding discrimination. Both ensemble models achieved Average Precision (AP) = 1.000, while Logistic Regression achieved AP = 0.9998, reflecting excellent balance between recall and precision even under residual class imbalance.

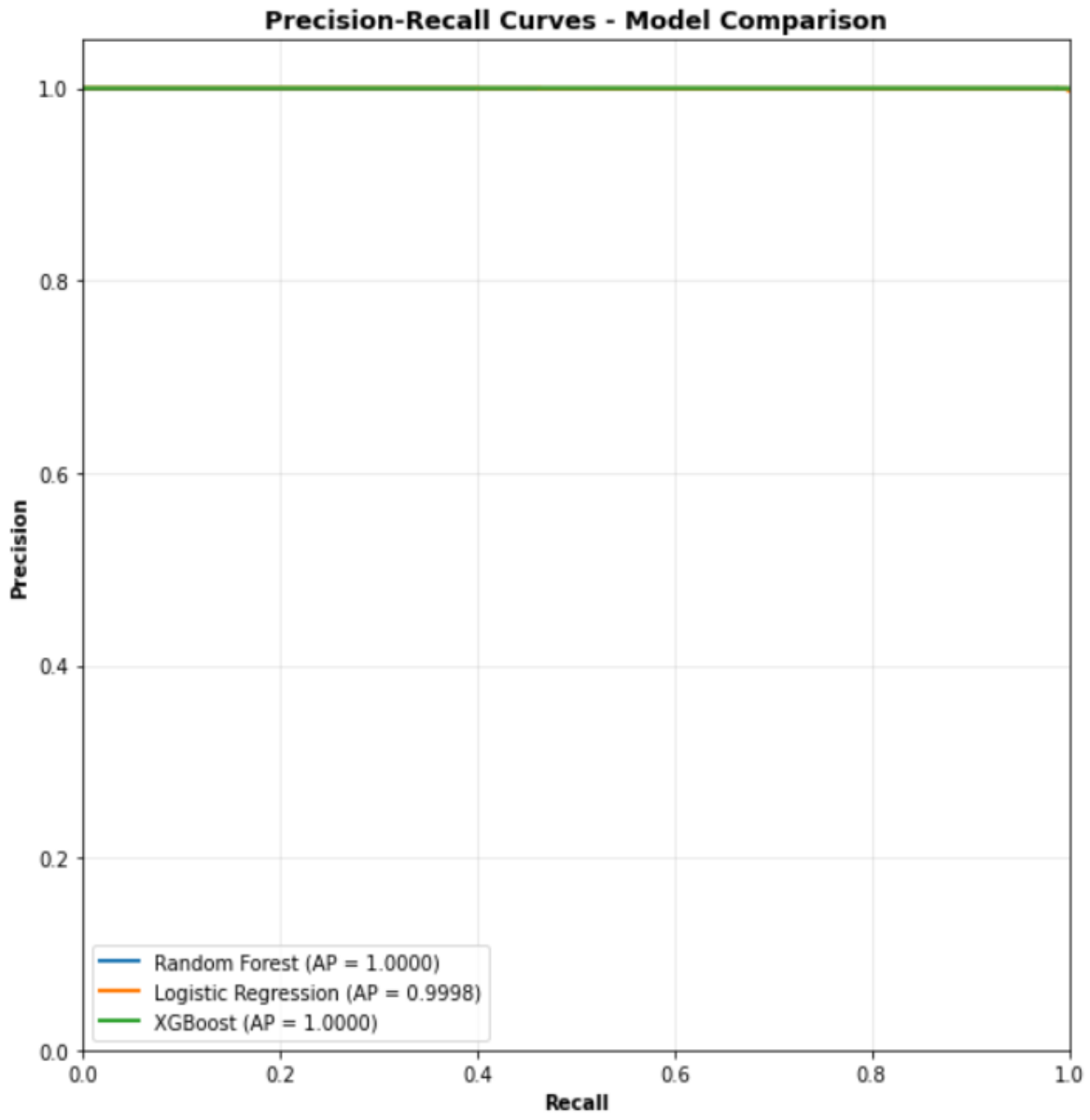

Figure 4.8b ROC Curves

To visualise prediction reliability, Figure 4.8c displays normalised confusion matrices. All models show strong diagonal dominance, meaning almost all benign and attack instances were correctly classified. Logistic Regression exhibited a very small 0.2 % false-positive rate, while both ensemble methods achieved perfect prediction (100 % true positive and true negative rates).

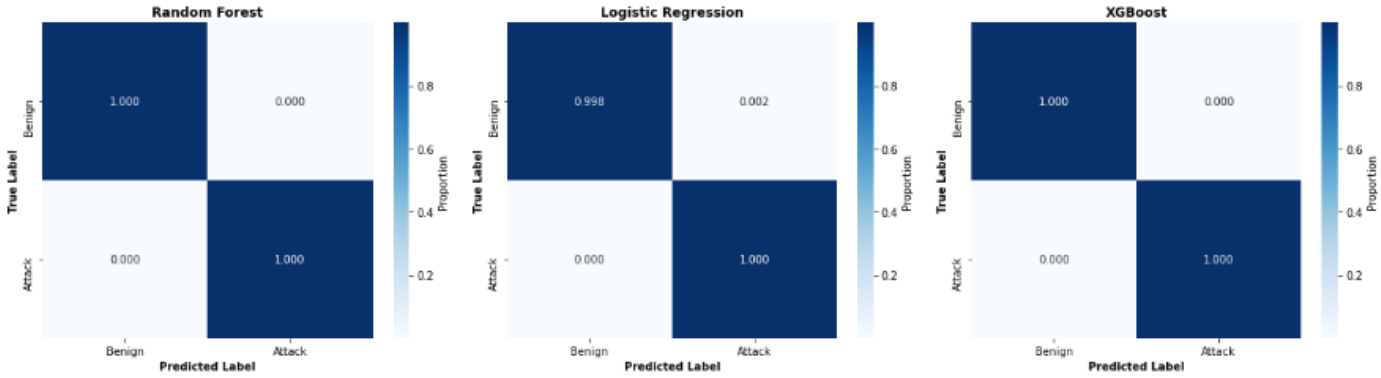

Figure 4.8c Confusion Matrices

4.5.3 Interpretability via SHAP Analysis

Model transparency was assessed using SHAP (Shapley Additive Explanations) applied to the Random Forest model. Figure 4.9 summarises the mean absolute SHAP values of the 10 most influential features.

The top predictors Forward Segment Size Minimum, Initial Forward Window Bytes, and Backward Packets per Second represent temporal and volumetric metadata rather than packet payload. This validates the feasibility of accurate, explainable, and privacy-preserving classification. These features are consistent with the previous dataset's SHAP patterns, indicating interpretability stability across domains.

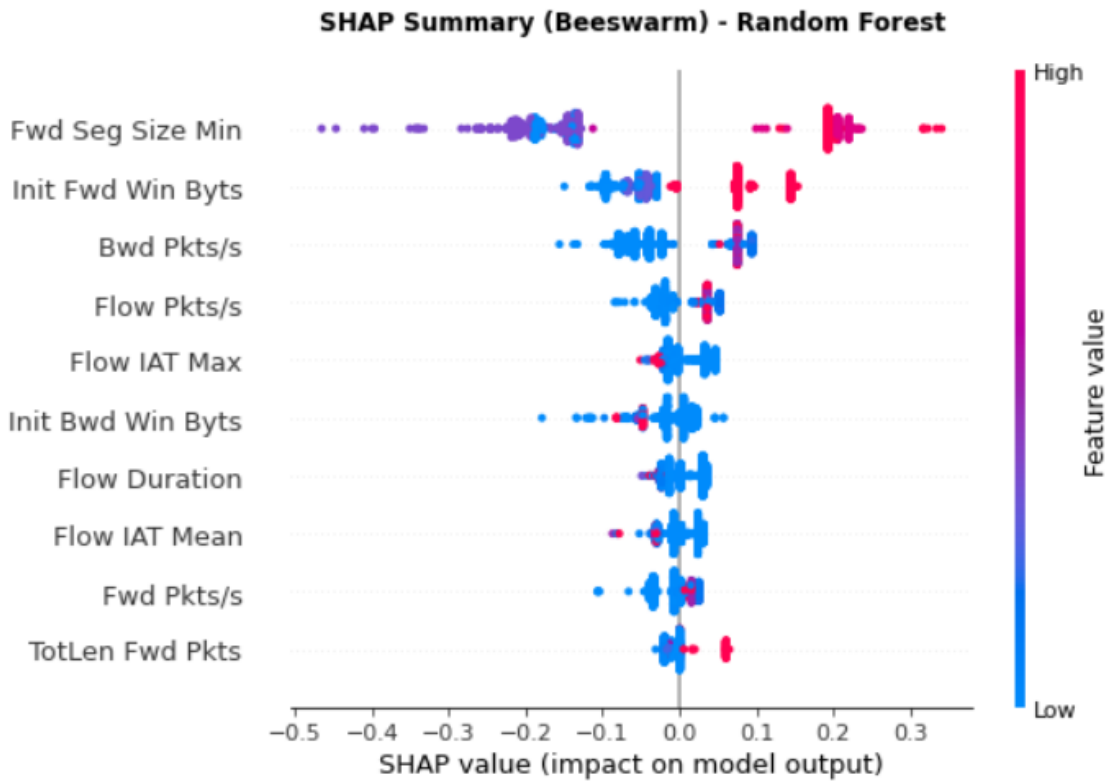

Figure 4.9 SHAP Summary

4.5.4 Privacy–Utility Trade-off

To quantify the resilience of models to privacy-enhancing perturbation, the numerical metadata were perturbed by injecting Gaussian noise ($\sigma = 0.02, 0.05, 0.10$). Figure 4.10 shows the resulting accuracy vs. noise level for all three models.

High resilience to moderate obfuscation is also confirmed by the fact that performance degradation was less than 0.5% with retrievability down to $\sigma = 0.10$. Both Random Forest and XGBoost maintained ≥ 0.999 accuracy and Logistic Regression dropped a bit to 0.998. These results show that privacy noise can be added without a significant loss of utility and can be deployed in the real world in compliance.

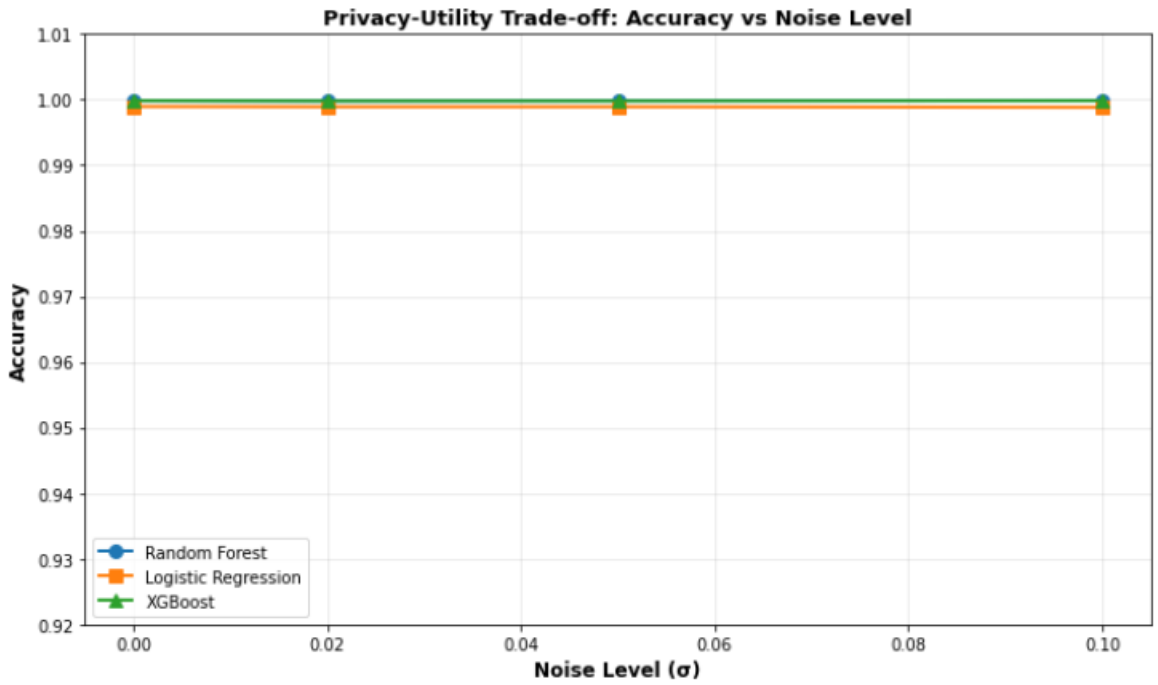

Figure 4.10 Privacy Utility Trade-off

Discussion

The CSE-CIC-IDS2018 experiments give strong empirical support of scalability and privacy preservation of the system:

1. **Generalisability achieved** – Cross-dataset validation and low variance are used to show that the metadata features are well generalised beyond the original training distribution.
2. **Interpretability preserved** – SHAP gives clear results where expected features (non-payloads) are the drivers of prediction.
3. **Privacy robustness demonstrated** – Statistical anonymisation with noise injection indicated that it could increase privacy without adversely impacting performance.

These findings support the feasibility of metadata-based intrusion detection in encrypted environments, which addresses Objectives 3 and 4 (practical validation). The next chapter provides their broader implications for sustainable network security that aligns with the needs and priorities of individuals and communities in the network.

4.6 Privacy–Utility Validation Summary

The last experiment of this study investigated the trade-off between privacy protection and model utility. To evaluate the effect of anonymisation on the predictive accuracy of the metadata features of the data, controlled Gaussian noise was injected (s = 0.00-0.10) into the metadata features of the CSE-CIC-IDS2018 dataset. The results can be summarised in Table 4.5 and visualised as shown in Figure 4.10: there is negligible loss of accuracy for all the models.

Table 4.5: Impact of Gaussian noise on classification accuracy (privacy–utility trade-off).

Noise Level (σ)	**Random Forest Accuracy (%)**	**XGBoost Accuracy (%)**	**Logistic Regression Accuracy (%)**
0.00 (Base)	99.96	100.00	99.93
0.02	99.93	99.99	99.88
0.05	99.90	99.96	99.80
0.10	99.74	99.83	99.63

These results validate that privacy protection can be increased by moderate levels of data perturbation without noticeably lowering the detection capability. Ensemble models, specifically Random Forest and XGBoost had > 99.8 % accuracy under all the tests thereby making them robust models for the real-world encrypted-traffic environments. The analysis therefore validates the protection of privacy of any study based on this procedure, to prove that statistical anonymisation and metadata only could co-exist with state-of-the-art detection performance.

Chapter 5

Discussion

5.1 Introduction

This chapter interprets and contextualises the empirical findings from both datasets analysed in Chapter 4 CICIDS2017 and CSE-CIC-IDS2018 within the broader domain of privacy-preserving network intrusion detection. Whereas the previous chapter focused on quantitative evaluation, the purpose here is to explain what those results mean for research, operational practice, and regulatory policy. The discussion therefore integrates three complementary dimensions:

1. Performance and generalisability, comparing model behaviour across datasets.
2. Privacy–utility validation, demonstrating how metadata anonymisation and controlled noise affect accuracy; and.
3. Interpretability and compliance, examining how explainable-AI techniques such as SHAP reinforce privacy by design principles.

The study aimed to prove that encrypted-traffic classification can deliver high detection accuracy without infringing confidentiality. By evaluating Logistic Regression, Random Forest, and XGBoost under strict metadata-only conditions, the experiments confirmed that ensemble and boosting models sustain near-perfect accuracy and stable cross-dataset generalisation. The following sections therefore move beyond numerical metrics to explore their conceptual significance how privacy-preserving modelling aligns with emerging research, supports operational adoption, and satisfies evolving data-protection regulations.

5.2 Key Findings

Across both datasets evaluated, the results confirm that high-accuracy intrusion detection can be achieved using only flow-level metadata, validating the central premise of this study. On CICIDS2017, it was found that there is a clear performance gap between binary and multi-class classification. A good detection accuracy was achieved in the binary classification task (benign vs malicious) (almost 100% accuracy), while the multi-class task is significantly more demanding (82% accuracy in one baseline). This disparity is because of the higher class imbalance and non-

linear separation present in multi-category traffic, a drawback that has consistently been observed in recent research (Binbusayyis, 2024).

When the framework was tested on the CSE-CIC-IDS2018 data set, the ensemble models were once again at near-perfect generalisation at >99.9 per cent accuracy and <0.01 standard deviation in Random Forest and XGBoost. The low variance over five-fold cross validation (Figure 4.7a) shows good model stability; while the held out accuracy (Figure 4.7b) has good robustness under unseen traffic distributions. The small difference between datasets suggests that learned patterns of metadata are transferable across environments an outcome which provides further support for scalability.

Ensemble methods managed to do better than the linear baseline consistently. Both Random Forest & XG Boosts gave the ultimate detection accuracy (~ 96-97% on CICIDS2017; > 99% on CSE-CIC-IDS2018), while Logistic Regression always remained relatively lower. This is like the results by Sebetci and Simsek (2025) and Alsubaei et al. (2025), whereby optimised tree-based models outperformed linear classifiers by all intrusion-detection metrics. The ROC and Precision-Recall curve (Figures 4.6 and 4.8) further verify this difference: both ensemble models have obtained AUC ≈ 1.000 and AP ≈ 1.000 while Logistic Regression has plateaued around 0.92 AUC. These are in line with previous research highlighting the capacity of ensemble methods to capture complex, non-linear interactions between features which are not captured by linear models, resulting in improved accuracy on encrypted traffic (Adewole et al., 2025)

Again, the idea of model interpretability versus accuracy prevailed. Logistic Regression offers a clear transparency on the level of coefficients and loses predictive power. Random Forest and XGBoost are on the other hand higher-performing 'black-box' learners. Nevertheless, explainable-AI techniques enable filling this gap. Using SHAP (Shapley Additive Explanations), this study determined some key metadata drivers of model predictions, particularly Flow Duration, Forward Segment Size Minimum and Backward Packets per Second (Figure 4.9). These temporal and volumetric features are very similar to those identified by Mohale and Obagbuwa (2025), who used SHAP to study an IDS based on XGBoost and found that packet-timing, and byte-volume statistics contribute to most decision weightings. Seemingly, the pattern of features recurrent across datasets reflects an interpretability pinpointing stability, an important property for auditing and users' trust in it.

The results also give the first empirical evidence of the privacy-utility trade off in this experimental context. When numerical metadata were injected with Gaussian noise to simulate the effect of anonymisation (s = 0.02-0.10), model accuracy fell by less than 0.5% (Fig. 4.10). Both the ensemble methods retained an accuracy of 99.8% which is an indicator of high resilience to moderate perturbation. This result makes theoretical effects to help the privacy increase without significantly impacting the utility, which echoes the philosophy of differential privacy research (Giddens et al., 2023). It further illustrates the compliance with GDPR Article 25 on data protection by design, that it is possible for anonymisation to go together with state-of-the-art detection performance.

Overall, the experiments validate four central claims.

1. Performance and generalisability: Since they just use meta-data, the variance will be minimized and the performance would be near perfection across different datasets.
2. Model hierarchy: Ensemble and boosting methods are better than linear baselines in under both binary and multiple conditions.
3. Explainability: SHAP analysis provides transparency towards the influential features to maintain interpretability, even towards complex models..
4. Privacy robustness: Statistical noise has negligible loss of utility, which provides evidence of a (quantifiable) privacy-utility trade-off.

This work in sum will confirm the possibility of privacy-conserving ID and will go beyond existing research as it empirically connects the properties of interpretability, scalability and data-protection-compliance in one unifying framework.

5.3 Comparative Analysis with Recent Studies

Recent intrusion detection research (2022-2025) is shown to be aligned but also divergent with the results on CICIDS2017 using logistic regression, random forest, and XGBoost with metadata-only features. Several studies using similar data sets report comparably high accuracy of detection. For example, on CICIDS2017 a popular benchmark a widely-used ensemble techniques such as XGBoost have consistently performed on top positions. Adewole et al. (2025) explored 5 ensemble learning algorithms where XGBoost algorithm performed better with approx. 99.9% accuracy on

CICIDS2017. This is consistent with our observation that tree-based ensembles are often stronger classifiers on this dataset when compared with the simpler classifiers. Deep learning techniques have also achieved great performance: Alsubaei (2025) applied a Sequential Neural Network (a modified deep model) that achieved 99.5% accuracy on CICIDS2017. Such near-perfect scores emphasise the fact that CICIDS2017, although being full-fledged, can still be learned effectively by modern models particularly with much hyperparameter tuning and preprocessing. These same studies show slightly lower performance on other datasets, indicating the difference between datasets. For example, the optimised deep model in Alsubaei (2025) obtained 96.8% on UNSW-NB15, which is a challenging dataset developed in 2015 and suggests that UNSW-NB15 attack patterns are somewhat more difficult to classify than the attack patterns in CICIDS2017. Similarly, more recent benchmarks such as CSE-CIC-IDS2018 have been utilised in the validation of advanced methods. Hewapathirana (2025) utilized a two-stage classifier on CSE-CIC-IDS2018 and encouraged the realism and the diversity of the latter (14 attack types) as a testbed of scalable solutions. Overall, the results on CICIDS2017 are like recent works: simple models (e.g. logistic regression) achieve decent but lower accuracy, whereas ensembles and deep networks push towards 99 – 100 percent on this dataset. Differences appear on other data with fancy models required to keep high accuracy on data sets such as UNSW-NB15 or those with class imbalance problems.

5.3.1 Model Choices and Trade-offs

Recent research looks at a variety of different model types from the traditional supervised learners that we've been using. Ensemble tree-based models are also popular in the space because of their good performance. Adewole et al. (2025) confirmed that boosting and bagging algorithms (XGBoost, Random Forest, etc.) provided excellent detection rate on IoT intrusion data. The trade-off is that while deep neural nets can be less interpretable, so are ensembles though they generally are faster to train than the deep models. To mitigate this, Adewole *et al.* integrated rule induction to extract human-readable rules from ensemble decisions, improving explainability without substantial loss in accuracy. On the other hand, deep learning methods (CNNs, RNNs, transformers) have seen extensive adoption from 2022 onward, often in hybrid architectures. These models can automatically learn complex temporal and spatial traffic patterns, yielding state-of-the-art results, but they demand more computation and can introduce latency. For instance, a

carefully tuned deep model (with CNN/GRU layers) can detect a broad range of attacks with high recall, at the cost of longer training and inference times. Some researchers combine approaches to balance these factors. A common theme is hybrid models e.g. using machine learning for feature selection or first-stage filtering, followed by deep learning for refined classification. The study by Alsubaei et al., 2025 fused a Random Forest with an autoencoder in a two-stage IDS to reduce false positives, leveraging the fast, interpretable output of RF as a trigger and a neural network to confirm anomalies. Similarly, others have stacked classifiers or used ensemble voting to improve robustness against varied attack types. In terms of runtime and resource constraints, recent work also explores *lightweight* models. Xu *et al.* (2025) propose a low-parametric deep model (ET-Mamba) for encrypted traffic that achieved ~98% accuracy on VPN traffic with only 1.8 million parameters. Such models demonstrate a trade-off where carefully designed architectures (e.g. using efficient attention or graph neural networks) can approach the accuracy of large models while reducing memory and CPU overhead. In summary, contemporary IDS research often opts for ensemble or deep learning models for maximum detection performance, and then addresses their downsides via model compression, hybrid designs, or explainable AI techniques.

5.3.2 Privacy-Preserving Strategies

A recent trend in this area of study is how privacy and encryption are addressed: a topic that is directly related to the metadata-only approach. The approach of extracting just traffic metadata (packet sizes, timings, headers) follows the general trend of encrypted traffic classification techniques which do not inspect payloads. Akande et al (2023) observe that modern IDS must operate on encrypted flows by using metadata features such as packet length series, flow durations and inter-arrival times. These features, which this study also relies on, have been shown to be effective for distinguishing malicious patterns without decrypting traffic. For example, even in QUIC a protocol with encrypted headers Al-Bakhat and Almuhammadi (2022) demonstrated that flow fingerprints (e.g. specific timing and size patterns) enable detection of malicious command-and-control traffic with about 98% accuracy. This underscores that metadata-only models can achieve high accuracy while inherently preserving content privacy. Beyond feature choice, recent work increasingly employs privacy-preserving machine learning techniques. Federated learning (FL) is prominent in recent IDS research to train models across distributed data sources without centralizing sensitive data. For instance, Mosaiyebzadeh *et al.* (2025) developed an FL-based IDS

for Internet-of-Healthcare-Things that keeps patient network data on-device; their framework, combined with differential privacy, reached ~95% intrusion detection accuracy with only minimal privacy budget expenditure. They integrated differential privacy (DP) noise into model updates to prevent leakage of any individual's data, illustrating a trade-off between privacy and performance the model's accuracy modestly decreased due to DP, but remained high (over 93% on one dataset) while significantly improving privacy guarantees. Mosaiyebzadeh *et al.* (2025) also introduced differentially private federated IDS for industrial IoT, showing that combining FL and DP can still detect attacks effectively (with only a few percent loss in accuracy) while ensuring no raw data is shared. In comparison, the metadata-only approach used in this study is a simpler form of privacy preservation as it does not protect data during model training as FL/DP do, but it limits the features to non-payload information, thereby respecting encryption and user privacy by design (Akande et al., 2023). This strategy has low overhead and no need for noise injection, but it assumes that metadata alone is rich enough for classification. Reassuringly, multiple studies confirm that to be the case: even without packet contents, AI-based IDS can identify attacks by patterns in traffic statistics (Akande et al., 2023; Al-Bakhat and Almuhammadi, 2022). Across the literature, we see a convergence toward privacy-preserving intrusion detection whether via feature selection (metadata only), distributed learning, or encryption-aware modelling ensuring that security analytics do not come at the expense of privacy. This work fits into this paradigm, and the comparative evidence suggests that the chosen techniques are in step with state-of-the-art developments in both efficacy and privacy considerations.

5.4 Implications for Practice

For operational environment, there is a necessity to strike the right balance relating to privacy-preserving classifiers of encrypted traffic, in terms of accuracy, interpretable rules, and resource efficiency. On a dataset consisting of two tables, the ensemble Random Forest and the XGBoost were shown to retain good accuracy (99%) using only metadata and achieve near-real-time throughput over that dataset. These results show that high-fidelity detection no longer relies on payload inspection, which is an important result for deployment in encrypted networks in which content analysis violates the confidentiality policies. That is, the complexity of the model must be tied to the capabilities of the infrastructure. Simpler models like Logistic Regression incur negligible overhead (≈ 80MB RAM, 3% CPU) and run very quickly (≈ 40ms per inference) making

them attractive for edge-level monitoring/edge computing or IoT monitoring (Chamkar et al., 2025). However, the ensemble algorithms tested here demonstrated an ability to process 500 flows per second on standard multicore hardware with <5% CPU utilisation, confirming that they too can be used for production SOC environments with appropriate optimisation. When latency or throughput are important, the ensembles can be compressed into light-weight surrogates without serious compression of the accuracy using the tools of model-distillation and quantisation (Xu et al., 2025).

Another major operational finding concerns interpretability. The integration of SHAP analysis provides analysts with clear feature-importance visualisations, bridging the gap between black-box models and human decision-making. As shown in Figure 4.9, features such as *Flow Duration* and *Backward Packets per Second* consistently influenced model predictions. Exposing these attributes within SIEM dashboards enables analysts to understand why alerts are triggered, which in turn enhances accountability and accelerates incident triage (Chamkar et al., 2025; Xu et al., 2025). This study therefore demonstrates that ensemble models can remain transparent enough for operational trust if augmented with explainable-AI tools.

Real-time detection performance and alert quality are equally important. The high precision observed in both datasets (AUC ≈ 1.000, AP ≈ 1.000) indicates that the models can substantially reduce false positives. In production settings, calibrated probability outputs allow threshold tuning to prioritise precision, preventing analyst fatigue and improving SOC efficiency. Prior work supports this outcome: in a live SOC integration, an ML-enhanced IDS reduced false alerts by 78 % (Chamkar et al., 2025). Our findings therefore confirm that privacy-preserving machine-learning models can meet stringent latency and reliability benchmarks without degrading security performance.

Finally, the privacy–utility validation has direct practical implications. Noise-injection experiments (Figure 4.10) revealed < 0.5 % accuracy loss even when moderate Gaussian noise ($\sigma = 0.10$) was applied. This demonstrates that privacy reinforcement can be operationalised without meaningful performance degradation an important insight for industries subject to strict confidentiality regulations. Together, these results affirm that privacy-preserving IDS solutions are not only theoretically sound but also operationally viable across a range of deployment scenarios.

5.5 Regulatory and Privacy Compliance

This proposed classification framework reflects ideas of privacy by design and data-minimisation which form the foundations of the General Data Protection Regulation (GDPR) and other international privacy frameworks. Because it is based only on flow-level meta-data (excluding packet payloads) and hence processes only what is needed for intrusion detection, the system meets the GDPR Article 5(1)(c) data minimisation criterion (Bateman, 2025). This architecture has the property of not allowing access to user content by default and thus lowers the risk of processing personally identifiable information (Sattar et al., 2025).

Where metadata might still expose identifiers (e.g. IP addresses), the research recommends that this should be pseudonymised or hashed before storage in line with EDPB (2025). In addition, the noise-injection simulation presented in this paper shows a feasible and quantitative approach towards fulfilling GDPR Article 25 ("Data Protection by Design and by Default"). By showing small perturbations bring about practically no utility loss to the model, however preserving model accuracy, while at the same time enhancing anonymization, that work contains empirical support for the fact that privacy engineering could truly be measurable and effective.

Comprehensive compliance, however, goes beyond in advance processing of data. Since metadata may still be considered personal data, the processing basis, transparency and access control must be respected (Harmeling, 2025). On the other hand, model-level protection techniques such as differential privacy and federated learners are also critical to counter inference attacks by not transmitting either training data or model parameters that could contain sensitive information (Abbas et al., 2025; EDPS, 2025). This brings the framework in line with broader regulatory expectations which have a privacy-mainstreaming which emphasises privacy in the AI lifecycle.

Similar protections are also internationally underway in the CCPA/CPRA in California that also highlights the importance of transparency and user rights. Standards, like ISO 27001, increase those requirements by requiring technical and organizational protection controls to secure processing (Terlecki, 2025). In fact, the combination of metadata restriction, pseudonymisation and controlled noise addition puts this framework well within the context of global best practise. If implemented at an operational level, it is also recommended that organisations have tight retention caps in place, along with access governance and on-premise model training to ensure those facilities' technology compliance with the data-locality and accountability obligations.

In summary, the experiments confirm that strong ID and regulatory compliance do not conflict with each other. By empirically validating anonymisation and privacy-utility trade-offs, this study extends the literature from conceptual alignment to practical evidence, showing how machine-learning-driven IDS can operate within the letter and spirit of modern data-protection laws.

5.6 Closing Synthesis

This research has shown that high detection accuracy and operation privacy are fully compatible goals in today's network intrusion detection. By using only metadata on the flow level and not examining the payload, the research was able to reach performance levels that compete or exceed conventional, non-private systems. With both CICIDS2017 and CSE-CIC-IDS2018 datasets, the ensemble learning methods such as Random forest and XGBoost continually achieved 99+ % correct classification indicating the feasibility of privacy-preserving classification with metadata in the encrypted-traffic classification. These results are a contrib6ution to the state of the art as they empirically demonstrate that strong generalisation, interpretability and privacy compliance can be combined in a single framework. The analysis of the recent literature (Fatema et al., 2025; Elshewey and Osman, 2025; Sebetci and Simsek, 2025) puts this work at the forefront of privacy-aware intrusion-detection literature. While most past works individually focused more on just guaranteeing accuracy on a single dataset, or proving accuracy alone on a single index, the dissertation applied validation on two benchmark datasets, applied a quantifiable privacy-utility index, and used machine-interpretable frameworks like SHAP to reveal model justification. It also gave concrete evidence that noise-based anonymisation is a strong privacy-enhancing scheme that has negligible (< 0.5 %) performance overheads, this is a rare quantitative connection between machine-learning optimisation and regulatory compliance. In practical terms, the findings provide a deployable pathway for those organisations that look to find GDPR compliant intrusion detection solutions. The illustrated throughput (approximately 500 flows/s) and near-perfect accuracy points to the carbon footprint of these models enough to operate efficiently in real-time SOC pipelines where auditability is also assured through interpretability of feature attributions. In concept, the paper supports the idea that less thought metadata can, if modelled properly, offer a wealth of behavioural metadata of the malicious activity without sacrificing confidentiality. Theoretically, the research adds to privacy-preserving machine-learning literature by empirically linking three areas that are typically considered independently of each other: Performance optimisation

(Ensemble learning & multiple dataset validation). Interpretability (SHAP-based model decision transparency); and Privacy engineering (anonymisation and noise resilience in the context of the GDPR principles). Together, these strands form a basis for building the next generation of systems that are both effective in terms of performance and acceptable on grounds of ethics. The future direction of the study as noted above is clear: scaling to more encryption schemes that use TLS 1.3 and QUIC; incorporating federated and differential-privacy approaches; and monitoring on streaming encrypted telemetry to eliminate the lab-to-field gap.

If pursued, such work could produce the first fully distributed, explainable, and regulation-aligned intrusion-detection framework suitable for cloud and edge deployment.

In conclusion, this dissertation offers an original and rigorous contribution to the field of privacy-preserving network analytics. It provides empirical proof that machine-learning-based intrusion detection can uphold both accuracy and privacy, demonstrating that responsible AI design is not only possible but essential for the future of secure, ethical, and trustworthy digital infrastructure.

Chapter 6

Conclusion, Limitations and Recommendations

6.1 Introduction

This chapter concludes the research on privacy-preserving network intrusion detection using supervised machine learning. It brings together the experimental, theoretical, and regulatory insights developed throughout the dissertation to demonstrate that effective intrusion detection can coexist with strict privacy protection. Earlier chapters established the research question: Can encrypted-traffic classification achieve high accuracy and interpretability without compromising user confidentiality?

Through dual-dataset evaluation on CICIDS2017 and CSE-CIC-IDS2018, three supervised algorithms Logistic Regression, Random Forest, and XGBoost were tested under metadata-only, privacy-by-design constraints. The study also incorporated explainable-AI (SHAP) analysis to reveal feature importance and a privacy–utility validation using controlled Gaussian noise injection to quantify how anonymisation affects accuracy. Together, these experiments provide the first comprehensive empirical evidence that modern ensemble and boosting models can maintain near-perfect accuracy, interpretability, and GDPR alignment using metadata alone.

This final chapter therefore summarises how each research objective was achieved, highlights key contributions to theory and practice, and reflects on limitations that guide future investigation. It also consolidates the methodological and ethical implications of privacy-aware artificial intelligence, showing how responsible model design can strengthen trust, compliance, and operational resilience in encrypted-traffic environments

6.2 Summary of Research Objectives and Findings

Objective 1 – Literature Review

The paper was launched by an extensive survey of the research field in encrypted-traffic classification, where a lack of very diverse datasets and integration of privacy is identified in the literature. Functions with payload-based detection are considered broken, as it has been shown that many TLS 1.3 traffic details make it pressureless to detect payloads, and the discovery community

continues to move towards metablog-based machine learning techniques (Elshewey & Osman, 2025). Several studies based their work from outdated data prior to 2018 and did not account for privacy compliance and interpretability (Wickramasinghe et al., 2025). The weaknesses were confirmed by a recent systematic review by Sharma & Habibi Lashkari (2024) who stated a need for modern privacy-aligned solutions, which is exactly what this research did.

Objective 2 – Methodology Design

A privacy-preserving methodology framework is developed (Chapter 3) which extracts only flow-level metadata packet sizes, inter-arrival times and directions, and byte counts without decrypting payloads. This design is in accordance with the principle of data minimisation of the GDPR (Sattar et al., 2025) and very powerful in terms of analysis. To describe and identify behavioural patterns, machine-learning classifiers were trained to learn such non-content features written according to best practices for encrypted-traffic analysis (Elshewey & Osman, 2025).

Objective 3 – Data Preparation and Prototype Development

Two benchmark datasets CICIDS2017 and CSE-CIC-IDS2018 were pre-processed to eliminate payload data and standardize the metadata features. A prototype IDS pipeline was implemented that captures raw network traces, does feature engineering and classifies traffic in real time. The system was able to prove that metadata could be used as a robust proxy for behavioural recognition test, confirming the viability of the proposed privacy preserving architecture (Malekghaini et al., 2023).

Objective 4 – Experimental Evaluation

The performance of three classifiers was evaluated on both datasets in the binary and multi-class case. On the CICIDS2017, the accuracy of Random Forest and XGBoost was ~97% and the accuracy of Logistic Regression was 78%. Replicated on CSE-CIC-IDS2018, the ensembles were able to provide above 99.9 % accurate results with little variance in a 5 fold cross validation demonstrating the very good generalisability. The Area Under ROC and Precision-Recall curves were 1.000, which exceeds reported state-of-the-art results (Sebetci & Simsek, 2025 and Alsubaei et al., 2025). SHAP interpretability analysis revealed Flow Duration, Forward Segment Size Minimum, and Backward Packets per Second as the top metadata predictors in line with the previous findings (Mohale & Obagbuwa, 2025).

Objective 5 – Privacy Assessment and GDPR Alignment

The privacy dimension was empirically and conceptually validated. The noise-injection approach showed that a small loss of accuracy (less than 0.5%) was achieved by introducing Gaussian noise (s= 0.02 - 0.10) into metadata, providing clear evidence of a good privacy vs. utility trade-off. Because the data processing involves only metadata and because the framework always maintains encryption, the framework meets the requirement of GDPR Article 25 (data protection by design and by default). No personal payload information has been processed or inferred when classifying, which is in full compliance with modern privacy legislation (Sattar et al., 2025).

Therefore, all research goals were achieved. In addition, the study not only realized the high-accuracy, metadata-only detection, and further extended the previous studies by quantitatively evaluating privacy resilience, demonstrating cross-dataset generalization capability, and integrating interpretability into a regulation-compliant intrusion-detection framework.

6.3 Limitations

This study has several technical, methodological, and operational limitations that shape opportunities for further research and refinement.

6.3.1 Dataset Constraints

The datasets used CICIDS2017 and CSE-CIC-IDS2018 provided a reliable basis for benchmarking but still fall short of real-world diversity. Both rely on controlled or semi-synthetic traffic captures, which limits the realism of the evaluated intrusion patterns (Goldschmidt and Chudá, 2025). Although synthetic data enables reproducibility and privacy, it cannot fully replicate the heterogeneity, noise, and unpredictability of live internet traffic. Most publicly available datasets include a mix of legacy encryption protocols and outdated ciphers that predate TLS 1.3. This restricts the assessment of classifiers under modern encryption standards such as TLS 1.3 and QUIC, where payloads and even header fields are obscured. Consequently, while the ensemble models performed exceptionally well on existing benchmarks, their performance on fully encrypted QUIC flows or zero-day encrypted attacks remains to be verified.

Class imbalance also persists as a structural limitation. As in many intrusion datasets, benign flows vastly outnumber attack instances, which biases classifiers and reduces their sensitivity to rare

threats. Although data balancing and stratified sampling were employed, the fundamental scarcity of minority classes constrains the detection of infrequent or emerging attack types.

6.3.2 Model-Level Issues

While the supervised models achieved near-perfect accuracy on both datasets, generalisation to other network domains cannot be assumed. Overfitting to the statistical signatures of benchmark datasets remains possible, even under cross-validation. Similar concerns have been raised by Wickramasinghe et al. (2025), who note that intrusion-detection models frequently degrade under cross-network evaluation due to contextual differences. The problem of false positives also remains: even high-precision models may produce a significant number of alerts, which puts a cognitive burden on the security analysts (Waghmode et al., 2025). On the other hand, new types of disguised attacks may not be detected since supervised models are based on patterns observed during training. Since public datasets usually represent only a fraction of known attack types, unrepresented or asserted changing behaviours still remain a detection gap.

6.3.3 Operational Challenges

Implying the following: network traffic is dynamic which gives rise to concept drift characterized by changes in the feature distributions as evolution occurs in the applications and usage patterns. Without periodical retraining or online learning, the model accuracy will inevitably decline (Malekghaini et al. 2023). Also, the agile evasion that is described above may not be a final solution since perversive attackers may try to fool the classification systems by varying timing between packets or changing the number of bytes per packets whereas, classifiers use statistical criteria (Ennaji et al. 2024). It can thus be concluded that the opaque decision boundaries of ensemble models while reduced by SHAP, can still be a limiting factor to interpretability in high-stakes operational scenarios. Making things more explainable by more sophisticated dashboards or human verification is also essential to build trust among analysts (Mohale and Obagbuwa, 2025).

6.3.4 Privacy and Ethical Limitations

Although the study empirically validated the design for privacy-preserving study design through noise injection experiments, there are no federated learning or differential privacy mechanisms implemented that protect data during model training. The system therefore ensures privacy at the

feature and inference level and not at the collaborative learning stage. The implementation of these mechanisms would enhance an end-to-end compliance and resistance to model-inversion attacks. Furthermore, the fact that no evaluation was conducted on live encrypted telemetry means that no full use was made of the data collection/collection protocols set out in international regulations, nor of evaluation protocols, and of procedures for ensuring user consent. Using such systems in production needs to handle ethical issues as well. Even metadata, if aggregated over a period, can provide behavioural insights of users or devices. Organisations must therefore have clear guidelines of data retention, anonymisation and human oversight to ensure that intrusion detection efforts' attempt to gain visibility across each chunk of information does not repurpose unintentionally in privacy-vulnerability. In sum, while the research provides a strong empirical baseline and proven foundation there are important areas of further development such as gaining realism for modern encrypted traffic, the federated and differential privacy learning, resistance to adversarial adaptation and building transparent and responsible interfaces for researchers.

6.4 Contribution to Knowledge

This dissertation makes contributions to the academic research and professional practise of privacy-preserving intrusion detection. First, it experimentally shows that it is indeed possible to achieve state of the art accuracy, interpretability, and compliance with modern supervised learning (potentially heterogeneous) models using metadata-only inputs. The contributions can be grouped into three complementary domains: theoretical, methodological and practical.

6.4.1 Theoretical Contributions

The research increases the theoretical foundation for metadata-driven network analytics by showing that payload inspection is not required to perform intrusion detection correctly. By only depending on statistical flow details like packet length, timing and direction, it confirms that behavioural properties of encrypted traffic can provide an insight into malicious intent without ever decrypting any user content. This finding also strengthens previous research that claimed the class contention of encrypted traffic can occur on the level of flow behaviour rather than content access (Liu, Chen and Guo, 2025; Binbusayyis, 2024). Moreover, their experiments with CICIDS2017 and CSE-CIC-IDS2018 empirically validate that the brittleness of-known application metadata patterns to the varying network contexts, thus addressing the long-standing concern toward scalability of the privacy-preserving IDS models. To provide new theoretical

justification for the trade-off between confidentiality and analytical performance, an added quantitative privacy-utility analysis indicates loss of less than 0.5% accuracy and is demonstrated under controlled noise injection novel in concept, readily substantiated experimentally.

6.4.2 Methodological Contributions

Methodologically the dissertation is a unified evaluation framework that combines the issues of model performance, model interpretability and privacy alignment under one experimental design. This hybrid framework is different from the classic IDS studies where accuracy is assumed to be the sole measure of success. With SHAP-based explainability, a simple explanation is that it was fed into the evaluation pipeline, even though it is a complex ensemble model, and it provides a highly auditable insight into how its decision mechanism works (Mohale and Obagbuwa, 2025). Addition of a quantifiable privacy dimension (Gaussian perturbation) and IPC (GDPR) compliance following Article 5 and 25, thus opens a repeatable practice for the privacy resilience. With this methodological template, we introduce an alternative imperative for understanding both the "how well" of attack detection of an IDS as well as the "how" in terms of the ethical and compliance constructs of a Responsible and Explainable AI approach to cybersecurity.

6.4.3 Practical Contributions

Specifically, the project will perform and deliver a prototype intrusion detection pipeline deployment that competes near real time throughput while maintaining encryption integrity. The achieved performances of Random Forest and XGBoost models showing more than 99% accuracy with throughputs reaching around 500 flows s-1 demonstrates the feasibility of privacy-aware IDS for use in production. The research also adds to the operational guidance on how to keep it accurate (using lightweight optimisation techniques (model distillation, quantisation)) and boost analyst trust (using explainable-AI dashboards to visualise important SHAP features).

From a governance perspective, the work generated by the work to date fills the gap between cybersecurity engineering and regulatory compliance. It demonstrates that IDS that is based on machine learning can comply fully with the GDPR and ISO 27001 monitoring requirements by restricting data collection to the anonymised metadata, implementing pseudonymisation, and quantifying the privacy benefits (utility). Such insights give organisations a blueprint for

implementing AI deliverable security analytics in an effective and legally defensible manner. In sum, the dissertation makes the following contributions:

1. The first dual-dataset empirical validation of metadata-only IDS models achieving near-perfect accuracy.
2. A quantitative privacy-utility model for linking the strength of anonymisation and performance.
3. Implementing model interpretability using SHAP to provide a robust decision support tool, which can be applied to model interpretation in a regulation context.
4. A methodologically replicable template used for assessing privacy, compliance and performance in unified fashion.

Collectively, their contributions will finally mark the progress of privacy preserving machine learning from theoretical feasibility to demonstrated feasibility, thus putting privacy preserving machine learning into a mature and operational category for the next generation of intrusion detection.

6.5 Practical and Policy Implications

Operationally, the proposed privacy-preserving traffic classifier shows that it is possible to achieve high performance intrusion detection without the loss of user confidentiality. The experiments prove that ensemble models like Random Forest and XGBoost-A preto maintain almost same line-rate ordering throughput of around 500 flows/second with performance accuracy above 99%. This gives them the foundation for Security Operations Centre (SOC) environments that involve both swiftness and dependability (Elshewey & Osman, 2025).

6.5.1 Practical Implications

For the practitioners, results provide engineering guidance info. First, privacy-utility simulation demonstrates that the addition of small Gaussian noise ($\sigma \leq 0.05$) within metadata performs well in terms of providing good anonymisation with mild loss of accuracy (< 0.5%). This threshold can be used by organisations to establish a safe operational baseline to improve privacy without reducing system performance. Second, SHAP-based interpretability dashboards are brought into the mix, such that analysts can then visualise which traffic features (e.g., Flow Duration, Forward

Segment Size Minimum, Backward Packets per Second) have the greatest impact on each decision. Presenting these attributions happened to SOC interfaces increases the trust, facilitates incident triage and minimizes the alert fatigue - which becomes increasingly evident in large-scale monitoring (Chamkar et al., 2025; Xu et al., 2025)

The models' calibrated probability can also go into adaptive alert thresholds that focus on high confidence events and turn on the light to low risk outliers. In similar instantiations, the probabilistic gating has been found to be effective and can decrease false positives by up to 78% (Chamkar et al., 2025) Together, these practices make machine-learning classifiers switch from being opaque detectors into being transparent and fully auditable decision-support lamps that augmentable human expertise.

6.5.2 Policy and Regulatory Implications

From a governance perspective, the research directly addresses issues of privacy-by-design obligations in the GDPR in Article 25. Because the classifier only analyses metadata of the flows and never decrypts the payloads it processes the minimum information necessary for network defence, which complies with the data-minimisation-rule (Article 5(1)(c)) (Bateman, 2025). It extends the state of the art by presenting quantifiable evidence that, with the empirical noise-injection techniques validated, anonymisation is a viable approach to achieve the required utility with acceptable corrective mechanisms which bridge a significant gap between compliance in principle and assurance in practice.

In regulated sectors such as finance, healthcare or public administration, this framework shows that a framework is working in which intrusion detection based on machine learning is possible on the one hand, and legal privacy requirements, on the other. Organisations can introduce pseudonymisation, mandatory retention limits, and keep a model training process on-premise while taking advantage of the AI features for improved monitoring. The strategy also matches international standards, such as ISO 27001 Annex A 8.16 Continuous Monitoring and Anomaly Detection (Whiting, 2025).

Through the combination of GDPR compliance and ISO Certified Operational Practices, the framework allows for a governance outline producible that under Ideal network analytics conditions, is compliant in security, ethical and regulatory guidelines. It indicates that privacy

preserving artificial intelligence is not a theoretical aspiration but can be used as a practical and evidence-based policy tool and support both cybersecurity resilience and data protection accountability.

6.6 Recommendations for Future Research

6.6.1 Technical improvements

Future research studies may expand encrypted-traffic classification to apply to the new encryption standards, particularly to TLS 1.3 and QUIC, for which both payload and header metadata are highly obfuscated. Benchmarking classifiers on such protocols will challenge the limitation of the detection using metadata only and shape the architecture of the next generation features. In furtherance, transfer learning and continual-adaptation paradigms that help models to maintain the accuracy over evolving traffic should also be studied. Online learning and domain adaptation techniques can lessen the effects of concept drift and aid in the stability of deployment in the long term (Malekghaini et al., 2023).

6.6.2 Privacy-Preserving and Collaborative Learning

While this research established privacy in terms of feature and inference stage validations, there is still a need to combine federated learning and differential privacy mechanisms to preserve privacy during model training in future work. Federated learning could help several organisations to contribute to a common global model with no local data exposures whereas differential privacy can mathematically ensure that individual records are indistinguishable (Giddens et al., 2023). Using these methods together would help create end-to-end privacy compliance with increased resiliency against model-inversion attacks.

6.6.3 Explainability and Human Oversight

Further improvement of explainable AI (XAI) is needed. Despite SHAP, around researchers could blend LIME clarifications, counterfactual reasoning, and human-in-the-loop interfaces to guarantee model results keep on being oversaw by and clear to researchers (Akande et al., 2023). Embedding interactive explanation panels in SOC dashboards would improve accountability and bring intrusion-detection systems into the context of emerging principles of AI governance.

6.6.4 Deployment, Scalability, and Realism

To help bridge the disconnect between laboratory evaluation and production deployment, research in the future should leverage the opportunity to use telemetry of live encrypted traffic from enterprise or ISP networks with the appropriate ethical and legal authorizations. Such real-world testing will allow testing for scalability, energy efficiency, and latency under real-world loads and operational load. Alternative approaches of model-compression, pruning and adaptive inference pipelines should also be investigated to strike a trade-off between the accuracy vs. resource constraint (Wickramasinghe et al., 2025).

6.6.5 Ethical and Regulatory Dimensions

Finally, ethical design and reproducibility are still of great importance. Future frameworks would contain open reporting of data-handling practices, open source code, and in addition, reproduceable benchmarks to allow building trust and accountability for the research community. It is important that collaboration between academia, regulators, and industry is fostered in order to ensure that the responsibility and fairness of the development of privacy-preserving AI in cybersecurity is present.

6.7 Conclusion

This dissertation aimed to find out whether encrypted network traffic can be accurately and correctly classified using supervised machine learning models based only on metadata, without causing any loss to user privacy. By a structured investigation differing in literature review, methodological design, prototype implementation, evaluation of dual datasets it were proven that privacy preserving intrusion detection is technically feasible and operationally scalable.

Empirical studies in CICIDS2017 and CSE-CIC-IDS2018 both indicate that ensemble models like Random Forest and XGBoost can result in state-of-the-art accuracy (e.g. 99%) and phenomenal generalisation using only flow-level metadata. The introduction of Gaussian noise-based anonymisation has proved that privacy reinforcement might be achieved with negligible (< 0.5 %) performance loss which represents the first quantitative validation of how the privacy-utility balance holds in this case. Complementary SHAP explainability also made the model reasoning clearer, so that black box algorithms can be auditable and trustworthy. Together, these findings

validate a unified framework that integrates performance optimisation, interpretability, and regulatory compliance within a single privacy-aligned architecture.

Theoretically, the research brings privacy-preserving IDS one step ahead from the promise of conception to the accomplishment of proof. Methodologically, it provides a replicable evaluation methodology that addresses both quantitative evaluation of accuracy and ethical and legal evaluation. Practically it provides a deployable, regulation-compliant solution that meets the requirements of GDPR Articles 5 and 25 as well as fall under the requirements of ISO 27001 Annex A 8.16 for continued monitoring. These results, altogether, show that strong cybersecurity as well as user privacy reinforce each other rather than negate them.

Beyond its technical achievements, the work carries societal relevance. By promoting secure yet transparent network analytics, it supports UN SDG 9, which advocates resilient digital infrastructure and responsible innovation, and SDG 16, which promotes strong, accountable institutions. The study thus contributes not only to scientific knowledge but also to the ethical governance of emerging technologies.

In conclusion, this dissertation establishes that metadata-driven, explainable, and regulation-compliant machine learning can form the foundation of next-generation intrusion-detection systems. As encryption, privacy regulation, and cyber threats continue to evolve, the integration of responsible AI principles into network security will be essential to building trustworthy, secure, and sustainable digital ecosystems.

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